SingleRAN

LTE FDD and NR Hybrid Dynamic Spectrum Sharing Feature Parameter Description

Issue Draft B

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Contents

1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	Base Station Model
Deleted the mutually exclusive relationship between Hybrid DSS Based on Asymmetric Bandwidth and uplink-interference-based inter-frequency handover from **4.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added impact relationships of PSD adaptation with "NR DU cell resource management between network slices" and "percentage-based dynamic RB sharing among operators". For details, see **4.3.2.3 Function Impacts**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

- Editorial changes
  - Added descriptions of the cell requirements for configuring the **NRDUCellGigaBand**.***LnrUlRbEstCorrectCoeff*** parameter. For details, see **7.3.3 Hardware**.
  - Added descriptions of configuration requirements, network impacts, and compatible RF modules for the "LTE and NR 3-ms frame offset difference" function when it is enabled together with Hybrid DSS Based on Asymmetric Bandwidth. For details, see **4.1.1.3.3 Common Functions Supported by Both LTE and NR**, **4.2.2 Impacts**, and **4.3.3 Hardware**.
  - Revised descriptions in this document.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 08 (2025-12-14).

- Technical changes

Change Description	Parameter Change	Base Station Model
Enabled the UMPThb to support Hybrid DSS Based on Asymmetric Bandwidth. For details, see **4.3.3 Hardware**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Enabled 16T16R RF modules to support Hybrid DSS Based on Asymmetric Bandwidth. For details, see **4.3.3 Hardware**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added incompatibility of BTS5929R with Hybrid DSS Based on Asymmetric Bandwidth. For details, see **4.3.3 Hardware**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added LNR independent configuration of uplink and downlink resource percentages. For details, see: • **5.1 Principles** • **5.2.1 Benefits** • **5.2.2 Impacts** • **5.3.2.1 Prerequisite Functions** • **5.3.2.2 Mutually Exclusive Functions** • **5.3.2.3 Function Impacts** • **5.3.3 Hardware** • **5.4.1.1 Data Preparation** • **5.4.1.2 Using MML Commands** • **5.4.2 Activation Verification**	Modified parameter on the LTE side: Added the **LNR_INDEP_UL_DL_RES_RATIO_SW** option to the **SpectrumCloud**.***SpectrumCloudExtSwitch*** parameter. Added parameters on the LTE side: • **SpectrumCloud**.***UlLtePreferredResRatio*** • **SpectrumCloud**.***DlLtePreferredResRatio***	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added support for simultaneous use of Hybrid DSS Based on Asymmetric Bandwidth and the LTE and NR flexible frame offset function. For details, see: • **4.1.1.3.3 Common Functions Supported by Both LTE and NR** • **4.1.2.3.3 Common Functions Supported by Both LTE and NR** • **4.3.1 Licenses** • **4.3.2.1 Prerequisite Functions** • **4.3.2.2 Mutually Exclusive Functions** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Modified parameter on the NR side: Added the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option to the **NRDUCellSpctShare**.***LnrSpectrumShareSwitch*** parameter. Modified parameter on the LTE side: Added the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option to the **SpectrumCloud**.***SpectrumCloudExtSwitch*** parameter.	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added support for LTE and NR Uplink Spectrum Pooling. For details, see: • **2.3 Features in This Document** • **2.4 Differences** • **3 Overview** • **7 LTE and NR Uplink Spectrum Pooling**	Added parameters on the NR side: • **NRDUCellGigaBand**.***GigaBandCellAlgoSw*** • **NRDUCellGigaBand**.***LnrPucchRbAdjProtDuration*** • **NRDUCellGigaBand**.***LnrUlRbEstCorrectCoeff*** Added parameters on the LTE side: • **GigaBandCell**.***GigaBandCellAlgoSw*** • **GigaBandCell**.***ServiceBasedOptMaxEffTime*** • **PUCCHCfg**.***NrPucchExtendedRbNum*** Modified parameter on the LTE side: Added the **GRP_NO_PREALLOCATE_TRIG_OPT_SW** option to the **AsParaGroup**.***AccelerationServOptExtSw*** parameter.	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Revised the constraints affecting cell activation due to license insufficiency. For details, see **4.3.1 Licenses**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added compatibility of 16T16R with Hybrid DSS Based on Asymmetric Bandwidth. For details, see: • **4.1.1.3.3 Common Functions Supported by Both LTE and NR** • **4.1.2 Hybrid DSS Based on Asymmetric Bandwidth (Multiple LTE Cells)** • **4.2.1 Benefits** • **4.3.2.2 Mutually Exclusive Functions** • **4.3.2.3 Function Impacts** • **4.3.3 Hardware** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands** • **5.3.3 Hardware** • **6.3.3 Hardware**	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship between Hybrid DSS Based on Asymmetric Bandwidth and LNR CRS muting. For details, see **4.3.2.3 Function Impacts**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Removed the mutually exclusive relationship between Hybrid DSS Based on Asymmetric Bandwidth and downlink adaptive selection of resource allocation type 0 or 1 from **4.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship between Hybrid DSS Based on Asymmetric Bandwidth and downlink adaptive selection of resource allocation type 0 or 1. For details, see **4.3.2.3 Function Impacts**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added configuration requirements for adjusting the compression degree of common PUCCH RBs in Hybrid DSS Based on Asymmetric Bandwidth scenarios. For details, see **4.3.5 Others**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added a mutually exclusive relationship between Hybrid DSS Based on Asymmetric Bandwidth and Static Multiple Beam 2.0. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship between Hybrid DSS Based on Asymmetric Bandwidth and intelligent beam weight optimization. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship between Hybrid DSS Based on Asymmetric Bandwidth and LTE flexible spectrum. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship between Hybrid DSS Based on Asymmetric Bandwidth and accurate uplink RB scheduling. For details, see **4.3.2.3 Function Impacts**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a switch for RedCap SSSG switching, which has a mutually exclusive relationship with Hybrid DSS Based on Asymmetric Bandwidth. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship between Hybrid DSS Based on Asymmetric Bandwidth and intra-frequency paging interference avoidance. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship between Hybrid DSS Based on Asymmetric Bandwidth and PUSCH coordinated power control. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added an impact relationship between Hybrid DSS Based on Asymmetric Bandwidth and the LTE and NR flexible frame offset function. For details, see **4.3.2.3 Function Impacts**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship of Hybrid DSS Based on Asymmetric Bandwidth with UL and DL Decoupling. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship between Hybrid DSS Based on Asymmetric Bandwidth and general precise interference isolation. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added impact relationships between Hybrid DSS Based on Asymmetric Bandwidth and the following functions: aggregation level adjustment for retransmissions upon DTXs and downlink CCE allocation without ratio constraint. For details, see **4.3.2.3 Function Impacts**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship between Hybrid DSS Based on Asymmetric Bandwidth and port sparseness. For details, see **4.3.2.3 Function Impacts**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added support of Hybrid DSS Based on Asymmetric Bandwidth for "cross-RAT configuration check in DSS scenarios". For details, see: • 4.1.1.1 Principles • 4.1.2.1 Principles • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter on the LTE side: Added the **DSS_CROSS_RAT_CFG_CHECK_SW** option to the **eNodeBResModeAlgo**.***ParamCfgProtectSw*** parameter. Modified parameter on the NR side: Added the **DSS_CROSS_RAT_CFG_CHECK_SW** option to the **gNBOamParam**.***CfgInterceptPolicySw*** parameter.	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship between Hybrid DSS Based on Asymmetric Bandwidth and user-rate-based dynamic network slice resource management. For details, see **4.3.2.3 Function Impacts**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship between Hybrid DSS Based on Asymmetric Bandwidth and user-rate-based dynamic QCI resource management. For details, see **4.3.2.3 Function Impacts**.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

- Editorial changes
  - Added descriptions of the configuration requirements for modifying the parameters related to common PUCCH RB resource optimization and ultimate PUCCH compression in Hybrid DSS Based on Asymmetric Bandwidth scenarios. For details, see **4.3.5 Others**.
  - Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding

(MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Applicable RAT

This document applies to LTE FDD and NR.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
LTE FDD	MRFD-171221	Hybrid DSS Based on Asymmetric Bandwidth(LTE FDD)	**4 Hybrid DSS Based on Asymmetric Bandwidth**
NR FDD	MRFD-171261	Hybrid DSS Based on Asymmetric Bandwidth(NR)	**4 Hybrid DSS Based on Asymmetric Bandwidth**
LTE FDD	MRFD-190221	DSS/Hybrid DSS Performance Boost(LTE FDD)	**5 DSS/Hybrid DSS Performance Boost**
NR FDD	MRFD-190261	DSS/Hybrid DSS Performance Boost(NR FDD)	**5 DSS/Hybrid DSS Performance Boost**
LTE FDD	MRFD-200222	DSS and NR Flexible Refarming (LTE FDD)	**6 DSS and NR Flexible Refarming**
NR FDD	MRFD-200262	DSS and NR Flexible Refarming (NR FDD)	**6 DSS and NR Flexible Refarming**
LTE FDD	MRFD-221225	LTE and NR Uplink Spectrum Pooling (LTE FDD)	**7 LTE and NR Uplink Spectrum Pooling**

RAT	Feature ID	Feature Name	Chapter/Section
NR FDD	MRFD-221265	LTE and NR Uplink Spectrum Pooling (NR FDD)	7 LTE and NR Uplink Spectrum Pooling

2.4 Differences

Differences Between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
Hybrid DSS Based on Asymmetric Bandwidth	Supported only in NR FDD	4 Hybrid DSS Based on Asymmetric Bandwidth
DSS/Hybrid DSS Performance Boost	Supported only in NR FDD	5 DSS/Hybrid DSS Performance Boost
DSS and NR Flexible Refarming	Supported only in NR FDD	6 DSS and NR Flexible Refarming
LTE and NR Uplink Spectrum Pooling	Supported only in NR FDD	7 LTE and NR Uplink Spectrum Pooling

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Hybrid DSS Based on Asymmetric Bandwidth	• Whether the NR PUCCH includes the common PUCCH differs between NSA networking and SA networking. • The setting notes for the SSB frequency-domain position differ between NSA networking and SA networking.	4 Hybrid DSS Based on Asymmetric Bandwidth
DSS/Hybrid DSS Performance Boost	None	5 DSS/Hybrid DSS Performance Boost
DSS and NR Flexible Refarming	None	6 DSS and NR Flexible Refarming
LTE and NR Uplink Spectrum Pooling	None	7 LTE and NR Uplink Spectrum Pooling

Differences Between High Frequency Bands and Low Frequency Bands

Function Name	Difference	Chapter/Section
Hybrid DSS Based on Asymmetric Bandwidth	Supported only in low frequency bands	**4 Hybrid DSS Based on Asymmetric Bandwidth**
DSS/Hybrid DSS Performance Boost	Supported only in low frequency bands	**5 DSS/Hybrid DSS Performance Boost**
DSS and NR Flexible Refarming	Supported only in low frequency bands	**6 DSS and NR Flexible Refarming**
LTE and NR Uplink Spectrum Pooling	Supported only in low frequency bands	**7 LTE and NR Uplink Spectrum Pooling**

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

3 Overview

In the early stages of 5G deployment, many operators have no dedicated NR spectrum resources or face low spectrum usage. To address this issue, LTE FDD and NR Flash Dynamic Spectrum Sharing (DSS) is introduced. This feature allows LTE FDD cells and NR FDD or NR supplementary downlink (SDL) cells to share time-frequency resources on the same spectrum segment. In this feature, the NR cell bandwidth is the same as the LTE cell bandwidth, and spectrum across the entire cell bandwidth can be shared. For more details, see *LTE FDD and NR Dynamic Spectrum Sharing*.

On the other hand, 3GPP specifications define large-bandwidth NR FDD cells to further improve the 5G network experience. However, the use of spectrum resources of large-bandwidth NR FDD cells is low in the early stages of 5G deployment due to the low penetration rate of 5G terminals. In view of this, Hybrid DSS Based on Asymmetric Bandwidth is introduced, which is applicable to operators that have both LTE FDD and NR FDD networks. This feature allows large-bandwidth NR FDD cells to dynamically share some of their spectrum resources with LTE FDD cells. In this feature, the NR cell bandwidth is greater than the LTE cell bandwidth. The amount of the shared spectrum is equal to the LTE cell bandwidth, and the NR cell bandwidth minus the amount of the shared spectrum is the amount of the NR-dedicated spectrum. For more details, see **4 Hybrid DSS Based on Asymmetric Bandwidth**.

To further improve spectrum utilization when LTE FDD and NR share spectrum, the following features have been introduced: DSS/Hybrid DSS Performance Boost, DSS and NR Flexible Refarming, and LTE and NR Uplink Spectrum Pooling. For more details, see **5 DSS/Hybrid DSS Performance Boost**, **6 DSS and NR Flexible Refarming**, and **7 LTE and NR Uplink Spectrum Pooling**.

4 Hybrid DSS Based on Asymmetric Bandwidth

4.1 Principles

3GPP specifications define large-bandwidth NR FDD cells to further improve the 5G network experience. However, the use of spectrum resources of large-bandwidth NR FDD cells is low in the early stages of 5G deployment due to the low penetration rate of 5G terminals. Hybrid DSS Based on Asymmetric Bandwidth allows LTE FDD and large-bandwidth NR FDD cells to dynamically share the same spectrum resources, improving spectrum utilization. It is applicable to operators that have both LTE FDD and large-bandwidth NR FDD networks.

For ease of description, LTE and NR are used to refer to LTE FDD and NR FDD, respectively.

This feature applies only to 3900 and 5900 series base stations.

Hybrid DSS Based on Asymmetric Bandwidth works in the following scenarios:

- Scenario 1: Spectrum sharing between an LTE cell and an NR cell. For details, see **4.1.1 Hybrid DSS Based on Asymmetric Bandwidth (Single LTE Cell)**.
- Scenario 2: Spectrum sharing between multiple LTE cells and an NR cell. For details, see **4.1.2 Hybrid DSS Based on Asymmetric Bandwidth (Multiple LTE Cells)**.

4.1.1 Hybrid DSS Based on Asymmetric Bandwidth (Single LTE Cell)

4.1.1.1 Principles

This function enables an LTE cell and an NR cell to dynamically share time-frequency resources on a shared spectrum segment based on their traffic volume requirements. The LTE and NR cells sharing spectrum resources are configured with different bandwidths. The NR cell bandwidth is greater than the LTE cell bandwidth, the LTE cell bandwidth is equal to the amount of the

shared spectrum, and the NR cell bandwidth minus the amount of the shared spectrum is equal to the amount of the NR-dedicated spectrum. Figure 4-1 illustrates spectrum sharing between LTE and NR cells with different bandwidths.

Figure 4-1 Spectrum sharing between LTE and NR cells with different bandwidths

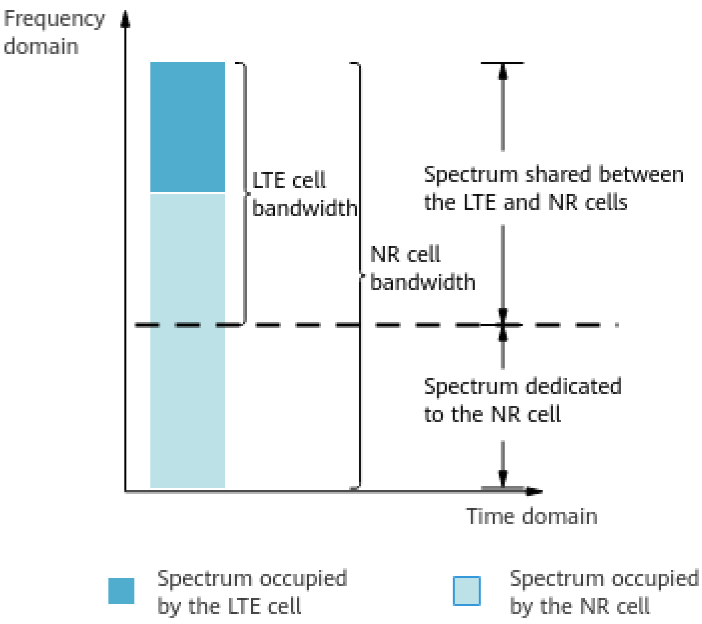


Table 4-1 lists the applicable cell bandwidths.

Table 4-1 Cell bandwidths to which Hybrid DSS Based on Asymmetric Bandwidth can be applied

LTE Cell Bandwidth	NR Cell Bandwidth
10 MHz	15 MHz
10 MHz	20 MHz
15 MHz	20 MHz
20 MHz	25 MHz
20 MHz	30 MHz
20 MHz	40 MHz

This function is enabled by turning on function switches, configuring NR-dedicated spectrum resources, and configuring spectrum sharing cell groups. In addition, the LTE and NR cells sharing spectrum resources must have aligned radio frames and subframes. (The alignment is ensured by configuring frame offsets and TA offsets.) **Table 4-2** describes parameter settings for enabling this function.

Table 4-2 Activation of Hybrid DSS Based on Asymmetric Bandwidth (single LTE cell)

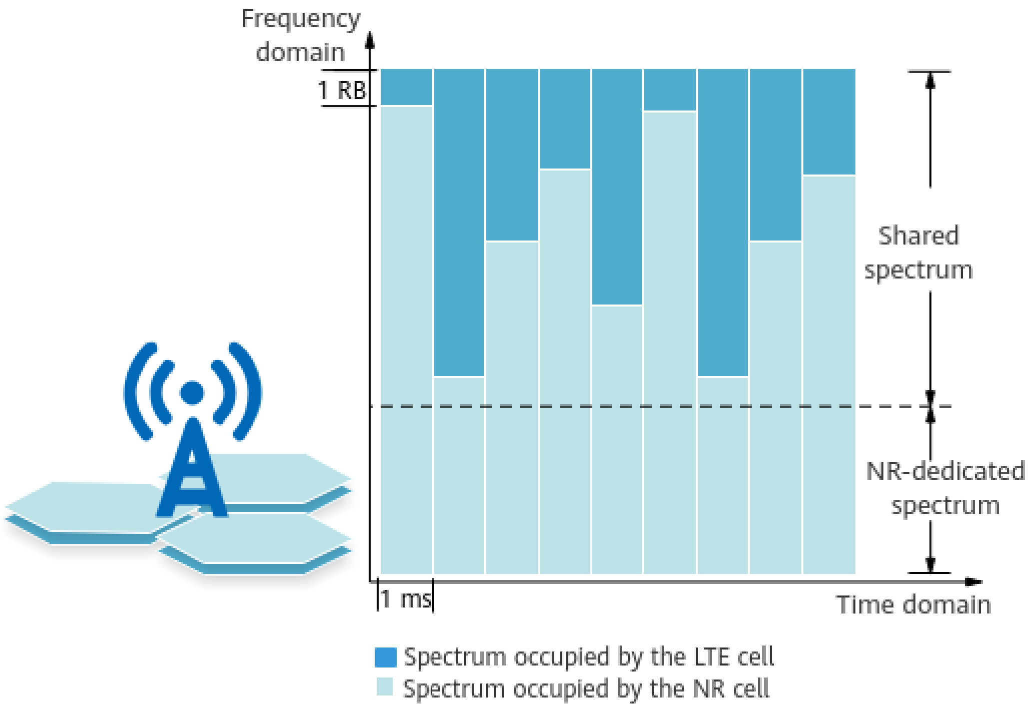
Operation	LTE Parameter Setting	NR Parameter Setting
Turning on function switches on the LTE and NR sides	Set the **SpectrumCloud.*SpectrumCloudSwitch*** parameter to **LTE_NR_SPECTRUM_SHR** and select the **LNR_SPECTRUM_SHR_ASYM_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter.	Select the **LTE_NR_FDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudSwitch*** parameter and the **LTE_NR_FDD_SPCT_SHR_ASYM_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** parameter.
Configuring NR-dedicated spectrum resources	-	Configure NR-dedicated spectrum resources at one end of the NR cell spectrum range. The amount of NR-dedicated uplink spectrum resources can be identical to or differ from that of NR-dedicated downlink spectrum resources. ● If only the **gNBDULteNrSpctShrCg.*NrReservedRbStartIndex*** and **gNBDULteNrSpctShrCg.*NrReservedRbEndIndex*** parameters are used to configure the dedicated spectrum, the same dedicated spectrum resources are configured in the uplink and downlink. ● If the **gNBDULteNrSpctShrCg.*NrUlReservedRbStartIndex*** and **gNBDULteNrSpctShrCg.*NrUlReservedRbEndIndex*** parameters are also configured, the uplink dedicated spectrum configuration takes the values of these two parameters, and the downlink dedicated spectrum configuration still takes the values of the **gNBDULteNrSpctShrCg.*NrReservedRbStartIndex*** and **gNBDULteNrSpctShrCg.*NrReservedRbEndIndex*** parameters.

Operation	LTE Parameter Setting	NR Parameter Setting
Configuring LTE and NR spectrum sharing cell groups	Add a planned LTE cell to an LTE spectrum sharing cell group by setting the **SpectrumCloud.*LteNrSpectrumShrCellGrpId*** and **LteNrSpctShrCellGrp.*LteNrSpectrumShrCellGrpId*** parameters.	1. Configure an association between the LTE spectrum sharing cell group and an NR spectrum sharing cell group by setting the **gNBDULteNrSpctShrCg.*NrSpctShrCellGrpId*** and **gNBDULteNrSpctShrCg.*LteSpctShrCellGrpId*** parameters. 2. Add a planned NR cell to the NR spectrum sharing cell group by setting the **NRDUCellSpctCloud.*NrDUCellId*** and **NRDUCellSpctCloud.*NrSpctShrCellGrpId*** parameters.
Configuring frame offsets	Set the **CellFrameOffset.*FrameOffset*** or **ENodeBFrameOffset.*FddFrameOffset*** parameter. If both parameters are configured, the value of the **CellFrameOffset.*FrameOffset*** parameter is used.	Set the **gNodeBParam.*FrameOffset*** or **gNBFreqBandConfig.*FrameOffset*** parameter. If both parameters are configured, the value of the **gNBFreqBandConfig.*FrameOffset*** parameter is used.
Configuring TA offsets	Set the **CellFrameOffset.*TaOffset*** parameter.	Set the **NRDUCell.*TaOffset*** parameter.

After this function is enabled, LTE and NR cells in the associated spectrum sharing cell groups can share spectrum resources on a shared spectrum segment. **Figure 4-2** shows the sharing mechanism.

- Time domain: Flash spectrum sharing is supported on a 1 ms basis, meaning that spectrum resources can be coordinated and scheduled every 1 ms.
- Frequency domain: Dynamic spectrum sharing is performed per RB. Spectrum resources are dynamically allocated to LTE and NR cells based on their traffic volumes.

Figure 4-2 Time-frequency resource sharing between LTE and NR cells



You are advised to enable "cross-RAT configuration check in DSS scenarios" by selecting both the **DSS_CROSS_RAT_CFG_CHECK_SW** option of the **eNodeBResModeAlgo**.***ParamCfgProtectSw*** parameter and the **DSS_CROSS_RAT_CFG_CHECK_SW** option of the **gNBOamParam**.***CfgInterceptPolicySw*** parameter. This function checks whether the configurations of NR and LTE cells that are configured to be activated in the same spectrum sharing group meet all requirements.

4.1.1.2 Key Technologies

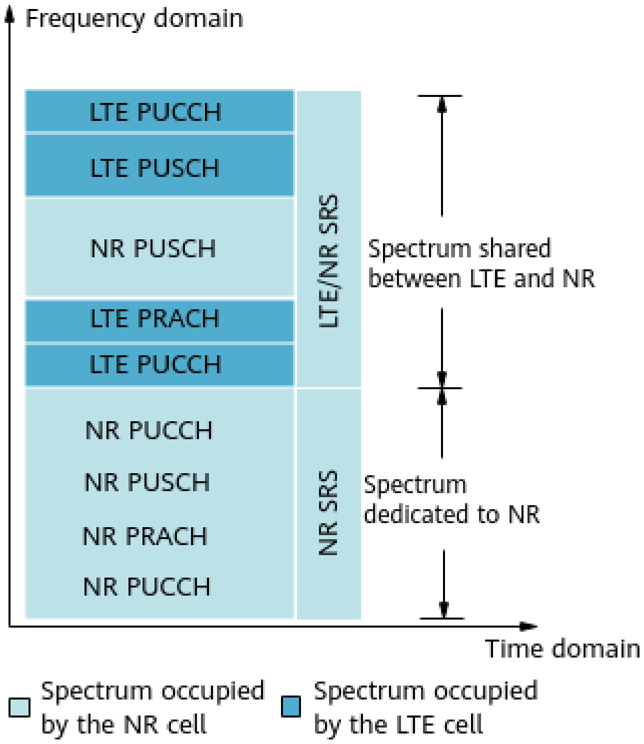
4.1.1.2.1 Coordinated Scheduling of Physical Channel Resources

Some channels will conflict with each other during spectrum sharing between LTE and NR cells in a spectrum sharing cell group. Such conflicts can be avoided by properly coordinating uplink and downlink physical channel resource scheduling, improving spectrum utilization.

Uplink Physical Channels

Figure 4-3 shows the default positions of uplink physical channels and signals after this function is enabled.

Figure 4-3 Default positions of uplink physical channels and signals



Uplink physical channel resources are coordinated and scheduled as follows:

- LTE and NR sounding reference signals (SRSs): SRSs can be configured either for both LTE and NR or for NR only. **Table 4-3** details the two configuration modes.

Table 4-3 LTE and NR SRS configuration modes

Configuration Mode	Parameter Setting	Effect
SRSs are configured for NR only.	The **LTE_UE_SRS_NOT_CONFIG_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter is selected.	SRS resources are allocated to NR but not to LTE.
SRSs are configured for both LTE and NR.[a]	The **LTE_UE_SRS_NOT_CONFIG_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter is deselected.	SRS resources are allocated to both LTE and NR.

Configuration Mode	Parameter Setting	Effect
a: In this scenario, LTE reserves subframes 0 and 5 for NR as cell-specific SRS subframes. When LTE UEs are scheduled in subframes during which NR SRSs are transmitted, the uplink MCS index decreases, affecting the uplink data rates of LTE UEs. LTE and NR SRS resource allocation optimization can be enabled by selecting the **LTE_NR_SRS_ALLOC_OPT_SW** option of the **LteNrSpctShrCellGrp.*LteNrSpctShrSwitch*** parameter. After the optimization, LTE reserves only subframe 0 for NR as the cell-specific SRS subframe, reducing the impact of NR SRS subframes on the uplink data rates of LTE UEs.		

- LTE PUCCH: The LTE PUCCH is configured in the shared spectrum. The position and the number of RBs for the LTE PUCCH adhere to the common configuration rules. The LTE PUCCH is not restricted by the deployment position of the NR PUCCH. For details, see the descriptions of the PUCCH in *Physical Channel Resource Management* in *eRAN Feature Documentation*. In this function, the LTE PUCCH can occupy a maximum of 16 RBs. This upper limit can prevent interference with SRSs and impacts on network performance.
- NR PUCCH:

  When the **NRDUCellBwp**.***UlBwp0ConfigPolicy*** parameter is set to **DEFAULT_DEDICATED_BANDWIDTH**, the NR PUCCH is deployed in the NR-dedicated spectrum. The number of RBs for the NR PUCCH is configured using the NR parameters listed in **Table 4-4**. In NSA networking, the NR PUCCH does not include the common PUCCH. In SA networking or NSA and SA hybrid networking, the NR PUCCH always includes the common PUCCH, which occupies four RBs. For details about the common PUCCH, see the descriptions of the PUCCH in *Channel Management* in *5G RAN Feature Documentation*.

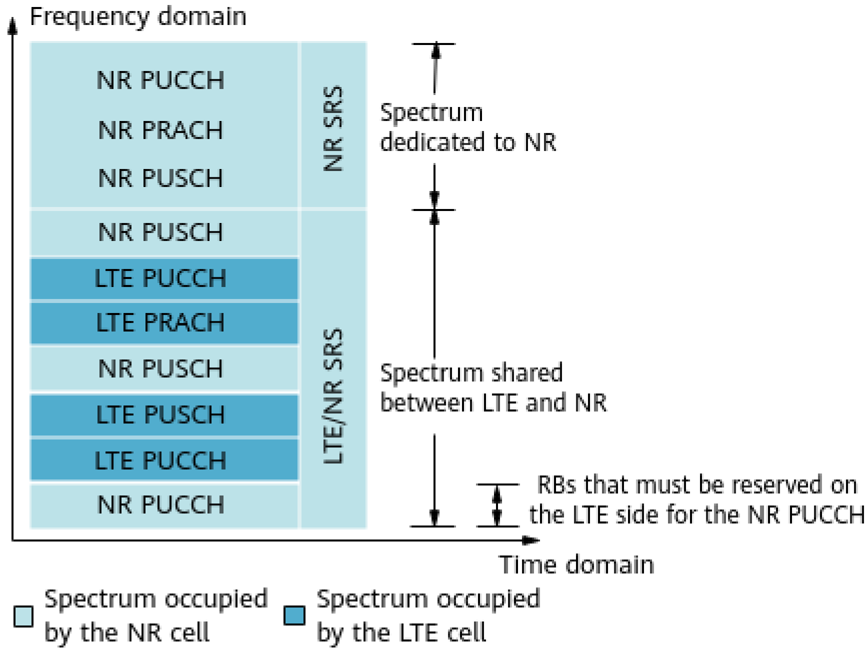
Table 4-4 Parameters for configuring the number of RBs for the NR PUCCH

Parameter	Function
NRDUCellPucch.*Format1RbNum*	Specifies the number of PUCCH format-1 RBs.
NRDUCellPucch.*Format3RbNum*	Specifies the number of PUCCH format-3 RBs.
NRDUCellPucch.*Format4RbNum*	Specifies the number of PUCCH format-4 RBs.
NRDUCellPucch.*CsiDedicatedRbNum*	Specifies the number of RBs reserved for format-3 periodic CSI.
NRDUCellPucch.*Format4CsiDedicatedRbNum*	Specifies the number of RBs reserved for format-4 periodic CSI.

When the **NRDUCellBwp**.***UlBwp0ConfigPolicy*** parameter is set to **FULL_BANDWIDTH**, one segment of the NR PUCCH is deployed in the NR-dedicated spectrum and the other segment is deployed at the symmetrical position in the shared spectrum. The number of RBs for the NR PUCCH is configured using the NR parameters listed in **Table 4-4**. In NSA networking, the NR PUCCH does not include the common PUCCH. In SA networking or NSA and SA hybrid networking, the NR PUCCH always includes the common PUCCH, which occupies four RBs. For details about the common PUCCH, see the descriptions of the PUCCH in *Channel Management* in *5G RAN Feature Documentation*.

Sufficient RBs must be reserved on the LTE side for the NR PUCCH in the shared spectrum, as shown in **Figure 4-4**.

Figure 4-4 RB reservation



- LTE PRACH: The LTE PRACH is configured adjacent to the LTE PUCCH at the lower end, and always occupies six RBs.
- NR PRACH: The NR PRACH is deployed in the NR-dedicated spectrum. The number of RBs for the NR PRACH is configured by a parameter. For details about the PRACH, see *Channel Management* in *5G RAN Feature Documentation*.
- LTE and NR PUSCHs: The total available shared PUSCH resources for LTE and NR equal the total shared spectrum resources minus the resources occupied by the LTE PUCCH and LTE PRACH. The total available shared PUSCH resources for LTE and NR are dynamically shared between them based on traffic requirements.

Downlink Physical Channels

When a resource conflict occurs between LTE and NR, resources are allocated to the channels and signals with a higher priority through coordinated scheduling. Conflicts between channels or signals of the same type are avoided in a static or dynamic way.

Frequency-domain positions of the NR SSB and system information on the NR PDSCH

The NR SSB and system information on the NR PDSCH can be positioned in the frequency domain within or beyond the NR-dedicated spectrum.

- When the NR SSB and system information on the NR PDSCH are positioned within the NR-dedicated spectrum, they do not conflict with the LTE system information, CRS/CSI-RS, or SS/PBCH. The following parameters related to the NR SSB frequency-domain position can be configured in a way that improves NR spectrum utilization:
  - **NRDUCell.*SsbDescMethod***: indicates the description method of the SSB frequency-domain position. In NSA networking, it is recommended that this parameter be set to **SSB_DESC_TYPE_NARFCN**, which indicates absolute frequency channel numbers. In SA networking, this parameter must be set to **SSB_DESC_TYPE_GSCN**, which indicates the Global Synchronization Channel Number (GSCN). This is because UEs have to determine the NR SSB frequency-domain position based on the GSCN before accessing the cell.
  - **NRDUCell.*SsbFreqPos***: The frequency-domain position indicated by this parameter is determined by the value of the **NRDUCell.*SsbDescMethod*** parameter. The specific parameter value for this function must be confirmed by Huawei engineers.

  Before configuring the preceding two parameters, you need to ensure that the number of RBs for common control resources (specified by the NR parameter **NRDUCellCoreset.*CommonCtrlResRbNum***) in the cell meets the requirements of Hybrid DSS Based on Asymmetric Bandwidth. The requirements vary depending on the number of symbols occupied by the NR PDCCH (specified by the NR parameter **NRDUCellPdcch.*OccupiedSymbolNum***). Table 4-5 details the requirements.

  Table 4-5 Required number of RBs for common control resources

Value of Occupied Symbol Number	Required Value of Common Control Resource RB Number
1SYM	**RB48**
2SYM	**RB24** or **RB48**

  The **SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** parameter must be selected before four SSB beams are used for cell coverage.

- When the NR SSB and system information on the NR PDSCH are positioned beyond the NR-dedicated spectrum, the NR SSB and LTE SS/PBCH avoid conflicts with each other in the same way as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see descriptions of coordinated scheduling of physical channel resources in *LTE FDD and NR Dynamic Spectrum Sharing*. In this case, you are advised to select both the **INIT_BWP_FULL_BW_SW** and **BWP0_DL_FULL_BW_SW** options of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter, so that the SSB is within BWP0. In addition, to ensure that scheduling resources are normal, select the **LNR_RES_ALLOC_DIRECTION_OPT_SW** option (mandatory) and the **HDSS_DL_PRB_ALLOCATE_OPT_SW** option (recommended) of the **LteNrSpctShrCellGrp**.***LteNrSpctShrSwitch*** parameter.

Yielding by the NR PDSCH to LTE CRS

In the shared spectrum, the NR PDSCH yields to LTE CRS through CRS rate matching. NR UEs scheduled in the shared spectrum must support CRS rate matching. For details about CRS rate matching, see section 5.1.4.2 "PDSCH resource mapping with RE level granularity" in 3GPP TS 38.214 V15.5.0. UEs that do not support CRS rate matching can be scheduled only in the dedicated spectrum.

How the NR PDSCH performs CRS rate matching depends on the number of LTE CRS ports in the following way:

- When the number of LTE CRS ports (specified by the LTE parameter **Cell.*CrsPortNum***) is 1, the **LTE_CRS_PORT_1_RM_SW** option of the NR parameter **gNBDULteNrSpctShrCg.*FddLteNrSpctShrSwitch*** can be selected so that the NR PDSCH performs CRS rate matching in the pattern of one LTE CRS port. CRS rate matching in the pattern of one LTE CRS port is supported when the cells use the independent power configuration mode (with the LTE parameter **SpectrumCloud**.***SpctShrMode*** set to **LTE_NR_PWR_INDEPENDENT**) or in certain scenarios where the cells use the spectrum power sharing mode (with the LTE parameter **SpectrumCloud**.***SpctShrMode*** set to **LTE_NR_PWR_DYN_SHR_WITH_SPCT**). If CRS rate matching in the pattern of one LTE CRS port is not enabled, the NR PDSCH performs CRS rate matching in the pattern of two LTE CRS ports even when only one CRS port is configured.
- When two or four CRS ports are configured, the NR PDSCH performs CRS rate matching in the pattern of two or four LTE CRS ports.

The **DL_RB_ESTIMATE_OPT_SW** option of the NR parameter **NRDUCellPdsch**.***DlPdschAlgoSwitch*** can be selected to enable the downlink scheduler to select a more appropriate number of RBs for NR UEs to be scheduled in the downlink. This prevents UE data from being truncated and increases the downlink user-perceived rate. For details, see *Downlink Scheduling* in *5G RAN Feature Documentation*.

Mutual avoidance of conflicts between LTE CSI-RS/CRS/DMRS/SS/PBCH and NR CSI-RS/TRS/PDCCH

Static coordinated scheduling is used for the LTE CSI-RS/CRS/DMRS/SS/PBCH and NR CSI-RS/TRS/PDCCH to avoid conflicts with each other.

Mutual avoidance of conflicts between NR UE-specific PDCCH and LTE PDCCH/PCFICH/PHICH

The conflict avoidance between the NR UE-specific PDCCH and LTE PDCCH/PCFICH/PHICH is related to the setting of the **UE_PDCCH_FULL_BANDWIDTH_CFG_SW** option of the **NRDUCellPdcch.*PdcchAlgoExtSwitch*** parameter. The UE-specific PDCCH is configured through NR parameters **NRDUCellPdcch.*SpctShrStartSymbol*** and **NRDUCellPdcch.*OccupiedSymbolNum***.

- When this option is selected, the NR UE-specific PDCCH and the LTE PDCCH/PCFICH/PHICH are transmitted on different symbols and therefore do not conflict with each other. The LTE PDCCH occupies at least one symbol (in the case of one or two LTE CRS ports) or two symbols (in the case of four LTE CRS ports), and the number of symbols occupied by the NR UE-specific PDCCH can be set to 1 or 2 by using NR parameters.

- When this option is deselected, the NR UE-specific PDCCH spans over the NR-dedicated spectrum, so that the NR UE-specific PDCCH and LTE PDCCH/PCFICH/PHICH are transmitted in different frequency-domain positions to avoid conflicts with each other. The NR UE-specific PDCCH can occupy one to three symbols, depending on parameter configurations.

Frequency-domain position of the NR common PDCCH

The NR common PDCCH can be positioned in the frequency domain within or beyond the NR-dedicated spectrum.

- When the NR common PDCCH is positioned within the NR-dedicated spectrum, it does not conflict with the LTE PDCCH/PCFICH/PHICH. The NR common PDCCH can be configured in two ways depending on the setting of the NR parameter **NRDUCellCoreset.*CommonCtrlResStartSymbol***: (1) When it is set to a valid value, the NR common PDCCH is separately configured through **NRDUCellCoreset.*CommonCtrlResStartSymbol*** and **NRDUCellCoreset.*CommonCtrlResRbNum***. (2) When it is set to an invalid value, the NR common PDCCH is configured together with the NR UE-specific PDCCH through **NRDUCellPdcch.*SpctShrStartSymbol***, **NRDUCellPdcch.*OccupiedSymbolNum***, and **NRDUCellCoreset.*CommonCtrlResRbNum***.
- When the NR common PDCCH is positioned beyond the NR-dedicated spectrum, the LTE CRS yields to the RRC messages on the NR PDSCH through static coordinated scheduling. In addition, the LTE CRS yields to the NR SSB and SIB1 on the PDSCH through parameter configuration. This configuration is the same as that in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see descriptions of coordinated scheduling of physical channel resources in *LTE FDD and NR Dynamic Spectrum Sharing*.

Mutual avoidance of conflicts between LTE and NR PDSCHs

Coordinated dynamic scheduling is used for the LTE and NR PDSCHs to avoid conflicts with each other. In addition, the LTE PDSCH is dynamically scheduled to yield to the NR CSI-RS/TRS/PDCCH, and the NR PDSCH is also dynamically scheduled to yield to the LTE CRS/CSI-RS/SS/PBCH/PHICH/PCFICH/PDCCH.

- The NR PDSCH can yield to LTE CSI-RS at the RE or subframe level. When the **LTE_CSI_RS_AVOID_POLICY_SW** option of the LTE parameter **LteNrSpctShrCellGrp.*LteNrSpctShrSwitch*** and the **LTE_CSI_RS_AVOID_SW** option of the NR parameter **gNBDULteNrSpctShrCg.*FddLteNrSpctShrSwitch*** are both selected, RE-level yielding is used. That is, the NR PDSCH avoids the REs occupied by the LTE CSI-RS. In other circumstances, subframe-level yielding is used. That is, the NR PDSCH avoids the subframes occupied by the LTE CSI-RS.

  When the TX/RX mode of the LTE and NR cells is 8T8R, the NR PDSCH cannot yield to LTE CSI-RS at the RE level.
- The NR PDSCH can yield to the LTE SS/PBCH through SS/PBCH rate matching at the RB or symbol level. You can select the **LTE_SS_PBCH_RM_OPT_SW** option of the NR parameter **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** to enable the NR PDSCH to

use one pattern for SS/PBCH rate matching. When this option is deselected, the NR PDSCH uses two patterns for SS/PBCH rate matching. This option can be set based on the network plan. When the NR PDSCH uses two rate matching patterns, the yielding is more accurate and at the same time more pattern resources are consumed, compared with when one rate matching pattern is used.

The position of the downlink additional DMRS on the NR PDSCH is specified by the NR parameter **NRDUCellPdsch.*DlAdditionalDmrsPos***. This parameter can be set to **POS1** to improve user experience.

LTE DRX information and UE-level control information have to yield to NR CSI-RS and TRS in the time domain and may therefore fail to be scheduled in time. If this occurs, control-plane performance indicators are affected. Adaptive DRX cycle extension can address this issue. This function is controlled by the **DSS_DRX_CYCLE_ADAPT_SW** option of the **SpectrumCloud**.***SpectrumCloudExtSwitch*** parameter. With this function, the DRX cycle is adaptively prolonged to reduce the probability of not scheduling such LTE signaling in a timely manner. This function may affect the UE power saving effect. In addition, on the LTE side, this function may decrease the packet transmission delay and change the user-perceived rate, BLER, and number of UEs scheduled in each TTI in both the uplink and downlink.

If the SSB is configured at the position indicated by SSB index 0 or 1 within the shared spectrum in Hybrid DSS Based on Asymmetric Bandwidth scenarios, there may be insufficient power for common channel transmission on the LTE and NR sides and insufficient bandwidth resources for the NR SSB. If the **NRDUCell**.***SsbTimePos*** parameter is set to **SSB12_PCI_MOD2_STAGGER** or **SSB_PCI_MOD3_STAGGER**, the SSB may be configured at the position indicated by SSB index 1. Therefore, if the SSB is configured within the shared spectrum in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the **NRDUCell**.***SsbTimePos*** parameter can only be set to **SSB23_PCI_MOD2_STAGGER**.

If the SSB is configured within the NR-dedicated spectrum in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the **NRDUCell**.***SsbTimePos*** parameter can be set to **SSB_PCI_MOD3_STAGGER**.

In Hybrid DSS Based on Asymmetric Bandwidth scenarios, the **PAGING_OSI_STAGGER_SW** option of the **NRDUCellPagingConfig**.***PagingSchAlgoSwitch*** parameter can be selected to enable paging-OSI staggering.

In Hybrid DSS Based on Asymmetric Bandwidth scenarios, the **TIME_DOMAIN_RMSI_STAGGER_SW** option of the **NRDUCellDlSch**.***DlSchAlgoSwitch*** parameter can be selected to enable time-domain RMSI staggering.

NR PDSCH support for rate matching on all RBs in LTE CRS symbols

The NR PDSCH supports rate matching on all RBs in LTE CRS symbols to eliminate the interference of neighboring LTE cells on the NR spectrum sharing cell.

If the LTE spectrum sharing cell and other neighboring LTE cells are configured with the same number of CRS ports, the **LTE_CRS_RATEMATCH_ALL_SYM_SW** option of the NR parameter **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** is selected to support this function. In this case, rate matching is performed based on the configured number of CRS ports.

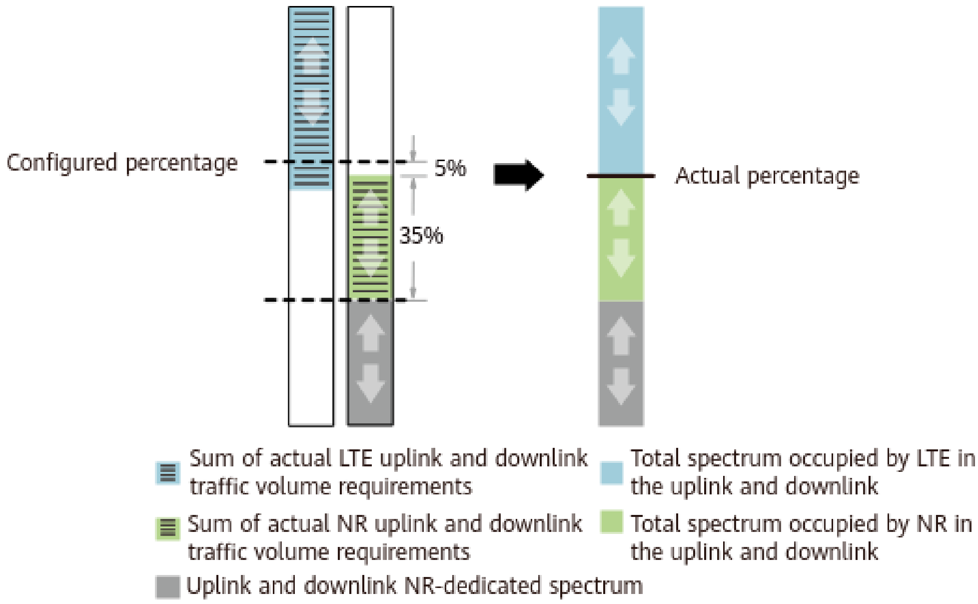
4.1.1.2.2 Flexible Spectrum Priority Mode

Configuration of the Percentages of Spectrum Resources to Be Preferentially Allocated

To ensure proper allocation of shared resources, the percentages of shared resources preferentially allocated to the LTE and NR cells for MBB services can be set based on their traffic volume requirements. The resources here refer to the PUSCH and PDSCH resources. The percentage for LTE is specified by the **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio*** parameter, and the percentage for NR is equal to 100% minus the value of this parameter. Shared spectrum resources are allocated based on these percentages as follows:

- When the percentages of actual traffic volume requirements of the LTE and NR cells to the total shared spectrum resources are both less than, or both exceed, the percentages of resources preferentially allocated, spectrum resources are allocated based on the percentages of resources preferentially allocated to each cell.
- When the actual traffic volume requirement of the cell of one RAT (RAT 1) exceeds the percentage of resources preferentially allocated to that RAT, and the actual traffic volume requirement of the cell of the other RAT (RAT 2) does not, the unused resources for RAT 2 can be occupied by RAT 1. For example, if the configured percentage of resources preferentially allocated to the LTE cell is 60% but the LTE cell requires 70%, and the configured percentage of resources preferentially allocated to the NR cell is 40% but the NR cell only requires 35%, then the unused 5% resources for the NR cell can be used by the LTE cell. That is, the LTE cell can use 65% of the shared spectrum resources, as shown in **Figure 4-5**.

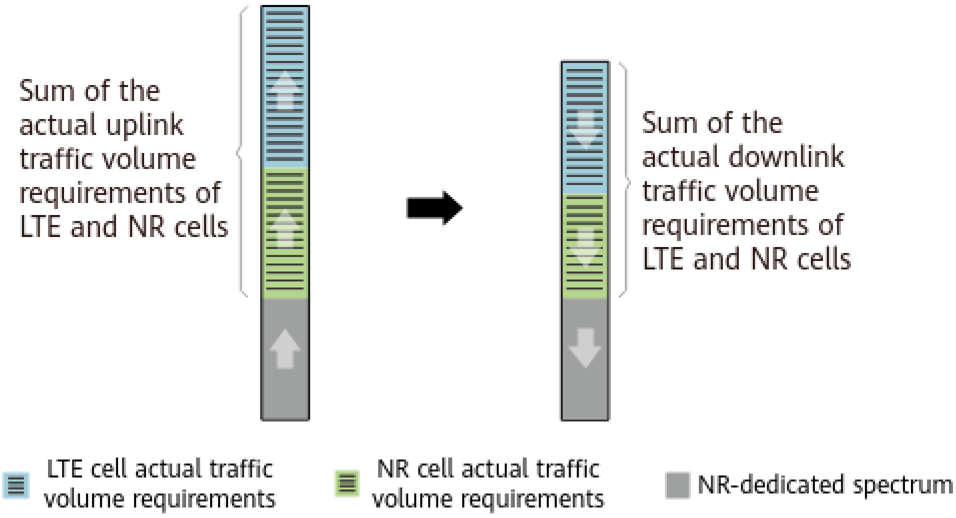
Figure 4-5 Example 1 of how the percentages of resources that are preferentially allocated work



In addition to the preceding allocation principles, the setting of the **LNR_RES_ALLOC_ADAPT_SW** option of the **LteNrSpctShrCellGrp.*LteNrSpctShrSwitch*** parameter also affects the actual allocation of spectrum resources.

- If this option is selected, the system allocates spectrum resources based on the percentage specified by the **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio*** parameter, for whichever direction (uplink or downlink) with the larger sum of traffic volume requirements of the LTE and NR cells. For the other direction, the system dynamically allocates spectrum resources based on the actual traffic volume requirements of the LTE and NR cells. For example, if the sum of the uplink traffic volume requirements of the LTE and NR cells is greater than the sum of the downlink traffic volume requirements of the LTE and NR cells, the uplink spectrum resources are allocated to the LTE and NR cells based on the percentage specified by the **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio*** parameter. The downlink spectrum resources are dynamically allocated to the LTE and NR cells based on their actual traffic volume requirements, as shown in **Figure 4-6**.

Figure 4-6 Example 2 of how the percentages of resources that are preferentially allocated work



- If this option is deselected, the system considers the uplink and downlink as a whole and allocates resources to the LTE and NR cells based on the value of the **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio*** parameter.

Table 4-6 lists the counters used for measuring the shared spectrum resources occupied by the LTE and NR cells in the uplink and downlink.

Table 4-6 Counters for measuring the number of uplink and downlink available RBs in LTE and NR cells

Counter ID	Counter Name
1526759050	**L.ChMeas.PRB.UL.Actual.Avail**
1526759052	**L.ChMeas.PRB.UL.PUSCH.Actual.Avail**

Counter ID	Counter Name
1526759051	**L.ChMeas.PRB.DL.Actual.Avail**
1911827161	**N.PRB.UL.Actual.Avail.Avg**
1911827162	**N.PRB.UL.PUSCH.Actual.Avail.Avg**
1911827160	**N.PRB.DL.Actual.Avail.Avg**

Inter-RAT Preferential Guarantee for GBR Services

If the percentage of resources preferentially allocated to a RAT is less than the percentage required by guaranteed bit rate (GBR) services of that RAT, experience of GBR services of that RAT will be affected. For example, if the percentage of resources preferentially allocated to LTE is 40% and the percentage required by GBR services is 60%, 20% of GBR service requirements cannot be satisfied, affecting GBR service experience.

The function of inter-RAT preferential guarantee for GBR services is introduced. This function defines the priority of GBR services as follows: The priority of LTE GBR services is higher than that of NR non-GBR services, and the priority of NR GBR services is higher than that of LTE non-GBR services. In the preceding example, the unsatisfied LTE GBR services can occupy the resources of NR non-GBR services. In this way, GBR service experience is preferentially guaranteed.

This function is enabled when the **GBR_PRI_ALLOC_SW** option of the LTE parameter **LteNrSpctShrCellGrp.*LteNrSpctShrSwitch*** is selected. For details about LTE GBR and non-GBR services, see Uplink Scheduling and Downlink Scheduling in *eRAN Feature Documentation*. For details about NR GBR and non-GBR services, see *QoS Management* in *5G RAN Feature Documentation*.

4.1.1.2.3 Flexible Power Allocation Mode

The power allocation mode can be flexibly configured using the LTE parameter **SpectrumCloud.*SpctShrMode***:

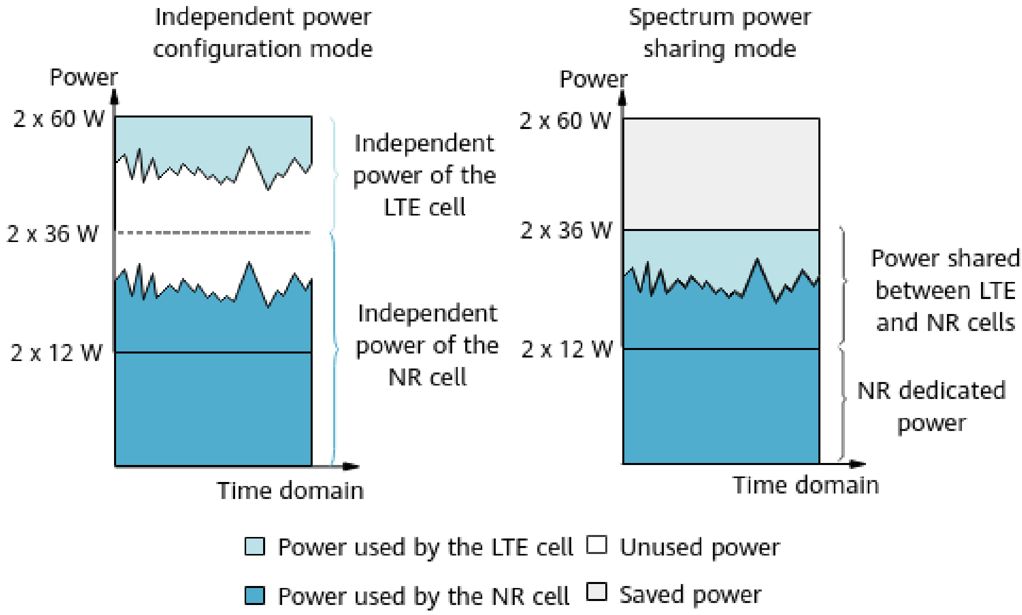
- When this parameter is set to **LTE_NR_PWR_DYN_SHR_WITH_SPCT**, spectrum power sharing mode is used. The LTE and NR cells share the same power on the shared spectrum, and the percentage of power allocated to a RAT is equal to the percentage of shared spectrum allocated to that RAT.
- When this parameter is set to **LTE_NR_PWR_INDEPENDENT**, independent power configuration mode is used. The LTE and NR cells use independently configured spectrum power.

Figure 4-7 uses a 2 x 60 W RF module serving a 20 MHz LTE cell and a 30 MHz NR cell to illustrate power allocation modes.

In independent power configuration mode, the maximum transmit power that can be configured for the LTE cell is 2 x 24 W, and that for the NR cell is 2 x 36 W. The LTE and NR cells each use 2 x 24 W power on the shared spectrum, and the NR cell exclusively uses 2 x 12 W power.

In spectrum power sharing mode, the power configured for the LTE and NR cells must be greater than or equal to the power in independent power configuration mode. The LTE cell can be configured with a transmit power ranging from 2 x 24 W to 2 x 40 W, and the NR cell can be configured with a transmit power ranging from 2 x 36 W to 2 x 60 W. The LTE and NR cells can minimally share the total power of 2 x 24 W on the shared spectrum, and the NR cell exclusively uses the 2 x 12 W power. The spectrum power sharing mode helps save the transmit power by 2 x 24 W (2 x 60 W – 2 x 36 W), compared with the independent power configuration mode.

Figure 4-7 Power allocation mode



Given the same cell power configuration (for example, 2x36 W for NR in both independent power configuration mode and spectrum power sharing mode), independent power configuration mode results in better capacity performance but a higher power consumption, whereas spectrum power sharing mode reduces power consumption but leads to capacity performance loss. Therefore, when the power is sufficient, independent power configuration mode is preferred. When the power is insufficient, spectrum power sharing mode can be used.

In spectrum power sharing mode, the PSD adaptation function can be used to reduce the PSD of the NR cell when the transmit power required by the NR cell exceeds the configured power. This ensures that the NR cell can use more spectrum resources, improving the spectral efficiency of UEs in the NR cell. This function is enabled when the **PWR_SPCT_DENSITY_ADAPT_SW** option of the NR parameter **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** is selected.

When spectrum power sharing mode is used, and the LTE and NR baseband processing units need to be connected to the RRU through their respective optical fibers, the **LNR_PWR_WITH_SPCT_OPT_SW** option of the LTE parameter **SpectrumCloud.*SpectrumCloudEnhSwitch*** must be selected. When independent power configuration mode is used, LTE and NR can use their respective optical fibers over the CPRI port without the need of selecting this option.

4.1.1.2.4 Co-carrier Co-CPRI Data

After Hybrid DSS Based on Asymmetric Bandwidth is enabled, the LTE and NR cells each occupies a portion of CPRI bandwidth. When LTE and NR share one CPRI optical fiber (the LTE or NR baseband processing unit is connected directly to the RRU by optical fiber) and the CPRI bandwidth is insufficient, the LTE and NR co-carrier co-CPRI data function can be enabled to meet CPRI bandwidth requirements. With this function, the LTE cell and the shared spectrum of the NR cell occupy only one portion of CPRI bandwidth. The LTE and NR co-carrier co-CPRI data function can be enabled by setting the LTE parameter **LteNrSpctShrCellGrp**.***LteNrCoCarrCoCpriDataSw*** to **ON**.

When the LTE and NR co-carrier co-CPRI data function is enabled, only the spectrum power sharing mode is supported. That is, the **SpectrumCloud.*SpctShrMode*** parameter must be set to **LTE_NR_PWR_DYN_SHR_WITH_SPCT**.

When the LTE and NR co-carrier co-CPRI data function and the CPRI compression function are both enabled, the CPRI compression ratios must be the same for the LTE and NR cells. For details about the configuration requirements, see **4.1.1.3 Simultaneous Use with Other Functions**.

4.1.1.3 Simultaneous Use with Other Functions

4.1.1.3.1 LTE Functions

Simultaneous Use with NB-IoT Deployed in LTE Guard Band Mode

For 3900 and 5900 series base stations, Hybrid DSS Based on Asymmetric Bandwidth can work together with NB-IoT deployed in LTE guard band mode. NB-IoT can only be deployed in the guard band on the spectrum occupied by the LTE cell.

In this scenario, the spectrum occupied by the NB-IoT cell and the spectrum occupied by the NR cell overlap, causing severe mutual interference. As a result, the NB-IoT cell or NR cell becomes unavailable. Specifically:

- If the NB-IoT cell is configured and activated prior to the NR cell, the NR cell cannot be activated.
- If the NR cell is configured and activated prior to the NB-IoT cell, the NB-IoT cell cannot be activated.

To eliminate the interference, spectrum resources need to be reserved when NB-IoT is deployed, and the reserved spectrum resources are unavailable for the NR cell. The reserved uplink and downlink RBs are configured through the **NRDUCellRbReserve.*RbRsvMode***, **NRDUCellRbReserve.*RbRsvType***, **NRDUCellRbReserve.*RbRsvStartIndex***, and **NRDUCellRbReserve.*RbRsvEndIndex*** parameters.

The reserved uplink and downlink RBs must be confirmed by Huawei engineers. **Table 4-7** lists the recommended number of uplink and downlink RBs to be reserved in different LTE and NR cell bandwidth scenarios. The recommended configuration preferentially ensures the NB-IoT cell performance while considering the NR cell throughput.

For details about NB-IoT deployed in LTE guard band mode, see *NB-IoT Basics (FDD)* in *eRAN Feature Documentation*.

Table 4-7 Recommended configurations to reserve spectrum resources in typical scenarios

LTE Cell Bandwidth	NR Cell Bandwidth	Number of Reserved Uplink and Downlink RBs
10 MHz	15 MHz	2
10 MHz	20 MHz	3
15 MHz	20 MHz	3
20 MHz	25 MHz	4
20 MHz	30 MHz	4
20 MHz	40 MHz	5

Simultaneous Use with LTE Downlink Semi-persistent Scheduling

If Hybrid DSS Based on Asymmetric Bandwidth is used together with LTE downlink semi-persistent scheduling, the upper limit on the proportion of RBs that can be allocated for LTE downlink semi-persistent scheduling is configurable (through the **SpectrumCloud.*DlSpsRestrictRatio*** parameter). This proportion together with the percentage of spectrum resources to be preferentially allocated to LTE MBB services (specified by the LTE parameter **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio***) determines the proportion of RBs allocated for LTE downlink semi-persistent scheduling as follows:

Proportion of RBs allocated for LTE downlink semi-persistent scheduling = (**SpectrumCloud.*DlSpsRestrictRatio*** x **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio***) x 100%

Simultaneous Use with WBB

If Hybrid DSS Based on Asymmetric Bandwidth is used together with WBB, it is recommended that the **WBB_MBB_CONTROL_OPT_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter be selected. When this option is selected, the eNodeB allocates resources to LTE WBB and MBB services based on the actually available LTE spectrum resources. The percentages of resources actually allocated to MBB services comply with the settings of the **CellWttxParaCfg.*MbbUserDlPrbUpLimit*** and **CellWttxParaCfg.*MbbUserUlPrbUpLimit*** parameters; the percentages of resources actually allocated to WBB services comply with the settings of the **CellWttxParaCfg.*WbbUserDlPrbUpLimit*** and **CellWttxParaCfg.*WbbUserUlPrbUpLimit*** parameters. For details about WBB, see *WBB* in *eRAN Feature Documentation*.

To ensure spectral efficiency, it is recommended that the following equations be met:

- **CellWttxParaCfg.*MbbUserDlPrbUpLimit*** + **CellWttxParaCfg.*WbbUserDlPrbUpLimit*** = 100%
- **CellWttxParaCfg.*MbbUserUlPrbUpLimit*** + **CellWttxParaCfg.*WbbUserUlPrbUpLimit*** = 100%

4.1.1.3.2 NR Functions

Simultaneous Use with Downlink Scheduling Bandwidth Adaptation in HDSS Scenarios

Hybrid DSS Based on Asymmetric Bandwidth can be used together with downlink scheduling bandwidth adaptation in HDSS scenarios (controlled by the **HDSS_DL_SCH_BW_ADAPT_SW** option of the **NRDUCellDlSch**.***DlSchAlgoEnhSwitch*** parameter). With the latter function, the base station adaptively determines whether to schedule a UE in the downlink within the NR-dedicated spectrum or full spectrum, based on information about multiple sets of CSI measurement. When the two functions are used together, the **NRDUCellDlSch**.***HdssAperCsiIntvlAdaptCoeff*** parameter can be used to control the impact of CSI measurement on the uplink overhead. In addition, the **NRDUCellDlSch**.***HdssSpctEffCoefficient*** parameter can be used to determine whether to allocate RBs within a non-interference range or across the entire bandwidth to UEs. Downlink scheduling bandwidth adaptation in HDSS scenarios is recommended when there are both LTE-only base stations and base stations with Hybrid DSS Based on Asymmetric Bandwidth enabled. Downlink scheduling bandwidth adaptation in HDSS scenarios requires that downlink interference randomization optimization in HDSS scenarios (controlled by the **HDSS_DL_INTRF_RANDOM_OPT_SW** option of the **NRDUCellDlSch**.***DlSchAlgoEnhSwitch*** parameter) be enabled. For details about these two functions, see *Interference Avoidance*.

4.1.1.3.3 Common Functions Supported by Both LTE and NR

Simultaneous Use with CPRI Compression

When the co-carrier co-CPRI data function and the CPRI compression function are both enabled, the CPRI compression ratios must be the same for the LTE and NR cells. The CPRI compression type of an LTE cell is specified by the **Cell.*CPRICompression*** parameter, and the CPRI compression type of an NR cell is specified by the **NRDUCellTrp.*CpriCompression*** parameter. **Table 4-8** describes the configuration requirements.

Table 4-8 Configuration requirements on the CPRI compression types of the LTE and NR cells

LTE Cell Bandwidth	TX/RX Mode	CPRI Compression Ratio	CPRI Compression Type of the LTE Cell	CPRI Compression Type of the NR Cell
10 MHz	2T2R/2T4R/4T4R/8T8R	No CPRI compression	NO_COMPRESSION	NO_COMPRESSION

LTE Cell Bandwidth	TX/RX Mode	CPRI Compression Ratio	CPRI Compression Type of the LTE Cell	CPRI Compression Type of the NR Cell
	2T2R/2T4R/4T4R/8T8R	2:1 CPRI compression	ENHANCED_COMPRESSION	2_COMPRESSION
	16T16R	2:1 CPRI compression[a]	ENHANCED_COMPRESSION	2_COMPRESSION
		3.2:1 CPRI compression[a]	3_2_COMPRESSION	3DOT2_COMPRESSION
	32T32R	2.2:1 CPRI compression	2_2_COMPRESSION	2DOT2_COMPRESSION
		2:1 CPRI compression[b]	ENHANCED_COMPRESSION	2_COMPRESSION
		3.2:1 CPRI compression[b]	3_2_COMPRESSION	3DOT2_COMPRESSION
15 MHz/20 MHz	2T2R/2T4R/4T4R/8T8R	No CPRI compression	NO_COMPRESSION	NO_COMPRESSION
	2T2R/2T4R/4T4R/8T8R	2:1 CPRI compression	NORMAL_COMPRESSION	2_COMPRESSION
	16T16R	2:1 CPRI compression[a]	ENHANCED_COMPRESSION	2_COMPRESSION
		3.2:1 CPRI compression[a]	3_2_COMPRESSION	3DOT2_COMPRESSION
	32T32R	2.2:1 CPRI compression	2_2_COMPRESSION	2DOT2_COMPRESSION
		2:1 CPRI compression[b]	NORMAL_COMPRESSION	2_COMPRESSION

LTE Cell Bandwidth	TX/RX Mode	CPRI Compression Ratio	CPRI Compression Type of the LTE Cell	CPRI Compression Type of the NR Cell
		3.2:1 CPRI compression[b]	3_2_COMPRESSION	3DOT2_COMPRESSION
a: This compression ratio is supported only when the interface board is a UBBPh3a. b: This compression ratio is supported only when the interface board is a UBBPh2/UBBPh3/UBBPh3a/UBBPh3b and the versions of NR and LTE are V100R020C10 or later.				

In Hybrid DSS Based on Asymmetric Bandwidth, the co-carrier co-CPRI data function needs to work together with CPRI compression in the following scenario:

The LTE cell bandwidth is 10 MHz, the NR cell bandwidth is greater than 10 MHz and less than or equal to 20 MHz, and the cell TX/RX mode is greater than 2T2R. The **BBP.*WM*** parameter is set to **SPEC_MODE3** for the baseband processing unit in the baseband equipment to which the LTE cell sector equipment is bound.

For details about CPRI compression, see *CPRI Compression* in *eRAN Feature Documentation* and *CPRI Compression* in *5G RAN Feature Documentation*.

Simultaneous Use with LTE and NR Flexible Frame Offset

In Hybrid DSS Based on Asymmetric Bandwidth scenarios, when the LTE and NR flexible frame offset function is enabled, the frame offset of the NR cell in a spectrum sharing group must be 3 ms later than that of the LTE cell in the same group. To achieve this, the "LTE and NR 3-ms frame offset difference" function must be enabled and the NR and LTE cell frame offsets must be configured accordingly.

The "LTE and NR 3-ms frame offset difference" function takes effect when the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of the NR parameter **NRDUCellSpctShare**.***LnrSpectrumShareSwitch*** parameter is selected and the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of the LTE parameter **SpectrumCloud**.***SpectrumCloudExtSwitch*** parameter is selected.

The "LTE and NR 3-ms frame offset difference" function can be enabled only in LTE FDD and NR spectrum sharing scenarios. When it is enabled, the **CellMbsfnSfEnhConfig.*SubframeAllocationMode*** parameter must be set to **NOT_CONFIG**. In addition, it is recommended that the SSB be configured within the NR-dedicated spectrum to reduce the impact of CRS puncturing.

The effective LTE and NR cell frame offsets are determined as follows:

- NR cell frame offset: If the **gNBFreqBandConfig**.***FrameOffset*** parameter is set, the frame offset equals the value of this parameter. Otherwise, the frame offset equals the value of the **gNodeBParam**.***FrameOffset*** parameter.

- LTE cell frame offset: If the **CellFrameOffset**.***FrameOffset*** parameter is set, the frame offset equals the value of this parameter. Otherwise, the frame offset equals the value of the **ENodeBFrameOffset**.***FddFrameOffset*** parameter.

For details about the LTE and NR flexible frame offset function, see *Cell ManagementCell Management* in *5G RAN Feature Documentation*.

4.1.2 Hybrid DSS Based on Asymmetric Bandwidth (Multiple LTE Cells)

Static Multiple Beam is one of the LTE FDD multi-sector solutions. Static Multiple Beam generates at least two beams by means of beamforming, with each beam serving a sector split cell, thereby maximizing network capacity. Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) enables the two or more LTE FDD sector split cells served by the beams generated by Static Multiple Beam to dynamically share the same spectrum resources with an NR FDD cell, improving spectrum utilization.

Static Multiple Beam can be used in Smart 8T8R scenarios, RuralLink antenna solution, LumiLink solution, and massive MIMO scenarios. In Smart 8T8R scenarios, it generates two beams, which correspond to two LTE FDD sector split cells. In RuralLink antenna solution and LumiLink solution, it generates three beams, which correspond to three LTE FDD sector split cells. In 16T16R massive MIMO scenarios, it generates two to three beams, which correspond to two to three LTE FDD sector split cells. In 32T32R massive MIMO scenarios, it generates two to four beams, which correspond to two to four LTE FDD sector split cells. For details about Static Multiple Beam in Smart 8T8R scenarios, see *Smart 8T8R (FDD)* in *eRAN Feature Documentation*. For details about Static Multiple Beam in RuralLink antenna solution, see *RuralLink Solution*. For details about Static Multiple Beam in LumiLink solution, see *LumiLink Solution*. For details about Static Multiple Beam in massive MIMO scenarios, see *Massive MIMO Basics (FDD)* in *eRAN Feature Documentation*.

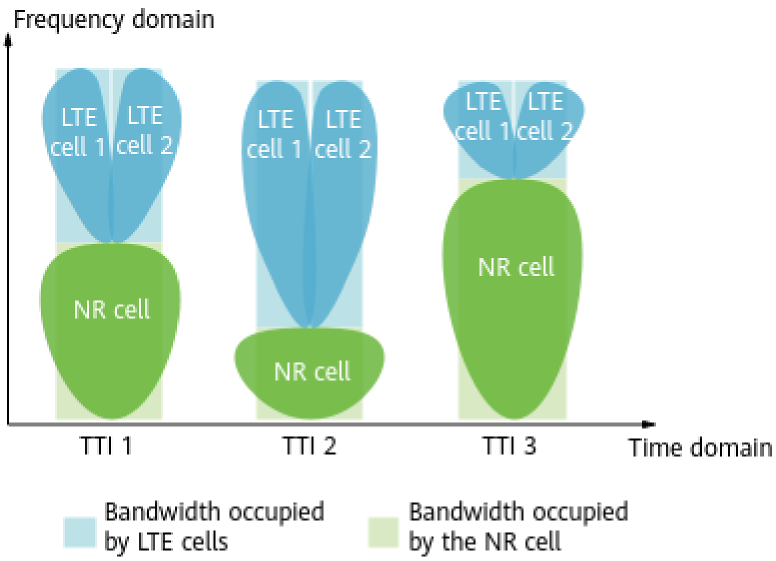
For ease of description, LTE and NR cells are used to refer to LTE FDD sector split cells served by the beams generated by Static Multiple Beam and NR FDD cells, respectively.

4.1.2.1 Principles

This function enables the multiple LTE FDD sector split cells served by the beams generated by Static Multiple Beam and an NR FDD cell to dynamically share time-frequency resources on a shared spectrum segment based on their traffic volume requirements. Figure 4-8 illustrates the working principles of this function. The LTE cell bandwidth must be 20 MHz, and the NR cell bandwidth must be 40 MHz. LTE and NR cells can share spectrum once they are added to associated spectrum sharing cell groups. Specifically:

- Time domain: Flash spectrum sharing is supported on a 1 ms basis, meaning that spectrum resources can be coordinated and scheduled every 1 ms.
- Frequency domain: Dynamic spectrum sharing is performed per RB. Spectrum resources are dynamically allocated to LTE and NR cells based on their traffic volumes. The same resources are allocated to the LTE cells.

Figure 4-8 Working principles of Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells)



This function is enabled by turning on function switches, configuring NR-dedicated spectrum resources, and configuring spectrum sharing cell groups. In addition, the LTE and NR cells sharing spectrum resources must have aligned radio frames and subframes. (The alignment is ensured by configuring frame offsets and TA offsets.) **Table 4-9** describes parameter settings for enabling this function.

Table 4-9 Activation of Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells)

Operation	LTE Parameter Setting	NR Parameter Setting
Turning on function switches on the LTE and NR sides	Set the **SpectrumCloud.*SpectrumCloudSwitch*** parameter to **LTE_NR_SPECTRUM_SHR** and select the **LNR_SPECTRUM_SHR_ASYM_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter for each LTE split cell.	Select the **LTE_NR_FDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudSwitch*** parameter and the **LTE_NR_FDD_SPCT_SHR_ASYM_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** parameter.

Operation	LTE Parameter Setting	NR Parameter Setting
Configuring NR-dedicated spectrum resources	-	Configure NR-dedicated spectrum resources at one end of the NR cell spectrum range. The amount of NR-dedicated uplink spectrum resources can be identical to or differ from that of NR-dedicated downlink spectrum resources. ● If only the **gNBDULteNrSpctShrCg.*NrReservedRbStartIndex*** and **gNBDULteNrSpctShrCg.*NrReservedRbEndIndex*** parameters are used to configure the dedicated spectrum, the same dedicated spectrum resources are configured in the uplink and downlink. ● If the **gNBDULteNrSpctShrCg.*NrUlReservedRbStartIndex*** and **gNBDULteNrSpctShrCg.*NrUlReservedRbEndIndex*** parameters are also configured, the uplink dedicated spectrum configuration takes the values of these two parameters, and the downlink dedicated spectrum configuration still takes the values of the **gNBDULteNrSpctShrCg.*NrReservedRbStartIndex*** and **gNBDULteNrSpctShrCg.*NrReservedRbEndIndex*** parameters.

Operation	LTE Parameter Setting	NR Parameter Setting
Configuring LTE and NR spectrum sharing cell groups	Add all planned LTE split cells to an LTE spectrum sharing cell group by setting the **SpectrumCloud.*LteNrSpectrumShrCellGrpId*** and **LteNrSpctShrCellGrp.*LteNrSpectrumShrCellGrpId*** parameters.	1. Configure an association between the LTE spectrum sharing cell group and an NR spectrum sharing cell group by setting the **gNBDULteNrSpctShrCg.*NrSpctShrCellGrpId*** and **gNBDULteNrSpctShrCg.*LteSpctShrCellGrpId*** parameters. 2. Add a planned NR cell to the NR spectrum sharing cell group by setting the **NRDUCellSpctCloud.*NrDUCellId*** and **NRDUCellSpctCloud.*NrSpctShrCellGrpId*** parameters.
Configuring frame offsets	Set the **CellFrameOffset.*FrameOffset*** or **ENodeBFrameOffset.*FddFrameOffset*** parameter. If both parameters are configured, the value of the **CellFrameOffset.*FrameOffset*** parameter is used.	Set the **gNodeBParam.*FrameOffset*** or **gNBFreqBandConfig.*FrameOffset*** parameter. If both parameters are configured, the value of the **gNBFreqBandConfig.*FrameOffset*** parameter is used.
Configuring TA offsets	Set the **CellFrameOffset.*TaOffset*** parameter. The LTE split cells must be configured with the same TA offset.	Set the **NRDUCell.*TaOffset*** parameter.

You are advised to enable "cross-RAT configuration check in DSS scenarios" by selecting both the **DSS_CROSS_RAT_CFG_CHECK_SW** option of the **eNodeBResModeAlgo**.***ParamCfgProtectSw*** parameter and the **DSS_CROSS_RAT_CFG_CHECK_SW** option of the **gNBOamParam**.***CfgInterceptPolicySw*** parameter. This function checks whether the configurations of NR and LTE cells that are configured to be activated in the same spectrum sharing group meet all requirements.

4.1.2.2 Key Technologies

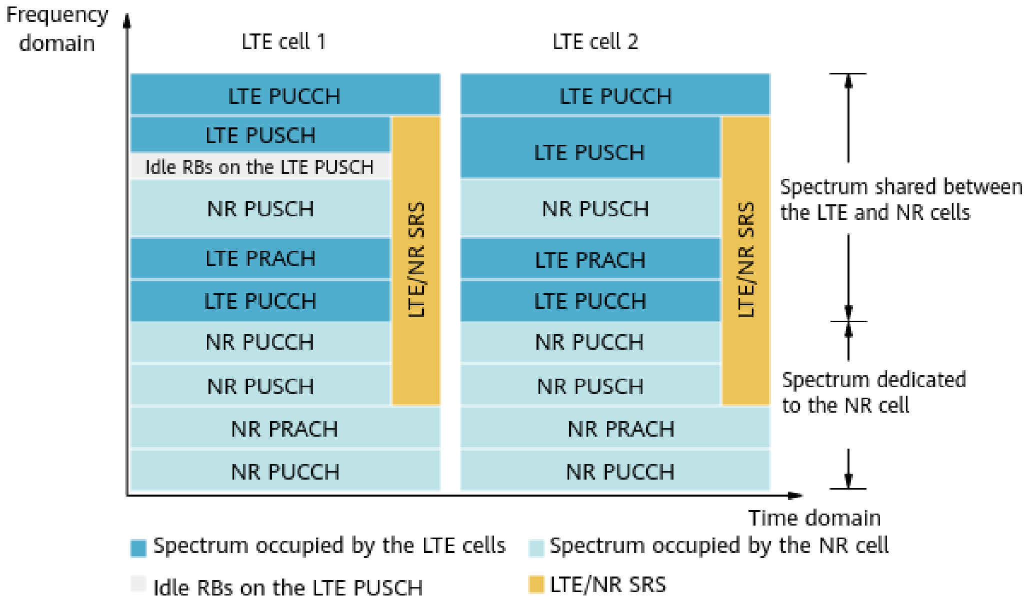
4.1.2.2.1 Coordinated Scheduling of Physical Channel Resources

Some channels will conflict with each other during spectrum sharing between LTE and NR cells in a spectrum sharing cell group. Such conflicts can be avoided by properly coordinating uplink and downlink physical channel resource scheduling, improving spectrum utilization.

Uplink Physical Channels

The LTE split cells in this function use the same spectrum resources through spatial multiplexing. **Figure 4-9** shows the positions of uplink physical channels when LTE and NR cells share spectrum.

Figure 4-9 Positions of uplink physical channels



The uplink physical channel resources of the LTE cells need to be merged before spectrum sharing with the NR cell. **Table 4-10** describes how the channel resources are merged.

Table 4-10 Uplink physical channel resource merging of LTE cells

Channel	Merging Mode
PUCCH PRACH	The combined PUCCH and PRACH resources of the LTE split cells are used as the total resources of these channels allocated to the LTE cells. If these resources are not fully occupied by the PUCCH and PRACH of the LTE cells, the idle resources can be used by the PUSCH of the LTE cells.
SRS	The largest amount of SRS time-frequency resources among the LTE split cells is considered as shared SRS resources allocated to the LTE and NR cells.
PUSCH	The largest amount of PUSCH resources among the LTE split cells is considered as the PUSCH resources allocated to these LTE cells.

Spectrum resources allocated to the NR cell are equal to the total shared spectrum resources minus the spectrum resources allocated to the LTE split cells. The spectrum resources allocated to the LTE split cells are the sum of the channel resources described in **Table 4-10**.

For details about the coordinated scheduling of uplink physical channel resources, see **4.1.1.2.1 Coordinated Scheduling of Physical Channel Resources**. The LTE split cells must have consistent settings.

Downlink Physical Channels

The downlink physical channel resources of the LTE cells need to be merged before spectrum sharing with the NR cell. **Table 4-11** describes how the channel resources are merged.

Table 4-11 Downlink physical channel resource merging of LTE cells

Channel	Merging Mode
CRS	Choose the LTE performance preferred or NR performance preferred policy based on the network plan. ● LTE performance preferred: LTE cells must have different PCIs as well as different PCI mod 3 values. In this case, LTE cells have two or three CRS patterns. The **LTE_CRS_RATEMATCH_ALL_SYM_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** parameter must be selected to enable rate matching on all RBs in the symbols where LTE CRS appears, so as to eliminate the interference of LTE cells on the NR cell. ● NR performance preferred: LTE cells must have different PCIs but the same PCI mod 3 value. In this case, LTE cells have the same CRS pattern. The **LTE_CRS_RATEMATCH_ALL_SYM_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** parameter must be deselected to allow the gNodeB to perform CRS rate matching based on one LTE CRS pattern.
SS/PBCH	The SS/PBCH time-frequency domain positions of the LTE cells are the same.
PDCCH	The largest number of PDCCH symbols among the LTE split cells is used as the number of PDCCH symbols of these LTE cells.
PDSCH	The largest amount of PDSCH resources among the LTE split cells is used as the amount of PDSCH resources of these LTE cells.

For details about the coordinated scheduling of downlink physical channel resources, see **4.1.1.2.1 Coordinated Scheduling of Physical Channel Resources**. The LTE split cells must have consistent settings.

4.1.2.2.2 Flexible Spectrum Priority Mode

The LTE split cells must be added to the same LTE spectrum sharing cell group, and must be configured with the same percentage of spectrum resources to be preferentially allocated and inter-RAT preferential guarantee policy for GBR services. For details about the flexible spectrum priority mode, see **4.1.1.2.2 Flexible Spectrum Priority Mode**.

4.1.2.2.3 Flexible Power Allocation Mode

The LTE split cells must be configured with the same power allocation mode. For details about the power allocation mode, see **4.1.1.2.3 Flexible Power Allocation Mode**.

4.1.2.2.4 Co-carrier Co-CPRI Data

The LTE split cells must be configured with the same co-carrier co-CPRI data policy. For details about co-carrier co-CPRI data, see **4.1.1.2.4 Co-carrier Co-CPRI Data**.

4.1.2.3 Simultaneous Use with Other Functions

4.1.2.3.1 LTE Functions

Simultaneous Use with LTE Downlink Semi-persistent Scheduling

When Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) is used together with LTE downlink semi-persistent scheduling, the upper limit on the proportion of RBs that can be allocated for LTE downlink semi-persistent scheduling is configurable (through the **SpectrumCloud.*DlSpsRestrictRatio*** parameter). It is recommended that this parameter be set to the same value, and that the **SpsSchSwitch** option of the **CellAlgoSwitch.*DlSchSwitch*** parameter be set consistently for the LTE cells. This proportion together with the percentage of spectrum resources to be preferentially allocated to LTE for MBB services (specified by the LTE parameter **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio***) determines the proportion of RBs allocated for LTE downlink semi-persistent scheduling as follows: The proportion of RBs allocated for LTE downlink semi-persistent scheduling = (**SpectrumCloud.*DlSpsRestrictRatio*** x **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio***/100)%.

Simultaneous Use with LTE PRACH Frequency-Domain Position Adaptation

With Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) enabled, more resources are allocated to LTE cells and fewer resources are allocated to the NR cell if PRACH frequency-domain position adaptation is enabled for only some LTE cells and the PRACH frequency-domain positions in LTE cells are different. Therefore, it is recommended that the **PrachFreqAdjSwitch** option of the **CellAlgoSwitch.*RachAlgoSwitch*** parameter be set consistently for the LTE cells.

Simultaneous Use with LTE Uplink Semi-persistent Scheduling and LTE TTI Bundling

When Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) is enabled, uplink semi-persistent scheduling is enabled in only some LTE cells and takes effect, and TTI bundling is enabled only in the rest LTE cells and takes effect, the resources for semi-persistent scheduling and those for TTI bundling are respectively merged for these LTE cells, and consequently the resources for the NR cell decrease. Therefore, it is recommended that the LTE cells use the same settings of the **TtiBundlingSwitch** and **SpsSchSwitch** options of the **CellAlgoSwitch.*UlSchSwitch*** parameter.

Simultaneous Use with WBB

When Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) is used together with WBB, it is recommended that the **WBB_MBB_CONTROL_OPT_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter be selected for each LTE cell. When this option is selected, the eNodeB allocates resources to LTE WBB and MBB services based on the actually available LTE spectrum resources. The percentages of resources actually allocated to MBB services comply with the settings of the **CellWttxParaCfg.*MbbUserDlPrbUpLimit*** and **CellWttxParaCfg.*MbbUserUlPrbUpLimit*** parameters; the percentages of resources actually allocated to WBB services comply with the settings of the **CellWttxParaCfg.*WbbUserDlPrbUpLimit*** and **CellWttxParaCfg.*WbbUserUlPrbUpLimit*** parameters. For details about WBB, see *WBB* in *eRAN Feature Documentation*.

To ensure spectral efficiency, it is recommended that the following equations be met: **CellWttxParaCfg.*MbbUserDlPrbUpLimit*** + **CellWttxParaCfg.*WbbUserDlPrbUpLimit*** =100%, **CellWttxParaCfg.*MbbUserUlPrbUpLimit*** + **CellWttxParaCfg.*WbbUserUlPrbUpLimit*** =100%.

4.1.2.3.2 NR Functions

Simultaneous Use with Downlink Scheduling Bandwidth Adaptation in HDSS Scenarios

When Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) is used together with downlink scheduling bandwidth adaptation in HDSS scenarios, the **NRDUCellDlSch**.***HdssAperCsiIntvlAdaptCoeff*** parameter must be set to the same value for all LTE cells.

4.1.2.3.3 Common Functions Supported by Both LTE and NR

Simultaneous Use with CPRI Compression

The LTE and NR cells with both Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) and CPRI compression enabled must have the same CPRI compression ratio. For details about the configuration requirements, see **4.1.1.3.3 Common Functions Supported by Both LTE and NR**.

Simultaneous Use with LTE and NR Flexible Frame Offset

In Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) scenarios, when the LTE and NR flexible frame offset function is enabled, the frame offset of the NR cell must be 3 ms later than that of each LTE cell. For details about the configuration requirements, see **4.1.1.3.3 Common Functions Supported by Both LTE and NR**.

4.2 Network Analysis

4.2.1 Benefits

This function increases cell downlink throughput. **Table 4-12** and **Table 4-13** list the gains provided by this function in NSA, SA, and NSA and SA hybrid networking compared with a typical static refarming solution.

Table 4-12 Downlink cell peak throughput gain (single LTE cell)

Cell TX Mode	Total Bandwidth	LTE Bandwidth[a]	NR Bandwidth[a]	LTE Cell Downlink Peak Throughput Gain[b]	NR Cell Downlink Peak Throughput Gain[b]
2T	15 MHz	10 MHz	5 MHz	90%–100%	250%–300%
	20 MHz	10 MHz	10 MHz	90%–100%	175%–200%
	20 MHz	15 MHz	5 MHz	90%–100%	355%–400%
	25 MHz	20 MHz	5 MHz	95%–100%	465%–500%
	30 MHz	20 MHz	10 MHz	95%–100%	250%–295%
	40 MHz	20 MHz	20 MHz	95%–100%	175%–200%
4T	15 MHz	10 MHz	5 MHz	90%–100%	240%–290%
	20 MHz	10 MHz	10 MHz	90%–100%	165%–190%
	20 MHz	15 MHz	5 MHz	90%–100%	350%–390%
	25 MHz	20 MHz	5 MHz	95%–100%	455%–485%
	30 MHz	20 MHz	10 MHz	95%–100%	240%–280%
	40 MHz	20 MHz	20 MHz	95%–100%	165%–190%
8T	15 MHz	10 MHz	5 MHz	95%–100%	240%–280%
	20 MHz	15 MHz	5 MHz	95%–100%	355%–400%
	20 MHz	10 MHz	10 MHz	95%–100%	165%–190%
	25 MHz	20 MHz	5 MHz	95%–100%	455%–485%
	30 MHz	20 MHz	10 MHz	95%–100%	240%–280%
	40 MHz	20 MHz	20 MHz	92%–100%	162%–190%
16T	15 MHz	10 MHz	5 MHz	90%–98%	240%–280%
	20 MHz	15 MHz	5 MHz	90%–98%	350%–400%
	20 MHz	10 MHz	10 MHz	90%–98%	160%–190%

Cell TX Mode	Total Bandwidth	LTE Bandwidth[a]	NR Bandwidth[a]	LTE Cell Downlink Peak Throughput Gain[b]	NR Cell Downlink Peak Throughput Gain[b]
	25 MHz	20 MHz	5 MHz	90%–98%	455%–485%
	30 MHz	20 MHz	10 MHz	90%–98%	240%–280%
32T	15 MHz	10 MHz	5 MHz	85%–95%	240%–280%
	20 MHz	15 MHz	5 MHz	85%–95%	345%–400%
	20 MHz	10 MHz	10 MHz	85%–95%	155%–190%
	25 MHz	20 MHz	5 MHz	85%–95%	455%–485%
	30 MHz	20 MHz	10 MHz	85%–95%	240%–280%
	40 MHz	20 MHz	20 MHz	83%–95%	152%–190%
a: LTE bandwidth and NR bandwidth in a static refarming solution when this function is not enabled b: LTE cell gain and NR cell gain provided by this function compared with a static refarming solution. If the **LTE_CRS_RATEMATCH_ALL_SYM_SW** or **LTE_CRS_RM_ALL_SYM_4PORT_SW** option of the NR parameter **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** is selected, the NR cell gain decreases by 15% to 30%, and the LTE cell gain is not affected.					

Table 4-13 Downlink cell peak throughput gain (multiple LTE cells)

Cell TX Mode	Total Bandwidth	LTE Bandwidth[a]	NR Bandwidth[a]	LTE Cell Downlink Peak Throughput Gain[b]	NR Cell Downlink Peak Throughput Gain[b]
8T LTE performance preferred	15 MHz	10 MHz	5 MHz	95%–100%	164%–180%
	20 MHz	15 MHz	5 MHz	95%–100%	200%–220%
	20 MHz	10 MHz	10 MHz	95%–100%	132%–140%
	25 MHz	20 MHz	5 MHz	95%–100%	220%–260%
	30 MHz	20 MHz	10 MHz	95%–100%	164%–180%
	40 MHz	20 MHz	20 MHz	92%–100%	100%–114%
8T NR performance preferred	15 MHz	10 MHz	5 MHz	76%–80%	240%–280%
	20 MHz	15 MHz	5 MHz	57%–60%	355%–400%
	20 MHz	10 MHz	10 MHz	76%–80%	165%–190%
	25 MHz	20 MHz	5 MHz	76%–80%	455%–485%

Cell TX Mode	Total Bandwidth	LTE Bandwidth[a]	NR Bandwidth[a]	LTE Cell Downlink Peak Throughput Gain[b]	NR Cell Downlink Peak Throughput Gain[b]
	30 MHz	20 MHz	10 MHz	76%–80%	240%–280%
	40 MHz	20 MHz	20 MHz	76%–80%	165%–190%
16T LTE performance preferred	15 MHz	10 MHz	5 MHz	90%–98%	164%–180%
	20 MHz	15 MHz	5 MHz	90%–98%	200%–220%
	20 MHz	10 MHz	10 MHz	90%–98%	127%–140%
	25 MHz	20 MHz	5 MHz	90%–98%	220%–260%
	30 MHz	20 MHz	10 MHz	90%–98%	164%–180%
16T NR performance preferred	15 MHz	10 MHz	5 MHz	72%–78%	230%–280%
	20 MHz	15 MHz	5 MHz	63%–68%	350%–400%
	20 MHz	10 MHz	10 MHz	72%–78%	160%–190%
	25 MHz	20 MHz	5 MHz	72%–78%	455%–485%
	30 MHz	20 MHz	10 MHz	72%–78%	230%–280%
32T LTE performance preferred	15 MHz	10 MHz	5 MHz	85%–95%	164%–180%
	20 MHz	15 MHz	5 MHz	85%–95%	200%–220%
	20 MHz	10 MHz	10 MHz	85%–95%	122%–140%
	25 MHz	20 MHz	5 MHz	85%–95%	220%–260%
	30 MHz	20 MHz	10 MHz	85%–95%	164%–180%
	40 MHz	20 MHz	20 MHz	83%–95%	100%–114%
32T NR performance preferred	15 MHz	10 MHz	5 MHz	68%–76%	220%–280%
	20 MHz	15 MHz	5 MHz	68%–76%	345%–400%
	20 MHz	10 MHz	10 MHz	68%–76%	155%–190%
	25 MHz	20 MHz	5 MHz	68%–76%	455%–485%
	30 MHz	20 MHz	10 MHz	68%–76%	220%–280%
	40 MHz	20 MHz	20 MHz	66%–76%	155%–190%

Cell TX Mode	Total Bandwidth	LTE Bandwidth[a]	NR Bandwidth[a]	LTE Cell Downlink Peak Throughput Gain[b]	NR Cell Downlink Peak Throughput Gain[b]
a: LTE bandwidth and NR bandwidth in a static refarming solution when this function is not enabled b: LTE cell gain and NR cell gain provided by this function compared with a static refarming solution. If the **LTE_CRS_RATEMATCH_ALL_SYM_SW** or **LTE_CRS_RM_ALL_SYM_4PORT_SW** option of the NR parameter **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** is selected, the NR cell gain decreases by 15% to 30%, and the LTE cell gain is not affected.					

Specific gains can only be achieved when certain conditions are met.

- The LTE cell gain can only be achieved when all of the following conditions are met:
  - The LTE cell is fully loaded and no online UEs exist in the NR cell.
  - The SSB period for the NR cell (specified by the NR parameter **NRDUCell.*SsbPeriod***) is set to **MS20(20)** or a larger value.
  - The NR cell SIB1 period (specified by the NR parameter **NRDUCell.*Sib1Period***) is set to **MS40(40)**.
  - The NR cell uses a single SSB beam.
- The NR cell gain can only be achieved when all of the following conditions are met:
  - The NR cell is fully loaded and no online UEs exist in the LTE cell.
  - The SSB period for the NR cell is set to **MS20(20)** or a larger value.
  - The NR cell SIB1 period is set to **MS40(40)**.
  - The NR cell uses a single SSB beam.

Actual gains achieved will be less than those listed in the table above if one of the preceding conditions is not met. However, gains will not be negatively impacted as a result.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

- The impacts on the LTE side are as follows:
  - NR reference channels interfere with LTE. The interference leads to LTE wideband quality-related indicators being inaccurately measured. The indicators include the SINR, RSRQ, wideband CQI, PMI, and RI. Consequently:
    - The average cell uplink and downlink IBLER and RBLER increase.

      Average cell uplink IBLER = (**L.Traffic.UL.SCH.QPSK.ErrTB.Ibler** + **L.Traffic.UL.SCH.16QAM.ErrTB.Ibler** + **L.Traffic.UL.SCH.64QAM.ErrTB.Ibler** +

L.Traffic.UL.SCH.256QAM.ErrTB.Ibler)/(**L.Traffic.UL.SCH.QPSK.TB** + **L.Traffic.UL.SCH.16QAM.TB** + **L.Traffic.UL.SCH.64QAM.TB** + **L.Traffic.UL.SCH.256QAM.TB**). Table 4-14 lists the related counters.

Average cell downlink IBLER = (**L.Traffic.DL.SCH.QPSK.ErrTB.Ibler** + **L.Traffic.DL.SCH.16QAM.ErrTB.Ibler** + **L.Traffic.DL.SCH.64QAM.ErrTB.Ibler** + **L.Traffic.DL.SCH.256QAM.ErrTB.Ibler**)/(**L.Traffic.DL.SCH.QPSK.TB** + **L.Traffic.DL.SCH.16QAM.TB** + **L.Traffic.DL.SCH.64QAM.TB** + **L.Traffic.DL.SCH.256QAM.TB**). Table 4-15 lists the related counters.

Average cell uplink RBLER = (**L.Traffic.UL.SCH.QPSK.ErrTB.Rbler** + **L.Traffic.UL.SCH.16QAM.ErrTB.Rbler** + **L.Traffic.UL.SCH.64QAM.ErrTB.Rbler** + **L.Traffic.UL.SCH.256QAM.ErrTB.Rbler**)/(**L.Traffic.UL.SCH.QPSK.TB** + **L.Traffic.UL.SCH.16QAM.TB** + **L.Traffic.UL.SCH.64QAM.TB** + **L.Traffic.UL.SCH.256QAM.TB**). Table 4-16 lists the related counters.

Average cell downlink RBLER = (**L.Traffic.DL.SCH.QPSK.ErrTB.Rbler** + **L.Traffic.DL.SCH.16QAM.ErrTB.Rbler** + **L.Traffic.DL.SCH.64QAM.ErrTB.Rbler** + **L.Traffic.DL.SCH.256QAM.ErrTB.Rbler**)/(**L.Traffic.DL.SCH.QPSK.TB** + **L.Traffic.DL.SCH.16QAM.TB** + **L.Traffic.DL.SCH.64QAM.TB** + **L.Traffic.DL.SCH.256QAM.TB**). Table 4-17 lists the related counters.

Table 4-14 Counters used for measuring the average cell uplink IBLER on the LTE side

Counter ID	Counter Name
1526728186	**L.Traffic.UL.SCH.QPSK.ErrTB.Ibler**
1526728188	**L.Traffic.UL.SCH.16QAM.ErrTB.Ibler**
1526728190	**L.Traffic.UL.SCH.64QAM.ErrTB.Ibler**
1526749528	**L.Traffic.UL.SCH.256QAM.ErrTB.Ibler**
1526727366	**L.Traffic.UL.SCH.QPSK.TB**
1526727367	**L.Traffic.UL.SCH.16QAM.TB**
1526727368	**L.Traffic.UL.SCH.64QAM.TB**
1526749524	**L.Traffic.UL.SCH.256QAM.TB**

Table 4-15 Counters used for measuring the average cell downlink IBLER on the LTE side

Counter ID	Counter Name
1526728180	**L.Traffic.DL.SCH.QPSK.ErrTB.Ibler**
1526728182	**L.Traffic.DL.SCH.16QAM.ErrTB.Ibler**
1526728184	**L.Traffic.DL.SCH.64QAM.ErrTB.Ibler**

Counter ID	Counter Name
1526739660	**L.Traffic.DL.SCH.256QAM.ErrTB.Ibler**
1526727354	**L.Traffic.DL.SCH.QPSK.TB**
1526727355	**L.Traffic.DL.SCH.16QAM.TB**
1526727356	**L.Traffic.DL.SCH.64QAM.TB**
1526739656	**L.Traffic.DL.SCH.256QAM.TB**

Table 4-16 Counters used for measuring the average cell uplink RBLER on the LTE side

Counter ID	Counter Name
1526728187	**L.Traffic.UL.SCH.QPSK.ErrTB.Rbler**
1526728189	**L.Traffic.UL.SCH.16QAM.ErrTB.Rbler**
1526728191	**L.Traffic.UL.SCH.64QAM.ErrTB.Rbler**
1526749530	**L.Traffic.UL.SCH.256QAM.ErrTB.Rbler**
1526727366	**L.Traffic.UL.SCH.QPSK.TB**
1526727367	**L.Traffic.UL.SCH.16QAM.TB**
1526727368	**L.Traffic.UL.SCH.64QAM.TB**
1526749524	**L.Traffic.UL.SCH.256QAM.TB**

Table 4-17 Counters used for measuring the average cell downlink RBLER on the LTE side

Counter ID	Counter Name
1526728181	**L.Traffic.DL.SCH.QPSK.ErrTB.Rbler**
1526728183	**L.Traffic.DL.SCH.16QAM.ErrTB.Rbler**
1526728185	**L.Traffic.DL.SCH.64QAM.ErrTB.Rbler**
1526739661	**L.Traffic.DL.SCH.256QAM.ErrTB.Rbler**
1526727354	**L.Traffic.DL.SCH.QPSK.TB**
1526727355	**L.Traffic.DL.SCH.16QAM.TB**
1526727356	**L.Traffic.DL.SCH.64QAM.TB**
1526739656	**L.Traffic.DL.SCH.256QAM.TB**

- The access success rate, handover success rate, and RRC connection reestablishment success rate decrease.

  Access success rate = **RRC Setup Success Rate**

  Handover success rate = (**L.HHO.IntraeNB.IntraFreq.ExecSuccOut** + **L.HHO.IntraeNB.InterFreq.ExecSuccOut** + **L.HHO.IntereNB.IntraFreq.ExecSuccOut** + **L.HHO.IntereNB.InterFreq.ExecSuccOut**)/(**L.HHO.IntraeNB.IntraFreq.ExecAttOut** + **L.HHO.IntraeNB.InterFreq.ExecAttOut** + **L.HHO.IntereNB.IntraFreq.ExecAttOut** + **L.HHO.IntereNB.InterFreq.ExecAttOut**). Table 4-18 lists the related counters.

  RRC connection reestablishment success rate = **L.RRC.ReEst.Succ**/**L.RRC.ReEst.Att**. Table 4-19 lists the related counters.

Table 4-18 Counters used for measuring the handover success rate

Counter ID	Counter Name
1526726997	**L.HHO.IntraeNB.IntraFreq.ExecSuccOut**
1526727000	**L.HHO.IntraeNB.InterFreq.ExecSuccOut**
1526727003	**L.HHO.IntereNB.IntraFreq.ExecSuccOut**
1526727006	**L.HHO.IntereNB.InterFreq.ExecSuccOut**
1526726996	**L.HHO.IntraeNB.IntraFreq.ExecAttOut**
1526726999	**L.HHO.IntraeNB.InterFreq.ExecAttOut**
1526727002	**L.HHO.IntereNB.IntraFreq.ExecAttOut**
1526727005	**L.HHO.IntereNB.InterFreq.ExecAttOut**

Table 4-19 Counters used for measuring the RRC connection reestablishment success rate

Counter ID	Counter Name
1526727086	**L.RRC.ReEst.Succ**
1526727085	**L.RRC.ReEst.Att**

- The bearer service drop rate increases.

  Bearer service drop rate = **Service Drop Rate**

- The PDCCH DTX proportion increases.

  PDCCH DTX proportion = (**L.ChMeas.PDCCH.DL.DTXNum.AggLvl1** + **L.ChMeas.PDCCH.DL.DTXNum.AggLvl2** + **L.ChMeas.PDCCH.DL.DTXNum.AggLvl4** + **L.ChMeas.PDCCH.DL.DTXNum.AggLvl8**)/ (**L.ChMeas.PDCCH.AggLvl1Num** + **L.ChMeas.PDCCH.AggLvl2Num** + **L.ChMeas.PDCCH.AggLvl4Num** + **L.ChMeas.PDCCH.AggLvl8Num**)

  Table 4-20 lists the related counters.

Table 4-20 Counters related to PDCCH DTX

Counter ID	Counter Name
526728664	**L.ChMeas.PDCCH.DL.DTXNum.AggLvl1**
1526728665	**L.ChMeas.PDCCH.DL.DTXNum.AggLvl2**
1526728666	**L.ChMeas.PDCCH.DL.DTXNum.AggLvl4**
1526728667	**L.ChMeas.PDCCH.DL.DTXNum.AggLvl8**
1526728668	**L.ChMeas.PDCCH.AggLvl1Num**
1526728669	**L.ChMeas.PDCCH.AggLvl2Num**
1526728670	**L.ChMeas.PDCCH.AggLvl4Num**
1526728671	**L.ChMeas.PDCCH.AggLvl8Num**

– After this function is enabled, there are fewer spectrum resources available for LTE than when the full bandwidth is available. This results in a fluctuation in the average uplink and downlink throughputs of the LTE cell, a decrease in the average uplink and downlink throughputs of UEs, and an increase in the uplink and downlink voice packet loss rates in the LTE cell.

- Average uplink cell throughput = **Cell Uplink Average Throughput**
- Average downlink cell throughput = **Cell Downlink Average Throughput**
- Average uplink UE throughput = **User Uplink Average Throughput**
- Average downlink UE throughput = **User Downlink Average Throughput**
- Uplink voice packet loss rate in a cell = **Uplink Packet Loss Rate (VoIP)**
- Downlink voice packet loss rate in a cell = **Downlink Packet Loss Rate (VoIP)**

- The impacts on the NR side are as follows:

  After this function is enabled, there are fewer spectrum resources available for NR than when the full bandwidth is available. This results in a fluctuation in the average uplink and downlink throughputs of the NR cell, and in a decrease in the average uplink and downlink throughputs of UEs.
  - Average uplink cell throughput = **Cell Uplink Average Throughput (DU)**
  - Average downlink cell throughput = **Cell Downlink Average Throughput (DU)**
  - Average uplink UE throughput = **User Uplink Average Throughput (DU)**
  - Average downlink UE throughput = **User Downlink Average Throughput (DU)**

  NOTE

  LTE synchronization signals, the PBCH, reference channels, and system information do not interfere with NR. As a result, NR wideband quality-related indicators are not affected.

  A change in the CRS bandwidth or power in a neighboring LTE cell will affect performance counters of the NR cell in both the uplink and downlink, such as the IBLER, RBLER, average MCS index, average cell throughput, and average UE throughput.
- In Hybrid DSS Based on Asymmetric Bandwidth scenarios, NR may be allocated incomplete RBGs and be unable to use these resources. If this occurs, RBs are wasted. The larger the bandwidth and RBG granularity, the greater the waste.
- Measurement for counters related to uplink interference is affected because available RBs are allocated to LTE and NR in real time when the two RATs share spectrum dynamically. As a result, the values of the following counters may change:
  - **L.UL.Interference.Avg**
  - **L.UL.Interference.Max**
  - **L.UL.Interference.Avg.PRB0** to **L.UL.Interference.Avg.PRB99**
  - **N.UL.NI.Avg**
  - **N.UL.NI.Max**
  - **N.UL.NI.Avg.PRB0** to **N.UL.NI.Avg.PRB105**
- This function involves the estimation and allocation of spectrum resources on both the LTE and NR sides, and therefore has the following impacts:
  - Increased board user-plane CPU usage
    - Average user-plane CPU usage of an eNodeB board = **L.Traffic.Board.UPlane.CPULoad.AVG**
    - Average user-plane CPU usage of a gNodeB board = **VS.NRBoard.UPlane.CPULoad.Avg**
  - Increased ping delay on the LTE side
- After this function is enabled on the LTE and NR sides, if MBSFN subframes are configured on TM9-dedicated carriers, the control format indicator (CFI) value and control channel element (CCE) usage change as the CFI range differs between MBSFN subframes and common subframes. The CCE usage

can be calculated using the following formula: CCE usage = (**L.ChMeas.CCE.CommUsed** + **L.ChMeas.CCE.ULUsed** + **L.ChMeas.CCE.DLUsed**)/**L.ChMeas.CCE.Avail**.

- On the NR side, if this function is enabled and SSB is positioned on the shared spectrum, fewer resources are available for the PDCCH so as to yield to SSB. As a result, the CCE allocation success rate and the user-perceived throughput decrease in both the uplink and downlink.
- When this function works together with the co-carrier co-CPRI data function, the LTE cell bandwidth is 10 MHz, the NR cell bandwidth is greater than 10 MHz and less than 20 MHz, and the LTE cell is established prior to the NR cell, the LTE cell will be reestablished to support the co-carrier co-CPRI data function.
- If the co-carrier co-CPRI data function is enabled and the UE is located near the cell center, the gNodeB implements time synchronization between the LTE and NR networks. This has the following impacts:
  - On the LTE side: The values of counters **L.TA.UE.Index0** to **L.TA.UE.Index6** slightly increase, and the values of counters **L.RA.TA.UE.Index0** to **L.RA.TA.UE.Index11** slightly fluctuate.
  - On the NR side:
    - The values of counters **N.RA.TA.UE.Index0** to **N.RA.TA.UE.Index12** slightly increase.
    - If CA is also enabled, the average uplink throughput (**Cell Uplink Average Throughput (DU)**) of both the PCell and SCells fluctuates.
    - If super uplink is also enabled, the average uplink throughput (**Cell Uplink Average Throughput (DU)**) of the SUL cell fluctuates.
- If the LTE TA offset (specified by the **CellFrameOffset.*TaOffset*** parameter) is changed from **0TS** to another value, the values of counters **L.TA.UE.Index0** to **L.TA.UE.Index15** and the values of counters **L.RA.TA.UE.Index0** to **L.RA.TA.UE.Index11** slightly fluctuate.
- If the four-beam function is enabled, the values of the following NR indicators decrease:
  - RRC connection setup success rate = **RRC Setup Success Rate (CU)**
  - QoS flow setup success rate of all services = **QoS Flow Setup Success Rate (CU)**
  - Handover success rate = (**L.HHO.IntraeNB.IntraFreq.ExecSuccOut** + **L.HHO.IntraeNB.InterFreq.ExecSuccOut** + **L.HHO.IntereNB.IntraFreq.ExecSuccOut** + **L.HHO.IntereNB.InterFreq.ExecSuccOut**)/(**L.HHO.IntraeNB.IntraFreq.ExecAttOut** + **L.HHO.IntraeNB.InterFreq.ExecAttOut** + **L.HHO.IntereNB.IntraFreq.ExecAttOut** + **L.HHO.IntereNB.InterFreq.ExecAttOut**)
  - Random access success rate = **N.RA.Contention.Resolution.Succ**/**N.RA.Contention.Att**
  - Average downlink UE throughput = **User Downlink Average Throughput (DU)**

  The value of the following NR indicator increases:

CCE allocation failure rate = 100% x (1 – (**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/**N.CCE.DL.AllocReq.Num**)

- When the NR SSB and system information on the NR PDSCH are positioned beyond the NR-dedicated spectrum, the impacts are as follows:
  - If the **LNR_RES_ALLOC_DIRECTION_OPT_SW** option of the **LteNrSpctShrCellGrp.*LteNrSpctShrSwitch*** parameter is selected, the CCE allocation failure rate will decrease. If this option is deselected, the CCE allocation failure rate will increase and KPIs such as the **QoS Flow Setup Success Rate (CU)** will be affected.
  - The processing delay at the gNodeB MAC layer will increase in the case of a high proportion of retransmissions upon DTXs.

  Uplink CCE allocation failure rate = 1 – (**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num**

  Downlink CCE allocation failure rate = 1 – (**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/**N.CCE.DL.AllocReq.Num**

  Processing delay at the MAC layer in the gNodeB = (**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num**)/1000
- If downlink scheduling bandwidth adaptation in HDSS scenarios is enabled for a cell with Hybrid DSS Based on Asymmetric Bandwidth enabled and there are surrounding LTE-only base stations, NR traffic channels of this cell will be less interfered with by surrounding LTE-only base stations and the NR downlink user-perceived rates will increase.
- If LTE and NR SRS resource allocation optimization is enabled, there may be a short period of time when NR UEs do not periodically report SRS or have SRS subframes migrated. During this period, the uplink and downlink user-perceived rates decrease.
- If the "LTE and NR 3-ms frame offset difference" function is enabled, the arrangement of subframes prioritized for one RAT changes and MBSFN subframes cannot be configured. As a result, the scheduling behavior in LTE and NR cells engaged in Hybrid DSS Based on Asymmetric Bandwidth may change, leading to changes in the average UE and cell throughputs in the uplink and downlink for both LTE and NR. In addition, this function affects the benefits brought by functions that depend on MBSFN subframes.

4.3 Requirements

4.3.1 Licenses

Hybrid DSS Based on Asymmetric Bandwidth requires both feature and capacity licenses, as listed in **Table 4-21** and **Table 4-22**, respectively.

Table 4-21 Feature licenses

RAT	Feature ID	Feature Name	Model	License Control Item	NE	Sales Unit
LTE FDD	MRFD-160222	LTE FDD and NR Flash Dynamic Spectrum Sharing (LTE FDD)	LT1S0LFNSS00	LTE FDD and NR Flash Dynamic Spectrum Sharing(LTE FDD)	eNodeB	per Cell
NR	MRFD-160262	LTE FDD and NR Flash Dynamic Spectrum Sharing (NR)	NR0S00FNSS00	LTE FDD and NR Flash Dynamic Spectrum Sharing (NR)	gNodeB	per Cell
LTE FDD	MRFD-171221	Hybrid DSS Based on Asymmetric Bandwidth(LTE FDD)	LT1SHDSSBOAB	Hybrid DSS Based on Asymmetric Bandwidth (LTE FDD)	eNodeB	per Cell
NR	MRFD-171261	Hybrid DSS Based on Asymmetric Bandwidth(NR)	NR0S0HDBAB00	Hybrid DSS Based on Asymmetric Bandwidth (NR)	gNodeB	per Cell

<table>
<tr><th>RAT</th><th>Feature ID</th><th>Feature Name</th><th>Model</th><th>License Control Item</th><th>NE</th><th>Sales Unit</th></tr>
<tr><td colspan="7">
For the license control items LT1S0LFNSS00 and LT1SHDSSBOAB:

If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:

When the license sales number pre-check function is enabled, the SpectrumCloud.SpectrumCloudSwitch parameter cannot be set to LTE_NR_SPECTRUM_SHR and the LNR_SPECTRUM_SHR_ASYM_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter cannot be selected due to license insufficiency, preventing cell activation failures.
When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified.

For the license control items NR0S00FNSS00 and NR0S0HDBAB00:

If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:

When the license sales number pre-check function is enabled, the LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter and the LTE_NR_FDD_SPCT_SHR_ASYM_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter cannot be selected due to license insufficiency, preventing cell activation failures.
When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified.

On the LTE side, the license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the eNBLicenseAlmThd.LicenseCheckCtrlSw parameter.
On the NR side, the license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.
</td></tr>
</table>

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

Table 4-22 Capacity licenses

RAT	Model	License Control Item	NE	Sales Unit
LTE FDD	LT1S5000RFSS	Spectrum Sharing License for 5000 Series RF Module(FDD)	eNodeB	per Band per RU
NR	NR0SSSLRFM 00	Spectrum Sharing License for 5000 Series RF Module (NR)	gNodeB	per Band per RU

The LTE and NR flexible frame offset function requires a capacity license, as listed in **Table 4-23**.

Table 4-23 Capacity license for the LTE and NR flexible frame offset function

RAT	Model	License Control Item	NE	Sales Unit
NR	NR0SFLCFOS0 0	Flexible Cell Frame Offset (NR FDD)	gNodeB	per RU

NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

- Prerequisite functions on the LTE side

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Normal CP	**Cell.*UlCyclicPrefix***	*Extended CP*	Set this parameter to **NORMAL_CP**.
LTE FDD	PRACH Frequency Offset Strategy	**RACHCfg.*PrachFreqOffsetStrategy***	*Random Access Control*	Set this parameter to **FULLY_AUTOMATIC**.
LTE FDD	CSI-RS Period	**CellCsiRsParaCfg.*CsiRsPeriod***	*Dedicated Carrier for TM9*	When the **CellCsiRsParaCfg.*CsiRsSwitch*** parameter is set to **FIXED_CFG** or **ADAPTIVE_CFG**, the following requirements must be met: ● If the NR SSB period (specified by the NR parameter **NRDUCell.*SsbPeriod***) is greater than or equal to **MS20(20)**, **CellCsiRsParaCfg.*CsiRsPeriod*** must be set to **ms10** or a larger value. ● If the NR SSB period is less than or equal to **MS10(10)**, **CellCsiRsParaCfg.*CsiRsPeriod*** must be set to **ms20** or a larger value.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	PHICH resource	**PHICHCfg.*PhichResource***	*Physical Channel Resource Management*	The dependency relationship exists only when the cell bandwidth is 10 MHz and requires this parameter to be set to **ONE_SIXTH**, **HALF**, or **ONE**.
LTE FDD	SRS Configuration Indicator	**SRSCfg.*SrsCfgInd***	*Physical Channel Resource Management*	Set this parameter to **BOOLEAN_TRUE**.
LTE FDD	SRS ACK/NACK simultaneous transmission	**SRSCfg.*AnSrsSimuTrans***	*Physical Channel Resource Management*	Set this parameter to **BOOLEAN_TRUE**.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	SRS subframe configuration	**SRSCfg.*SrsSubframeCfg***	*Physical Channel Resource Management*	When SRS automatic neighboring cell measurement is enabled (the **NCellSrsMeasPara.*SrsAutoNCellMeasSwitch*** parameter set to **ON**), the **SRSCfg.*SrsSubframeCfg*** parameter must be set to **SC2**. When SRS automatic neighboring cell measurement is disabled, the **SRSCfg.*SrsSubframeCfg*** parameter must be set to **SC3**, **SC7**, **SC14**, or **SC0**. When the "LTE and NR 3-ms frame offset difference" function is enabled, the **SRSCfg.*SrsSubframeCfg*** parameter must be set to **SC6**, **SC13**, **SC14**, or **SC0**.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Dynamic adjustment to cell-specific SRS subframe configuration	**SrsSubframeRecfSwitch** option of the **CellAlgoSwitch.*SrsAlgoSwitch*** parameter	*Physical Channel Resource Management*	Select this option of this parameter only when the **LTE_UE_SRS_NOT_CONFIG_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter is deselected.
LTE FDD	Uplink adaptive HARQ	**CellUlschAlgo.*AdaptHarqSwitch***	*Uplink Scheduling*	Set this parameter to **ADAPTIVE_HARQ_SW_ON** or **ADAPTIVE_HARQ_SW_SEMI_ON**.

- Prerequisite functions on the NR side

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Basic functions of downlink scheduling	None	*Downlink Scheduling*	The **NRDUCellPdsch.*DlDmrsMaxLength*** parameter must be set to **1SYMBOL**.
NR FDD	Search Space Zero	**NRDUCellPdcch.*SearchSpaceZero***	*Channel Management*	The **NRDUCellPdcch.*SearchSpaceZero*** parameter must be set to **DEFAULT**.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	SSB beam adjustment	**SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch*** parameter	None	SSB beam adjustment must be enabled before Hybrid DSS Based on Asymmetric Bandwidth is enabled for NR LumiLink cells. Hybrid DSS Based on Asymmetric Bandwidth can take effect in NR LumiLink cells only if the **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_20M** for the cells.
NR FDD	LumiLink solution (NR)	**EASY_OMNI_SITE_MODE_SW** option of the **NRDUFddCellTrpBeam**.***ScenarioModeSw*** parameter	*LumiLink Solution*	
NR FDD	Downlink interference randomization optimization in HDSS scenarios	**HDSS_DL_INTRF_RANDOM_OPT_SW** option of the **NRDUCellDlSch**.***DlSchAlgoEnhSwitch*** parameter	*LTE FDD and NR Hybrid Dynamic Spectrum Sharing*	Downlink scheduling bandwidth adaptation in HDSS scenarios (controlled by the **HDSS_DL_SCH_BW_ADAPT_SW** option of the **NRDUCellDlSch**.***DlSchAlgoEnhSwitch*** parameter) requires downlink interference randomization optimization in HDSS scenarios to be enabled.

4.3.2.2 Mutually Exclusive Functions

- Mutually exclusive functions on the LTE side
  - CIoT functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	eMTC introduction	**EMTC_SWITCH** option of the **CellEmtcAlgo.*EmtcAlgoSwitch*** parameter	*eMTC*
NB-IoT	LTE in-band deployment	**CellRbReserve.*RbRsvMode*** set to **NB_RESERVED** or **NB_DEPLOYMENT**	*NB-IoT Basics (FDD)*
NB-IoT	NR guard band deployment	**Prb**.***DeployMode*** set to **GUARD_BAND** and **Prb**.***NrCellGbCfgInd*** set to **CFG**	*NB-IoT Basics (FDD)*

– Inter-RAT coordination functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	GSM and LTE Spectrum Concurrency	**SpectrumCloud.*SpectrumCloudSwitch*** set to **GL_SPECTRUM_CONCURRENCY**	*GSM and LTE Spectrum Concurrency*
LTE FDD	UMTS and LTE Spectrum Sharing	**SpectrumCloud.*SpectrumCloudSwitch*** set to **UL_SPECTRUM_SHARING**	*UMTS and LTE Spectrum Sharing*
LTE FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA	**SpectrumCloud.*SpectrumCloudSwitch*** set to **DC_HSDPA_BASED_UL_SPECTRUM_SHR**	*UMTS and LTE Spectrum Sharing Based on DC-HSDPA*
LTE FDD	UMTS and LTE Dynamic Power Sharing	**UMTS_LTE_DYN_POWER_SHARING_SW** option of the **CellDynPowerSharing.*DynamicPowerSharingSwitch*** parameter	*UMTS and LTE Dynamic Power Sharing*
LTE FDD	Dynamic Power Sharing Between LTE Carriers	**LTE_DYN_POWER_SHARING_SW** option of the **CellDynPowerSharing.*DynamicPowerSharingSwitch*** parameter	*Dynamic Power Sharing Between LTE Carriers*

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Extreme power sharing	**LTE_EXTREME_POWER_SHARING_SW** option of the **CellDynPowerSharing.*DynamicPowerSharingSwitch*** parameter	*Dynamic Power Sharing Between LTE Carriers*
LTE FDD	GSM and LTE Dynamic Power Sharing	**CellAlgoSwitch.*GLPwrShare***	*GSM and LTE Dynamic Power Sharing*
LTE FDD	LTE and NR SRS resource allocation optimization[a]	**LTE_NR_SRS_ALLOC_OPT_SW** option of the **LteNrSpctShrCellGrp.*LteNrSpctShrSwitch*** parameter	*LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
LTE FDD	Absence of SRS configuration for LTE UEs[a]	**LTE_UE_SRS_NOT_CONFIG_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter	*LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
a: This function cannot work with the "LTE and NR 3-ms frame offset difference" function (controlled by the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of the **SpectrumCloud**.***SpectrumCloudExtSwitch*** parameter).			

– Energy saving functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Multi-RAT Carrier Joint Intelligent Shutdown (UL/GUL)	**InterRatCellShutdown**.***ForceShutdownSwitch***	*Multi-RAT Coordinated Energy Saving*
LTE FDD	32T beam mode shift for energy saving	**BEAM_MODE_SHIFT_SW** option of the **SectorSplitGroup**.***SplitCellGroupOptSwitch*** parameter	*Energy Conservation and Emission Reduction*
LTE FDD	RF channel dynamic muting[a]	**RF_CHN_DYN_MUTING_SW** option of the **CellRfChnDynMuting.*RfChnDynMutingAlgoSwitch*** parameter	*Energy Conservation and Emission Reduction*

<table>
<tr><th>RAT</th><th>Function Name</th><th>Function Switch</th><th>Reference (eRAN Feature Documentation)</th></tr>
<tr><td>LTE FDD</td><td>LNR coordinated channel shutdown[b]</td><td>CellRfShutdown.MultiRatJointChnShutdownSw</td><td>Multi-RAT Coordinated Energy Saving</td></tr>
<tr><td colspan="4">a: In 4T and 8T scenarios where the SymbolPwrSaving.MultiRatSymbolShutdownSw parameter is set to ON, Hybrid DSS Based on Asymmetric Bandwidth can be enabled together with RF channel dynamic muting. In other scenarios, the two functions cannot work together. For these functions to work together in 4T2P and 8T2P scenarios, the LTE and NR bandwidth combinations must be one of the following: LTE 10 MHz + NR 15 MHz, LTE 10 MHz + NR 20 MHz, and LTE 15 MHz + NR 20 MHz. In addition, the NR and LTE PDCCHs cannot be positioned in the same symbol, which means that all of the following conditions must be met:
● The frequency-domain position of the NR common PDCCH is within the shared spectrum.
● The UE_PDCCH_FULL_BANDWIDTH_CFG_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter is selected.
● If the PDCCH_RATEMATCH_SW option of the NRDUCELLPDSCH.RateMatchSwitch parameter is selected, the spectrum occupied by the UE-specific PDCCH does not overlap with the NR-dedicated spectrum.
b: When Hybrid DSS Based on Asymmetric Bandwidth is enabled, LNR coordinated channel shutdown can take effect in 2T and 4T scenarios, but not in 8T or massive MIMO scenarios. In 2T and 4T scenarios, both NR and LTE must work in V100R020C10 or later; for details about the configuration requirements, see section "Cells" for LNR coordinated channel shutdown in Multi-RAT Coordinated Energy Saving.</td></tr>
</table>

– MIMO functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Static Multiple Beam 2.0	**SMB2_SW** option of the **SectorSplitGroup**.***SplitCellGroupOptSwitch*** parameter	*Smart 8T8R (FDD)*
LTE FDD	Enhanced extreme power sharing	**LTE_ENH_EXT_POWER_SHARING_SW** option of the **CellDynPowerSharing**.***DynamicPowerSharingSwitch*** parameter	*Smart 8T8R (FDD)*

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Centralized power sharing	**LTE_CENTRAL_POWER_SHARING_SW** option of the **CellDynPowerSharing**.***DynamicPowerSharingSwitch*** parameter	*Smart 8T8R (FDD)*
LTE FDD	Unequal downlink port power[a]	**PDSCHCfg.*TxPowerOffsetAnt2*** **PDSCHCfg.*TxPowerOffsetAnt3***	*Smart 8T8R (FDD)*
LTE FDD	Power optimization for Static Multiple Beam[a]	**8T_SMB_PWR_OPT_SW** option of the **PDSCHCfg**.***TxChnPowerCfgSw*** parameter	*Smart 8T8R (FDD)*
LTE FDD	Uplink SRS-based frequency selective scheduling[a]	**CellUlschAlgo.*UlSrsFreqSelSchSinrThld*** set to a value other than **255**	*Smart 8T8R (FDD)*
LTE FDD	Virtual 4T4R	**Virtual4T4RSwitch** option of the **CellAlgoSwitch.*EmimoSwitch*** parameter	*Virtual 4T4R (FDD)*
LTE FDD	Downlink turbo pilot	**SectorSplitGroup.*TurboPilotPowerOffset*** set to a non-zero value, and **Cell.*TxRxMode*** set to **8T8R**, **16T16R**, or **32T32R**	*Massive MIMO Enhancements (FDD)*
LTE FDD	Intelligent beam shaping[b]	**MM_INTELLIGENT_BEAM_SHAPING_SW** option of the **SectorSplitGroup**.***SectorSplitSwitch*** parameter	*Massive MIMO Enhancements (FDD)*
LTE FDD	Intelligent beam scheduling[b]	**JOINT_SCHEDULING_SWITCH** option of the **SectorSplitGroup**.***IntelligentBeamSchSwitch*** parameter	*Massive MIMO Enhancements (FDD)*

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Intelligent beam weight optimization[b]	**INTELLIGENT_BEAM_WEIGHT_OPT_SW** option of the **SectorSplitGroup**.***SectorSplitSwitch*** parameter	*Massive MIMO Enhancements (FDD)*
a: This mutually exclusive relationship exists only when Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) is enabled. b: This function cannot be enabled together with "absence of SRS configuration for LTE UEs" for an activated LTE cell engaged in Hybrid DSS Based on Asymmetric Bandwidth. ("Absence of SRS configuration for LTE UEs" is controlled by the **LTE_UE_SRS_NOT_CONFIG_SW** option of the **SpectrumCloud**.***SpectrumCloudEnhSwitch*** parameter.)			

– Multi-RAT coordination functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	GSM and LTE Buffer Zone Optimization	**AvoidGeranInterfSwitch** option of the **CellAlgoSwitch**.***AvoidInterfSwitch*** parameter	*GSM and LTE Buffer Zone Optimization*
LTE FDD	UL Refarming Zero Bufferzone	**AvoidUtranInterfSwitch** option of the **CellAlgoSwitch**.***AvoidInterfSwitch*** parameter	*UL Refarming Zero Bufferzone*
LTE FDD	UMTS and LTE Zero Bufferzone	**UMTS_LTE_ZERO_BUFFER_ZONE_SW** option of the **ULZeroBufferZone**.***ZeroBufZoneSwitch*** parameter	*UMTS and LTE Zero Bufferzone*
LTE FDD	CDMA and LTE Zero Bufferzone	**AvoidCDMAInterfSwitch** option of the **CellAlgoSwitch**.***AvoidInterfSwitch*** parameter	*CDMA and LTE Zero Bufferzone*

– Coverage improvement functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Cell radius greater than 100 km	**Cell.*CellRadius***	*Extended Cell Range*
LTE FDD	Superior Uplink Coverage	**CellAlgoExtSwitch.*UlCoverageEnhancementSw***	*Superior Uplink Coverage (FDD)*

– Interference mitigation functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Super combined cell	**SuperCombCellSwitch** option of the **CellAlgoSwitch.*SfnAlgoSwitch*** parameter	*Super Combined Cell (FDD)*
LTE FDD	Dynamic TDM eICIC	**CellAlgoSwitch.*EicicSwitch***	*TDM eICIC (FDD)*
LTE FDD	Channel isolation for PIM interference avoidance	**PIM_INTERF_CHN_SHUTDOWN_SW** option of the **PimInterfSmartAvoid.*PimAlgoSwitch*** parameter	*Interference Detection and Suppression*
LTE FDD	Intelligent PIM interference avoidance[a]	**PIM_INTERF_SMART_AVOID_SW** option of the **PimInterfSmartAvoid.*PimAlgoSwitch*** parameter	*Interference Detection and Suppression*
LTE FDD	Adaptive SFN/SDMA (FDD)[b]	**CellAlgoSwitch.*SfnUlSchSwitch*** set to **ADAPTIVE** or **CellAlgoSwitch.*SfnDlSchSwitch*** set to **ADAPTIVE**	*SFN*
LTE FDD	SFN algorithm	**CellAlgoSwitch.*SfnAlgoSwitch***	None

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
a: Intelligent PIM interference avoidance cannot be enabled when the **EMTC_SWITCH** option of the **CellEmtcAlgo**.***EmtcAlgoSwitch*** parameter is selected and the **SpectrumCloud**.***SpectrumCloudSwitch*** parameter is set to **LTE_NR_SPECTRUM_SHR**. b: When RRUs are combined or RRUs and pRRUs are combined to serve an SFN cell, adaptive SFN/SDMA (FDD) cannot work with Hybrid DSS Based on Asymmetric Bandwidth.			

– Power management functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Coordinated scheduling-based power control	**CellCspcPara.*CellCspcSwitch***	*CSPC*
LTE FDD	Cell Power Limit	**PDSCHCfg.*EmfPowerLimitSwitch*** set to **ON**, and **PDSCHCfg.*CellPowerLimit*** set to a value other than **0**	*On-Demand TX Power Allocation Under EME*

– Basic functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	UL CRA	**UL_COORD_RES_ALLOC_SWITCH** option of the **UlCsAlgoPara.*UlCsSw*** parameter	*Uplink Coordinated Scheduling*
LTE FDD	UL CPC	**UL_COORD_PC_SWITCH** option of the **UlCsAlgoPara.*UlCsSw*** parameter	*Uplink Coordinated Scheduling*
LTE FDD	Manual RB blocking	**CellRbReserve.*RbRsvMode*** set to **RB_MASKING** or **RB_MASKING_WITH_SRS**	*Downlink Scheduling*

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Flexible PUCCH configuration [a]	**PucchFlexCfgSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter	*Physical Channel Resource Management*
LTE FDD	ePDCCH	**CellPdcchAlgo.*EpdcchAlgoSwitch***	*Physical Channel Resource Management*
LTE FDD	Extended PHICH	**PHICHCfg.*PhichDuration***	*Physical Channel Resource Management*
a: When Hybrid DSS Based on Asymmetric Bandwidth is enabled, the settings of the **CellRbReserve.*RbRsvMode***, **CellRbReserve.*RbRsvType***, **CellRbReserve.*RbRsvStartIndex***, and **CellRbReserve.*RbRsvEndIndex*** parameters can be used as an alternative to the flexible PUCCH configuration function.			

– Mobility management functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	High speed mobility[a]	**Cell.*HighSpeedFlag***	*High Speed Mobility*
LTE FDD	Ultra high speed mobility[a]	**Cell.*HighSpeedFlag***	*High Speed Mobility*
a: This mutually exclusive relationship exists only when the **LTE_NR_SRS_ALLOC_OPT_SW** option of the **LteNrSpctShrCellGrp.*LteNrSpctShrSwitch*** parameter is selected.			

– Functions related to RAN services

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	In-band relay	**InBandRelayDeNbSwitch**, **InBandRelayReNbSwitch**, and **InBandRelayDistantDeploySw** options of the **CellAlgoSwitch.*RelaySwitch*** parameter	*Relay*
LTE FDD	Uplink Relay Subframe Ratio[a]	**CellRelayAlgoPara**.***UlRelaySubframeRatio***	*Relay*
LTE FDD	Downlink Relay Subframe Ratio[a]	**CellRelayAlgoPara**.***DlRelaySubframeRatio***	*Relay*
LTE FDD	eMBMS	**CellMBMSCfg.*MBMSSwitch***	*eMBMS*
a: When Hybrid DSS Based on Asymmetric Bandwidth and out-of-band relay are both enabled, the values of both **CellRelayAlgoPara**.***UlRelaySubframeRatio*** and **CellRelayAlgoPara**.***DlRelaySubframeRatio*** must be less than the value of **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio***.			

– Other functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	LTE flexible spectrum	**Cell**.***CustomizedBandWidthCfgInd*** set to **FLEX_CFG**	*LTE Flexible Spectrum (FDD)*
LTE FDD	TM4 and TM6 Adaptation	**CL_TM_ADAPTATION_OPT_SW** option of the **CellMimoParaCfg.*MimoSwitch*** parameter	*Low-Band Booster*
LTE FDD	Flexible bandwidth based on overlapping carriers	**DdCellGroup.*DdBandwidth***	*Flexible Bandwidth based on Overlap Carriers (FDD)*
LTE FDD	PUSCH FH	**CellUlschAlgo.*UlHoppingType***	None

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	LTE flexible bandwidth	**CellLteFlexBw.*LteFlexBwSwitch***	None
LTE FDD	Short TTI	**SHORT_TTI_SW** option of the **CellShortTtiAlgo.*SttiAlgoSwitch*** parameter	*Short TTI (FDD)*
LTE FDD	Zero Guard Band Between Contiguous Intra-Band Carriers	**CONTIG_INTRA_BAND_CARR_SW** option of the **ContigIntraBand-Carr.*ContigIntraBand-CarrSw*** parameter	*Seamless Intra-Band Carrier Joining (FDD)*
LTE FDD	Uplink data transmission path selection[a]	**NSA_DC_UL_PATH_SELECTION_SW** option of the **NsaDcMgmtConfig.*NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*
LTE FDD	CSI-RS Period[b]	**CellCsiRsParaCfg.*CsiRsPeriod***	None
LTE FDD	CSI-RS subframe selection[c]	**CSIRS_SUBFRAME_SELECT_SW** option of the **SpectrumCloud.*LnrSpectrumShrOptSwitch*** parameter	None
a: This mutually exclusive relationship exists only when the **LTE_UE_SRS_NOT_CONFIG_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter is selected. b: When the "LTE and NR 3-ms frame offset difference" function (controlled by the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of the **SpectrumCloud**.***SpectrumCloudExtSwitch*** parameter) is enabled, the **CellCsiRsParaCfg**.***CsiRsPeriod*** parameter cannot be set to **ms5** or **ms10**. c: The "LTE and NR 3-ms frame offset difference" function cannot work with this function.			

- Mutually exclusive functions on the NR side
  – MIMO functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Inter-gNodeB Distributed MIMO	**INTER_GNB_DL_D_MIMO_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Distributed Antenna Solutions (FDD)*
NR FDD	PUSCH coordinated power control	**PUSCH_COORD_PWR_CTRL_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch*** parameter	*Smart Coverage (FDD)* *Smart 8T8R (FDD)* *Smart Massive MIMO (FDD)*
NR FDD	Coordinated scheduling	**DL_COORDINATED_SCH_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Smart Coverage (FDD)* *Smart 8T8R (FDD)* *Smart Massive MIMO (FDD)*
NR FDD	MU-MIMO measurement enhancement	**MU_MIMO_MEAS_ENH_SW** option of the **NRDUCellDlMimo**.***DLMuMimoSchSupplementSw*** parameter	*Smart 8T8R (FDD)*
NR FDD	Frequency selective scheduling	**FREQ_SEL_SCH_SW** option of the **NRDUCellDlSch**.***DlSchAlgoSwitch*** parameter	*Smart Coverage (FDD)* *Smart 8T8R (FDD)*
NR FDD	UE-specific PDCCH symbol number fast adaptation	**UE_PDCCH_SYM_NUM_FAST_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter	*Smart Coverage (FDD)* *Smart 8T8R (FDD)* *Smart Massive MIMO (FDD)*
NR FDD	Intelligent Precise MU Pairing[a]	**INTEL_PRCS_DL_MU_MIMO_PAIR_SW** option of the **NRDUCellFeatureSw**.***MimoFeatureSwitch*** parameter	*Smart 8T8R (FDD)*

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
a: If the **INTEL_PRCS_DL_MU_MIMO_PAIR_SW** option of the **NRDUCellFeatureSw**.***MimoFeatureSwitch*** parameter is selected and the **EASY_OMNI_SITE_MODE_SW** option of the **NRDUFddCellTrpBeam**.***ScenarioModeSw*** parameter is also selected for a cell, the **HDSS_DL_SCH_BW_ADAPT_SW** option of the **NRDUCellDlSch**.***DlSchAlgoEnhSwitch*** parameter must be deselected for the cell.			

– NSA networking functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Uplink single-side transmission[a]	**SINGLE_UL_NSA_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Basics*
NR FDD	NSA DC enhanced uplink power control[a]	**NSA_DC_ENH_UL_POWER_CONTROL_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter	None
a: This mutually exclusive relationship exists only when the cell TX/RX mode is 16T16R or 32T32R.			

– UCN functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch.*CompSwitch*** parameter	*CoMP*
NR FDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE**	*Cell Combination*

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Joint Transmission Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **JOINT_MODE**	*Cell Combination*
NR FDD	Inter-TRP handover in CA SCells[a]	**SCELL_INTER_TRP_HO_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*Cell Combination*
NR FDD	Inter-TRP SSB staggering[a]	**INTER_TRP_SSB_STAGGER_SWITCH** option of the **NRDUCellMultiTrp**.***MultiTrpAlgoSwitch*** parameter	*Cell Combination*
NR FDD	Inter-TRP CSI-RS staggering based on TRP ID mod 2 values[a]	**NRDUCellCsirs**.***InterTrpCsirsStaggerMode*** set to **MOD2_STAGGER**	*Cell Combination*
a: This mutually exclusive relationship exists only when **NRDUCell**.***NrDuCellNetworkingMode*** is set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** is set to **DPS_MODE**.			

– Coverage improvement functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	SSB power aggregation	**SSB_POWER_AGG_OPT_SW** option of the **NRDUCellChnPwr**.***ChnPwrAlgoSwitch*** parameter	*Turbo Coverage (FDD)*
NR FDD	VMIMO in DAS-based indoor coverage	**NRDUCellAlgoSwitch**.***VmimoSwitch***	*Indoor Coverage*

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
Low-frequency NR TDD	UL and DL Decoupling[a]	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	*UL and DL Decoupling*
a: When UL and DL Decoupling is enabled in SUL and NR FDD co-deployment mode, Hybrid DSS Based on Asymmetric Bandwidth cannot be enabled.			

– Interference mitigation functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	General precise interference isolation	**UNIV_PRECISE_INTRF_ISOLATE_SW** option of the **NRDUCellIntrfIdent**.***ChnMeasCtrlSw*** parameter	*Interference Avoidance*
NR FDD	Intra-frequency paging interference avoidance	**INTRA_FREQ_PAGING_AVOID_SW** option of the **NRDUCellPagingConfig**.***PagingSchAlgoSwitch*** parameter	*Interference Avoidance*
NR FDD	Channel power reduction for PIM interference avoidance	**PIM_INTERF_CHN_PWR_REDUCE_SW** option of the **NRDUCellPimIntrfAvd**.***PimAlgoSwitch*** parameter	*Interference Avoidance*
NR FDD	Channel isolation for PIM interference avoidance	**PIM_INTERF_CHN_SHUTDOWN_SW** option of the **NRDUCellPimIntrfAvd**.***PimAlgoSwitch*** parameter	*Interference Avoidance*
NR FDD	Intelligent PIM interference avoidance	**PIM_INTERF_SMART_AVOID_SW** option of the **NRDUCellPimIntrfAvd**.***PimAlgoSwitch*** parameter	*Interference Avoidance*

– Energy saving functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Energy saving based on flexible frequency-domain scheduling[a]	**FLEX_FREQ_SCH_ENERGY_SAVING_SW** option of the **NRDUCellAlgoSwitch.*PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*
NR FDD	Common-channel dynamic shutdown in TTI-level channel shutdown	**NRDUCellPowerSaving**.***ComChnDynShutdownSwitch*** parameter and the **TTI_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.*PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*
NR FDD	Intelligent power control energy saving	**SMART_PWR_CTRL_ENERGY_SAVE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*
NR FDD	Dynamic RF channel shutdown[b]	**RF_CHN_DYN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter	*Energy Conservation and Emission Reduction*
NR FDD	LNR coordinated channel shutdown[c]	**MULTI_RAT_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Multi-RAT Coordinated Energy Saving*
NR FDD	SSSG switching	**UE_PWR_SAVING_ENHANCE_SW** and **SSSG_SWITCHING_SW** options of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*UE Power Saving*

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
a: This mutually exclusive relationship exists when the PSD adaptation function is enabled. b: Among the sub-functions of dynamic RF channel shutdown, only data-channel dynamic shutdown can be enabled (by setting **NRDUCellPowerSaving**.***ComChnDynShutdownSwitch*** to **OFF**) together with Hybrid DSS Based on Asymmetric Bandwidth, but only in 4T4R or 8T8R scenarios. c: When Hybrid DSS Based on Asymmetric Bandwidth is enabled, LNR coordinated channel shutdown can take effect in 2T and 4T scenarios where the versions of NR and LTE are V100R020C10 or later, but not in 8T or massive MIMO scenarios.			

– Basic functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Enhanced paging capacity mode 1	**NRDUCellPagingConfig**.***PagingCapacityMode*** set to **ENHANCE_MODE_1**	*5G Networking and Signaling*
NR FDD	PUCCH RB adaptation	**PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter	*Channel Management*
NR FDD	PDCCH symbol number adaptation	**UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter	*Channel Management*
NR FDD	Ultimate PUCCH compression	**PUCCH_COMPRESSION_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter	*Channel Management*
NR FDD	Basic functions of downlink scheduling[a]	None	*Downlink Scheduling*

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Rate-matching-pattern-configuration-free PDCCH rate matching[b]	**PDCCH_NO_PATTERN_RATE_MATCH_SW** option of the **NRDUCellPdcch.*PdcchAlgoSwitch*** parameter	*Downlink Scheduling*
NR FDD	SSB Period[c]	**NRDUCell.*SsbPeriod***	*Channel Management*
a: Hybrid DSS Based on Asymmetric Bandwidth is mutually exclusive with the **NRDUCellPdsch.*DlAdditionalDmrsPos*** parameter (set to **POS2**) in the basic functions of downlink scheduling. b: This mutually exclusive relationship exists when Hybrid DSS Based on Asymmetric Bandwidth is enabled or when rate matching is enabled on all RBs in the symbols where LTE CRS appears (by selecting the **LTE_CRS_RATEMATCH_ALL_SYM_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** parameter). c: When the "LTE and NR 3-ms frame offset difference" function (controlled by the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of the **NRDUCellSpctShare**.***LnrSpectrumShareSwitch*** parameter) is enabled, the **NRDUCell**.***SsbPeriod*** parameter cannot be set to **MS5**.			

– Spectrum management functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Flexible bandwidth	**FLEXIBLE_BANDWIDTH_SW** option of the **NRDUCellFlexBw**.***FlexBwAlgoSwitch*** parameter	*Flexible Bandwidth (FDD)*
NR FDD	UE bandwidth adaptation[a]	**UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter	*Scalable Bandwidth*
a: If the **NRDUCellBwp**.***UlBwp0ConfigPolicy*** parameter is set to **FULL_BANDWIDTH**, the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter must be deselected.			

– Functions related to RAN services

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Multiple BWP1s for RedCap[a]	**NRDUCellRedCap**.***RedCapUeMaxAvailBand-width***	*RedCap*
NR FDD	RedCap SSSG switching	**REDCAP_SSSG_SWITCHING_SW** option of the **NRDUCellUePwrSav-ing**.***UePowerSavingSwitch*** parameter **REDCAP_SSSG_SWITCH_EXT_SW** option of the **NRDUCellUePwrSav-ing**.***UePowerSavingSwitch*** parameter	*RedCap*
NR FDD	Adaptive TBoMS	**TBOMS_SW** option of the **NRDUCellRedCap**.***RedCapCoverageAlgoSw*** parameter	*RedCap*
NR FDD	PEI[b]	**PEI_SW** option of the **NRDUCellUePwrSav-ing**.***IdleUePwrSavingS-witch*** parameter	*RedCap*
a: When Hybrid DSS Based on Asymmetric Bandwidth is enabled in an NR FDD cell, the **NRDUCellRedCap**.***RedCapUeMaxAvailBandwidth*** parameter must be set to **20M**. b: The PEI function cannot be enabled if Hybrid DSS Based on Asymmetric Bandwidth is enabled and the cell bandwidth is less than or equal to 10 MHz.			

– Other functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	Customized backup power saving policy	**NRDUCellPowerSav-ing**.***BackupPwrSaving-Policy*** set to **CUSTOMIZED_BACKUP_POLICY**	*Green Site*
NR FDD	Inter-base-station downlink joint transmission	**INTER_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	None

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	SSB beam adjustment[a]	**SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** parameter	None
NR FDD	Dynamic voltage adjustment	**DYN_VOLT_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	None
NR TDD	LTE TDD and NR spectrum sharing	**LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudSwitch*** parameter	None
NR FDD	DSS inter-RAT PDCCH spatial multiplexing[b]	**DSS_INTER_RAT_PDCCH_SPC_MUX_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch*** parameter	*LTE FDD and NR Dynamic Spectrum Sharing*
NR FDD	Intrf-free Band Spct Eff Weight Factor[c]	**NRDUCellIntrfIdent**.***NoIntrfBandSpctEffWtFactor*** set to **1000**	None
NR FDD	R16 type 2[d]	**R16_TYPE2_SW** option of the **NRDUCellPdschPrecode.*DlCodebookTypeSwitch*** parameter	None

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
a: This mutually exclusive relationship exists when the SSB spectrum exceeds the range of NR-dedicated spectrum in Hybrid DSS Based on Asymmetric Bandwidth scenarios. b: This mutually exclusive relationship exists only when **NRDUCell**.***NrDuCellNetworkingMode*** is set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** is set to **DPS_MODE**. c: This function does not work with downlink scheduling bandwidth adaptation in HDSS scenarios (controlled by the **HDSS_DL_SCH_BW_ADAPT_SW** option of the **NRDUCellDlSch**.***DlSchAlgoEnhSwitch*** parameter). d: The **R16_TYPE2_SW** option of the **NRDUCellPdschPrecode.*DlCodebookTypeSwitch*** parameter must be deselected for a cell if the **LTE_NR_FDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudSwitch*** parameter is selected and the TX/RX mode of the cell is not 4T4R, 8T8R, 16T16R, or 32T32R.			

4.3.2.3 Function Impacts

- Function impacts on the LTE side
  - CIoT functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
NB-IoT	LTE guard band deployment	**Prb**.***DeployMode***	*NB-IoT Basics (FDD)*	• For 3900 and 5900 series base stations, when Hybrid DSS Based on Asymmetric Bandwidth is used together with NB-IoT deployed in LTE guard band mode, uplink and downlink RBs are reserved to reduce the interference between NB-IoT and NR. For details, see **4.1.1.3.1 LTE Functions**. The RB reservation decreases the spectrum resources available for the NR cell, reducing the NR cell throughput. • For DBS3900 LampSite and DBS5900 LampSite, Hybrid DSS Based on Asymmetric Bandwidth cannot be used together with NB-IoT deployed in LTE guard band mode.

– Inter-RAT coordination functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	LTE spectrum coordination	**SpectrumCoordinationSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter	*LTE Spectrum Coordination*	The number of uplink RBs available for LTE decreases, which reduces the proportion of UEs for which LTE spectrum coordination takes effect.
LTE FDD	LTE FDD and NR Flash Dynamic Power Sharing	**LTE_NR_DYN_POWER_SHARING_SW** option of the **CellDynPowerSharing.*DynamicPowerSharingSwitch*** parameter	*LTE and NR Power Allocation*	When LTE FDD and NR Flash Dynamic Power Sharing is enabled, the power sharing capability of the LTE spectrum sharing cell is lower than that of common cells. As a result, the power sharing benefits decrease.
LTE FDD	LNR CRS muting	**LNR_CRS_MUTE_SW** option of the **CellAlgoExtSwitch**.***LnrZeroBufferzoneSw*** parameter	*LTE and NR Zero Bufferzone*	In Hybrid DSS Based on Asymmetric Bandwidth scenarios, LNR CRS muting takes effect in a lower proportion of cases and consequently offers fewer benefits than in NR-only scenarios.

– CoMP functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	UL CoMP	**CellAlgoSwitch.*UplinkCompSwitch***	*UL CoMP*	LTE UEs perform measurement on the entire bandwidth. When the LTE CRS transmission avoids the NR SSB, LTE RSRP measurement results become inaccurate. Consequently, the number of cell edge users (CEUs) in the LTE cell decreases, that is, the number of UEs for which UL CoMP takes effect decreases.
LTE FDD	DL CoMP with TM9	**Tm9JtSwitch** option of the **CellAlgoSwitch.*DlCompSwitch*** parameter	*DL CoMP (FDD)*	DL CoMP with TM9 does not take effect when it is used together with Hybrid DSS Based on Asymmetric Bandwidth, as its triggering conditions cannot be met.

– MIMO functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Uplink MU-MIMO	**UlVmimoSwitch** option of the **CellAlgoSwitch.*UlSchSwitch*** parameter	*MIMO*	The number of uplink RBs available for LTE decreases, which increases the pairing success rate of uplink MU-MIMO. Consequently, the BLER increases.
LTE FDD	Downlink MU-MIMO	**EmimoMuMimoSwitch** and **4TxTM9MuMimoSwitch** options of the **CellAlgoSwitch**.***EmimoSwitch*** parameter	*eMIMO (FDD)*	The number of downlink RBs available for LTE decreases, which increases the pairing success rate of downlink MU-MIMO. Consequently, the BLER increases.
LTE FDD	Inter-carrier power sharing	**INTER_CARRIER_PWR_SHR_SWITCH** option of the **SectorSplitGroup.*IntelligentBeamSchSwitch*** parameter	*Massive MIMO Enhancements (FDD)*	When Hybrid DSS Based on Asymmetric Bandwidth is enabled, the power that can be shared between carriers decreases.
LTE FDD	Dynamic power sharing for TM4 UEs	**DYNAMIC_TM4_PWR_SHR_SW** option of the **SectorSplitGroup.*IntelligentBeamSchSwitch*** parameter	*Massive MIMO Enhancements (FDD)*	After Hybrid DSS Based on Asymmetric Bandwidth is enabled, dynamic power sharing for TM4 UEs cannot take effect.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	8T UE pairing enhancement	**8T_TM9_ENH_MU_SW** option of the **CellMimoParaCfg.*MimoSwitch*** parameter	*Smart 8T8R (FDD)*	When Hybrid DSS Based on Asymmetric Bandwidth is enabled, the number of downlink RBs available to LTE decreases, and therefore the pairing success rate of downlink MU-MIMO increases if 8T UE pairing enhancement is enabled. As a result, interference among UEs increases and the downlink cell IBLER increases.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Cell Beam Azimuth Offset	**eUCellSectorEqm**.***CellBeamAzimuthOffset*** set to a value other than **0**	*Smart 8T8R (FDD)*	In Hybrid DSS Based on Asymmetric Bandwidth scenarios, if the azimuth offset of beams in an LTE cell is adjusted (by setting the **eUCellSectorEqm**.***CellBeamAzimuthOffset*** parameter to a value other than **0**), LTE CRS full-symbol rate matching can be used to mitigate the impact of the change in the LTE cell coverage on the NR cell as a result of the azimuth offset adjustment. (LTE CRS full-symbol rate matching is controlled by the **LTE_CRS_RATEMATCH_ALL_SYM_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch*** parameter).

– Interference mitigation functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	eNodeB-based interference randomization	**CellAlgoSwitch.*InterfRandSwitch***	*ICIC*	A decrease in the number of RBs available for LTE reduces the benefits provided by eNodeB-based interference randomization.
LTE FDD	Uplink interference randomization	**CellUlschAlgo.*UlRbAllocationStrategy*** set to **FS_INRANDOM_ADAPTIVE**	*ICIC*	After Hybrid DSS Based on Asymmetric Bandwidth is enabled, the number of RBs available for LTE decreases. As a result, the benefits provided by uplink interference randomization decrease.

– Basic functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Accurate uplink RB scheduling	**UL_RB_PRECISE_SCH_SW** option of the **CellAlgoSwitch.*UlSchExtSwitch*** parameter	*Uplink Scheduling*	Accurate uplink RB scheduling may produce fewer benefits when it is enabled together with Hybrid DSS Based on Asymmetric Bandwidth.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	UL Multi-Cluster	**UlMultiClusterSwitch** option of the **CellAlgoSwitch.*UlSchExtSwitch*** parameter	*Uplink Scheduling*	The number of uplink RBs available for LTE decreases, which reduces the benefit provided by UL Multi-Cluster.
LTE FDD	Uplink frequency selective scheduling	None	*Uplink Scheduling*	The number of uplink RBs available for LTE decreases, which reduces the benefit provided by uplink frequency selective scheduling.
LTE FDD	Uplink short-interval SPS	**CellUlschAlgo.*IntvlOfUlSpsWithSkipping***	*Uplink Scheduling*	The number of uplink RBs available for LTE decreases, which reduces the benefit provided by uplink short-interval SPS to UEs.
LTE FDD	SRS Configuration Indicator	**SRSCfg.*SrsCfgInd***	*Physical Channel Resource Management*	As LTE must share SRS resources with NR, the LTE cell triggers the expansion of the cell-specific SRS subframes in advance, which prolongs the SRS period of LTE UEs.

– Energy saving functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	DRX	**CellDrxPara.*DrxAlgSwitch***	*DRX and Signaling Control*	The number of RBs available for LTE decreases, leading to a decrease in the average number of scheduling times. As a result, the average number of UEs that enter DRX mode, as well as the frequency of UEs entering and exiting DRX mode increase.
LTE FDD	Dynamic DRX	**CellAlgoSwitch.*DynDrxSwitch***	*DRX and Signaling Control*	The number of RBs available for LTE decreases, leading to a decrease in the average number of scheduling times. As a result, the average number of UEs that enter DRX mode, as well as the frequency of UEs entering and exiting DRX mode increase.
LTE FDD	Dynamic voltage adjustment	**CellAlgoSwitch.*DynAdjVoltSwitch***	*Energy Conservation and Emission Reduction*	Dynamic voltage adjustment does not take effect when enabled together with Hybrid DSS Based on Asymmetric Bandwidth, as its triggering conditions cannot be met.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	RF channel intelligent shutdown	**CellRfShutdown**.***RfShutdownSwitch***	*Energy Conservation and Emission Reduction*	RF channel intelligent shutdown does not take effect when enabled together with Hybrid DSS Based on Asymmetric Bandwidth.
LTE FDD	8T beam mode shift for energy saving	**BEAM_MODE_SHIFT_SW** option of the **SectorSplitGroup**.***SplitCellGroupOptSwitch*** parameter	*Energy Conservation and Emission Reduction*	After the switching to one beam through 8T beam mode shift for energy saving, the LTE side will experience a higher PRB usage and start to preempt NR resources. This may affect NR user experience.
LTE FDD	Port sparseness	**PORT_SPARSENESS_SW** option of the **CellPwrSavingAlgo**.***PwrSavingAlgoSwitch*** parameter	*Energy Conservation and Emission Reduction*	Port sparseness may provide fewer benefits when it is in effect together with Hybrid DSS Based on Asymmetric Bandwidth.

– Functions related to RAN services

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Out-of-band relay	**OutOfBandRelaySwitch** option of the **CellAlgoSwitch.*RelaySwitch*** parameter	*Relay*	When Hybrid DSS Based on Asymmetric Bandwidth and out-of-band relay are both enabled, the following impacts will occur: ● NR occupies some scheduling resources when LTE and NR dynamically share resources. This means that fewer resources are available for LTE in Hybrid DSS Based on Asymmetric Bandwidth scenarios than in LTE-only scenarios. As a result, a donor eNodeB (DeNB) has fewer resources, which further affects the performance of the corresponding relay BTSs (ReBTSs). The following methods can be used to mitigate the impacts:

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
				– Increasing the percentage of resources preferentially allocated to LTE (using the **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio*** parameter) to ensure the performance of relay UEs to a certain extent – Adjusting the uplink and downlink relay subframe ratios as well as the percentage of resources preferentially allocated to LTE to ensure available resources for relay services on the LTE side ● In Hybrid DSS Based on Asymmetric Bandwidth scenarios,

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
				common LTE UEs and common NR UEs have fewer available resources and their performance is affected by LTE relay UEs with a higher priority. The impact severity depends on the uplink and downlink relay subframe ratios as well as the percentage of resources preferentially allocated to LTE. ● In extreme scenarios (for example, signaling storms and concurrent access of excessive UEs on the NR side) with a large proportion of system information, control information, or signaling, fewer resources are available for relay UEs. As a

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
				result, the performance of relay UEs and delay-related indicators deteriorate. ● When services with QoS guarantee requirements are running on the NR side, there may be no available NR resources in LTE relay subframes, affecting NR performance. To ensure NR performance, the uplink and downlink relay subframe ratios can be reduced. ● Hybrid DSS Based on Asymmetric Bandwidth increases the number of messages exchanged between the LTE and NR sides. When the number of cells or UEs served by a board is too large, LTE and NR cells will be congested. In

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
				this case, resources are allocated to LTE and NR based on static configurations, which has a great impact on the performance of DeNBs and ReBTSs. The access of relay UEs and common LTE UEs can be guaranteed, but their user experience cannot be guaranteed.
LTE FDD	Uplink semi-persistent scheduling	**SpsSchSwitch** option of the **CellAlgoSwitch.*UlSchSwitch*** parameter	*VoLTE*	The number of uplink RBs available for LTE decreases, which reduces the benefit provided by uplink semi-persistent scheduling.
LTE FDD	Downlink semi-persistent scheduling	**SpsSchSwitch** option of the **CellAlgoSwitch.*DlSchSwitch*** parameter	*VoLTE*	The number of downlink RBs available for LTE decreases, which reduces the benefit provided by downlink semi-persistent scheduling.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	TTI bundling	**TtiBundlingSwitch** option of the **CellAlgoSwitch.*UlSchSwitch*** parameter	*VoLTE*	The number of uplink RBs available for LTE decreases. Consequently, the maximum number of UEs that can enter the TTI bundling state decreases.
LTE FDD	LCS	**ENodeBAlgoSwitch.*LcsSwitch***	*LCS*	The PRS sent for LCS causes interference to NR, increasing the BER on the NR side. As a result, the performance of the NR cell deteriorates. Therefore, it is not recommended that LCS be enabled together with Hybrid DSS Based on Asymmetric Bandwidth.

– Mobility management functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Intra-RAT mobility load balancing	**CellAlgoSwitch.*MlbAlgoSwitch***	*Intra-RAT Mobility Load Balancing*	If the PRB usage of LTE changes or the number of RBs available for LTE changes, the intra-RAT MLB policy changes.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	High speed mobility	**Cell.*HighSpeedFlag***	*High Speed Mobility*	SFN is required for downlink AFC of high speed mobility to deliver optimal benefits in Hybrid DSS Based on Asymmetric Bandwidth scenarios.
LTE FDD	Ultra high speed mobility	**Cell.*HighSpeedFlag***	*High Speed Mobility*	SFN is required for downlink AFC of ultra high speed mobility to deliver optimal benefits in Hybrid DSS Based on Asymmetric Bandwidth scenarios.

– Carrier management functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Uplink CA	**CaUl2CCSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter	*Carrier Aggregation*	The number of uplink RBs available for LTE decreases, which reduces the throughput of UEs for which FDD uplink CA takes effect.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Downlink CA	None	*Carrier Aggregation*	The number of downlink RBs available for LTE decreases, which reduces the throughput of UEs for which FDD downlink CA takes effect.
LTE FDD	Downlink 3CC aggregation	**CaDl3CCSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter	*Carrier Aggregation*	When downlink 3CC aggregation is used together with Hybrid DSS Based on Asymmetric Bandwidth, and the percentage of resources preferentially allocated to LTE is set to **0**, LTE UEs cannot access the network. It is recommended that the PUCCH configuration be optimized.
LTE FDD	Flexible CA	**MultiCarrierFlexCaSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter	*Carrier Aggregation*	The number of uplink and downlink RBs available for LTE decreases, and the selected serving cell combination may change as a result.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Downlink massive CA	**DlMassiveCaSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter	*Carrier Aggregation*	LTE cells with Hybrid DSS Based on Asymmetric Bandwidth enabled are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured, which decreases the LTE network throughput.

– Other functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	RAN sharing with common carrier	**ENodeBSharingMode.*EnodeBSharingMode*** set to **SHARED_FREQ** or **HYBRID_SHARED**	*RAN Sharing*	When RAN sharing with common carrier is enabled together with Hybrid DSS Based on Asymmetric Bandwidth, the number of available RBs for an LTE operator is calculated as follows: (Total number of RBs – Number of RBs shared with NR) x RB allocation proportion configured for the operator.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Compact bandwidth	**Cell.*CustomizedBandWidthCfgInd***	*Compact Bandwidth (FDD)*	When the **Cell**.***CustomizedBandWidthCfgInd*** parameter is set to **CFG** and the **Cell.*TxRxMode*** parameter is set to **16T16R** or **32T32R**, Hybrid DSS Based on Asymmetric Bandwidth cannot work with compact bandwidth.
LTE FDD	Flexible TM4 MU-MIMO	**FLEX_TM4_MU_MIMO_SW** option of the **CellMimoParaCfg.*MimoSwitch*** parameter	*Low-Band Booster*	A decrease in the number of downlink RBs available for LTE will increase the pairing success rate of downlink MU-MIMO but also increase the BLER.
LTE FDD	Uplink single-side in-band interference impact mitigation	**UlInterfSuppressCfg.*UlSingleSideIntrfImpMitSw***	None	A decrease in the number of RBs available for LTE reduces the benefits provided by uplink single-side in-band interference impact mitigation.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Physical Resource Slicing	**PHY_RES_SLICING_SWITCH** option of the **CellAlgoExtSwitch.*OperatorFunSwitch*** parameter	*Physical Resource Slicing (FDD)*	When Hybrid DSS Based on Asymmetric Bandwidth is enabled together with Physical Resource Slicing, the number of available PRBs in each LTE PRB group varies as follows: ● If resources are reserved based only on SPIDs, the number of available PRBs in an LTE PRB group equals (Total spectrum resources – Number of PRBs shared with the non-LTE side) x Proportion of reserved uplink or downlink PRBs for the PRB group. ● If resources are reserved based on both operators and SPIDs, the number of available PRBs in an LTE PRB group equals (Total spectrum resources – Number of PRBs shared

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
				with the non-LTE side) x Proportion of reserved uplink or downlink PRBs for the corresponding operator x Proportion of reserved uplink or downlink PRBs for the PRB group.
LTE FDD	Downlink maximum cell transmit power increase	**EDlMaxTXPwrSwitch** option of the **CellAlgoSwitch**.***DlPcAlgoSwitch*** parameter	None	In Hybrid DSS Based on Asymmetric Bandwidth scenarios, if the function of downlink maximum cell transmit power increase is enabled, the PSDs of LTE and NR will exceed the required values. As a result, cells may fail to be activated.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Load simulation	**CellSimuLoad**.***SimuLoadMode***	None	Only global and subframe-level load simulation can be used in Hybrid DSS Based on Asymmetric Bandwidth scenarios. Global load simulation is enabled when the **CellSimuLoad**.***SimuLoadMode*** parameter is set to **GLOBAL_SIMU_LOAD**. Subframe-level load simulation is enabled when the **CellSimuLoad**.***SimuLoadMode*** parameter is set to **SUBFRAME_SIMU_LOAD**. If different load simulation modes are used in multiple LTE split cells that share spectrum with an NR cell, load simulation will not achieve expected results.

- Function impacts on the NR side
  - Inter-RAT coordination functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	NR inter-carrier dynamic power sharing	**NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch.*DynPowerSharingSwitch*** parameter	*NR Inter-Carrier Dynamic Power Allocation*	When Hybrid DSS Based on Asymmetric Bandwidth is enabled, the power that can be shared between NR carriers decreases.
NR FDD	LTE FDD and NR Flash Dynamic Power Sharing	**LTE_NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***DynPowerSharingSwitch*** parameter	*LTE and NR Power Allocation*	When LTE FDD and NR Flash Dynamic Power Sharing is enabled, the power sharing capability of the NR spectrum sharing cell is lower than that of common cells. As a result, the power sharing benefits decrease.

– Energy saving functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Power saving scheduling	**POWER_SAVING_SCHEDULE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	When both Hybrid DSS Based on Asymmetric Bandwidth and power saving scheduling take effect, the gains of power saving scheduling decrease, and the LTE capacity may decrease. If the **gNBPdcpParamGroup**.***DlDataPdcpSplitMode*** parameter is set to **SCG_AND_MCG** in this scenario, the downlink throughput of NR cells decreases.
NR FDD	Energy saving based on flexible frequency-domain scheduling	**FLEX_FREQ_SCH_ENERGY_SAVING_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	When both Hybrid DSS Based on Asymmetric Bandwidth and energy saving based on flexible frequency-domain scheduling take effect, the gains of energy saving based on flexible frequency-domain scheduling decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	RF channel intelligent shutdown	**RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.*PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	RF channel intelligent shutdown does not take effect when enabled together with Hybrid DSS Based on Asymmetric Bandwidth, as its triggering conditions cannot be met.

– MIMO functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	3D coverage pattern	**NRDUFddCellTrpBeam.*CoverageScenario*** set to **SCENARIO_202**, **SCENARIO_205**, **SCENARIO_207**, **SCENARIO_208**, **SCENARIO_240**, **SCENARIO_241**, or **SCENARIO_242**	*Beam Management*	Four SSB beams take effect in Hybrid DSS Based on Asymmetric Bandwidth scenarios only when the **NRDUFddCellTrpBeam.*CoverageScenario*** parameter is set to **SCENARIO_202**, **SCENARIO_205**, **SCENARIO_208**, **SCENARIO_240**, **SCENARIO_241**, or **SCENARIO_242** and the **SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** parameter is selected. Otherwise, only two SSB beams take effect as follows: ● In 8T8R scenarios, the coverage scope is as follows: – When **NRDUFddCellTrpBeam.*CoverageScenario*** is set to **SCENARIO_202** or **SCENARIO_205**, the

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				coverage scope is the same as that in the scenario indicated by the value **SCENARIO_201**. – When **NRDUFddCellTrpBeam**.***CoverageScenario*** is set to **SCENARIO_208**, the coverage scope is the same as that in the scenario indicated by the value **SCENARIO_207**. ● In 16T16R and 32T32R scenarios, the coverage scope is the same as that in the scenario indicated by the value **SCENARIO_220**.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	PDCCH CCE ratio adjustment in common slots	**PDCCH_COMM_SLOT_RATIO_ADJ_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter	*Smart Coverage (FDD)* *Smart 8T8R (FDD)* *Smart Massive MIMO (FDD)*	In Hybrid DSS Based on Asymmetric Bandwidth scenarios, PDCCH CCE ratio adjustment in common slots can take effect only if the NR cell bandwidth is greater than or equal to 10 MHz.
NR FDD	Distributed MIMO	**INTRA_GNB_DL_D_MIMO_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Distributed Antenna Solutions (FDD)*	After Hybrid DSS Based on Asymmetric Bandwidth is enabled, robust weight for joint transmission in Intra-gNodeB Distributed MIMO does not take effect.

– Coverage improvement functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Super uplink with SUL and NR FDD co-deployment	**FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch.*SuperUplinkSwitch*** parameter	*Super Uplink*	When Hybrid DSS Based on Asymmetric Bandwidth is enabled together with super uplink with SUL and NR FDD co-deployment, the amount of uplink spectrum resources that can be allocated to implement super uplink decreases if the LTE FDD or NR traffic volume increases. As a result, the uplink throughput of the SUL carrier decreases. If uplink semi-persistent scheduling, eMTC, or compact bandwidth is also enabled in the LTE cell, super uplink with SUL and NR FDD co-deployment cannot take effect as these functions periodically or continuously occupy spectrum resources. When the NR TDD and NR FDD cells enabled with super uplink with SUL and NR FDD co-deployment are configured on the same

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				baseband processing unit, Hybrid DSS Based on Asymmetric Bandwidth cannot be enabled together with super uplink with SUL and NR FDD co-deployment.
NR FDD	RuralLink antenna solution (NR)	**EASY_OMNI_SITE_MODE_SW** option of the **NRDUFddCellTrpBeam**.***ScenarioModeSw*** parameter	*RuralLink Solution*	The RuralLink antenna solution can take effect together with Hybrid DSS Based on Asymmetric Bandwidth only if each of the following conditions is met: ● The NR cell bandwidth is 15 MHz or 20 MHz. ● The **SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch*** parameter is selected.

– Interference mitigation functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Interference avoidance for indoor cells	**INDOOR_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***IndoorOutdoorCooperateSw*** parameter	*3D Networking Experience Improvement*	After Hybrid DSS Based on Asymmetric Bandwidth is enabled, the gains brought by the function of interference avoidance for indoor cells decrease because the start position for PDSCH scheduling is no longer fixed and some of the positions for PDSCH scheduling in indoor and outdoor cells are already different on the NR side. If the positions for PDSCH scheduling are completely different between indoor and outdoor cells on the NR side after Hybrid DSS Based on Asymmetric Bandwidth is enabled, enabling the function of interference avoidance for indoor cells leads to occasional decreases in the throughputs of these cells but improves the downlink spectral

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				efficiency of the entire network.

– Basic functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Downlink power control	None	*Power Control*	After Hybrid DSS Based on Asymmetric Bandwidth is enabled, the downlink power cannot be increased.
NR FDD	DC component rate matching	**DC_COMPONENT_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	After Hybrid DSS Based on Asymmetric Bandwidth is enabled, configurations for DC component rate matching become invalid.
NR FDD	PDCCH rate matching	**PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	When both PDCCH rate matching and Hybrid DSS Based on Asymmetric Bandwidth are enabled, the MCS index decreases. As a result, the downlink UE throughput and cell capacity may decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Downlink adaptive selection of resource allocation type 0 or 1	**TYPE0_TYPE1_ADAPTIVE_SCH_SW** option of the **NRDUCellPdsch**.***DlPdschAlgoSwitch*** parameter	*Downlink Scheduling*	If Hybrid DSS Based on Asymmetric Bandwidth is enabled together with downlink adaptive selection of resource allocation type 0 or 1, the **LNR_RES_ALLOC_DIRECTION_OPT_SW** option of the **LteNrSpctShrCellGrp**.***LteNrSpctShrSwitch*** parameter must be selected. Before enabling downlink adaptive selection of resource allocation type 0 or 1 in Hybrid DSS Based on Asymmetric Bandwidth scenarios, you are advised to select the **DL_RB_ESTIMATE_OPT_SW** option of the **NRDUCellPdsch**.***DlPdschAlgoSwitch*** parameter to prevent UE data packets from being truncated.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Unbalanced transmit power configurations between channels	**NRDUCellCoverage.*TransmitPowerOffsetChn0***, **NRDUCellCoverage.*TransmitPowerOffsetChn1***, **NRDUCellCoverage.*TransmitPowerOffsetChn2***, or **NRDUCellCoverage.*TransmitPowerOffsetChn3*** set to a non-zero value	*Cell Management*	When the function of unbalanced transmit power configurations between channels is used together with Hybrid DSS Based on Asymmetric Bandwidth, and the spectrum power sharing mode is used, the number of downlink available RBs in the NR cell decreases.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	LTE and NR flexible frame offset	**LNR_FRAME_OFFSET_SW** option of the **RRUEXTINFO**.***RruOptlFeatSw*** parameter	*Cell Management*	Assume that the LTE and NR flexible frame offset function is enabled. Hybrid DSS Based on Asymmetric Bandwidth can take effect only when the "LTE and NR 3-ms frame offset difference" function is enabled. (The "LTE and NR 3-ms frame offset difference" function is controlled by the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of the NR parameter **NRDUCellSpctShare**.***LnrSpectrumShareSwitch*** and the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of the LTE parameter **SpectrumCloud**.***SpectrumCloudExtSwitch***).

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	SSB Time Position	**NRDUCell**.***SsbTimePos***	*Channel Management*	If Hybrid DSS Based on Asymmetric Bandwidth is enabled and the SSB frequency-domain position (specified by the **NRDUCell**.***SsbFreqPos*** parameter) of an NR cell is within the LTE cell bandwidth, the NR cell will have fewer resources available for the PDCCH, which decreases the CCE allocation success rate and the user-perceived throughput of the NR cell.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	SSB Time Position	**NRDUCell**.***SsbTimePos***	*Channel Management*	Assume that Hybrid DSS Based on Asymmetric Bandwidth is enabled, **NRDUCellCoreset**.***CommonCtrlResStartSymbol*** is set to **NULL**, **NRDUCellPdcch**.***OccupiedSymbolNum*** is set to **1SYM**, and the result of Rounddown[(Frequency corresponding to **NRDUCell**.***SsbFreqPos*** – Frequency corresponding to **NRDUCell**.***DlNarfcn*** + 6 x NRB x **NRDUCell**.***SubcarrierSpacing***)/(12 x **NRDUCell**.***SubcarrierSpacing***)] is less than the result of (**gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex*** of the group identified by **NRDUCellSpctCloud**.***NrSpctShrCellGrpId*** + 22) or greater than the result of (**gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex*** of the group identified by **NRDUCellSpctCloud**.***NrSpctShrCe***

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				llGrpId – 22). The setting of **NRDUCell**.***SsbTimePos*** must meet the following requirements: ● If the **SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch*** parameter is selected, the **NRDUCell**.***SsbTimePos*** parameter must be set to **DEFAULT**. ● If the **SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch*** parameter is deselected, the **NRDUCell**.***SsbTimePos*** parameter must be set to **DEFAULT** or **SSB23_PCI_MOD2_STAGGER**. NRB indicates the number of RBs corresponding to the cell bandwidth specified in Table

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				5.3.2-1 in 3GPP TS 38.104.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	SSB Time Position	**NRDUCell**.***SsbTimePos***	*Channel Management*	Assume that Hybrid DSS Based on Asymmetric Bandwidth is enabled, **NRDUCellCoreset**.***CommonCtrlResStartSymbol*** is set to **NULL**, **NRDUCellPdcch**.***OccupiedSymbolNum*** is set to **2SYM**, and the result of Rounddown[(Frequency corresponding to **NRDUCell**.***SsbFreqPos*** – Frequency corresponding to **NRDUCell**.***DlNarfcn*** + 6 x NRB x **NRDUCell**.***SubcarrierSpacing***)/(12 x **NRDUCell**.***SubcarrierSpacing***)] is less than the result of (**gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex*** of the group identified by **NRDUCellSpctCloud**.***NrSpctShrCellGrpId*** + 10) or greater than the result of (**gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex*** of the group identified by **NRDUCellSpctCloud**.***NrSpctShrCe***

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				llGrpId – 10). The setting of **NRDUCell**.***SsbTimePos*** must meet the following requirements: ● If the **SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch*** parameter is selected, the **NRDUCell**.***SsbTimePos*** parameter must be set to **DEFAULT**. ● If the **SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch*** parameter is deselected, the **NRDUCell**.***SsbTimePos*** parameter must be set to **DEFAULT** or **SSB23_PCI_MOD2_STAGGER**. NRB indicates the number of RBs corresponding to the cell bandwidth specified in Table

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				5.3.2-1 in 3GPP TS 38.104.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	SSB Time Position	**NRDUCell**.***SsbTimePos***	*Channel Management*	Assume that Hybrid DSS Based on Asymmetric Bandwidth is enabled, the result of Rounddown[(Frequency corresponding to **NRDUCell**.***SsbFreqPos*** – Frequency corresponding to **NRDUCell**.***DlNarfcn*** + 6 x NRB x **NRDUCell**.***SubcarrierSpacing***)/(12 x **NRDUCell**.***SubcarrierSpacing***)] is greater than or equal to the result of (**gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex*** of the group identified by **NRDUCellSpctCloud**.***NrSpctShrCellGrpId*** + 22) and less than or equal to the result of (**gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex*** of the group identified by **NRDUCellSpctCloud**.***NrSpctShrCellGrpId*** – 22), and any of the following conditions is met: ● The **NRDUCellCoreset**.***CommonC***

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				trlResStartSymbol parameter is set to **NULL** and the **NRDUCellPdcch**.***OccupiedSymbolNum*** parameter is set to **1SYM**. ● The **NRDUCellCoreset**.***CommonCtrlResStartSymbol*** parameter is set to a value other than **NULL** and the **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to **RB48**. If the **SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch*** parameter is deselected, the **NRDUCell**.***SsbTimePos*** parameter must be set to **DEFAULT**. NRB indicates the number of RBs corresponding to the cell bandwidth specified in Table 5.3.2-1 in 3GPP TS 38.104.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	SSB Time Position	**NRDUCell**.***SsbTimePos***	*Channel Management*	Assume that Hybrid DSS Based on Asymmetric Bandwidth is enabled, the result of Rounddown[(Frequency corresponding to **NRDUCell**.***SsbFreqPos*** – Frequency corresponding to **NRDUCell**.***DlNarfcn*** + 6 x NRB x **NRDUCell**.***SubcarrierSpacing***)/(12 x **NRDUCell**.***SubcarrierSpacing***)] is greater than or equal to the result of (**gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex*** of the group identified by **NRDUCellSpctCloud**.***NrSpctShrCellGrpId*** + 10) and less than or equal to the result of (**gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex*** of the group identified by **NRDUCellSpctCloud**.***NrSpctShrCellGrpId*** – 10), and any of the following conditions is met: ● The **NRDUCellCoreset**.***CommonC***

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				trlResStartSymbol parameter is set to **NULL** and the **NRDUCellPdcch**.***OccupiedSymbolNum*** parameter is set to **2SYM**. ● The **NRDUCellCoreset**.***CommonCtrlResStartSymbol*** parameter is set to a value other than **NULL** and the **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to **RB24**. If the **SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch*** parameter is deselected, the **NRDUCell**.***SsbTimePos*** parameter must be set to **DEFAULT**. NRB indicates the number of RBs corresponding to the cell bandwidth specified in Table 5.3.2-1 in 3GPP TS 38.104.

– Network slicing functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	When both the NR DU cell resource management between network slices function and Hybrid DSS Based on Asymmetric Bandwidth are enabled, the number of RBs available for a network slice group is calculated using the following formula: Number of RBs available for a network slice group = Available bandwidth of the NR cell (not the bandwidth configured for the NR cell) x Percentage of RBs configured for the network slice group. When both the NR DU cell resource management between network slices function and PSD adaptation are enabled, the downlink cell throughput may decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	User-rate-based dynamic network slice resource management	**DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch*** parameter	*Network Slicing*	When both "user-rate-based dynamic network slice resource management" and Hybrid DSS Based on Asymmetric Bandwidth take effect, dynamic resource adjustment is inaccurate, resulting in decreased user-perceived rates of UEs in a network slice group.

– Resource control functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	RFSP-group-based RB resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellCommonSw**.***RfspAlgoSwitch*** parameter	*Resource Control in NSA Networking*	When RFSP-group-based RB resource control is enabled together with Hybrid DSS Based on Asymmetric Bandwidth, the number of RBs available for an RFSP group equals the available bandwidth (not configured bandwidth) of the NR cell multiplied by the RB proportion configured for the RFSP group.
NR FDD	QCI-based resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*	When QCI-based resource control is enabled together with Hybrid DSS Based on Asymmetric Bandwidth, the number of RBs available for a QCI group equals the available bandwidth (not configured bandwidth) of the NR cell multiplied by the RB proportion configured for the QCI group.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	User-rate-based dynamic QCI resource management	**DYNAMIC_QCI_GROUP_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*	When both "user-rate-based dynamic QCI resource management" and Hybrid DSS Based on Asymmetric Bandwidth take effect, dynamic resource adjustment is inaccurate, resulting in decreased user-perceived rates of UEs in a QCI group.

– Functions related to RAN services

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Aggregation level adjustment for retransmissions upon DTXs	**DTX_RETRANS_AGG_LVL_ADJ_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter	*VoNR*	Aggregation level adjustment for retransmissions upon DTXs may not take effect when it is enabled together with Hybrid DSS Based on Asymmetric Bandwidth and the bandwidth of the NR-dedicated spectrum is less than 20 MHz. In this case, it is recommended that the **UE_PDCCH_FULL_BANDWIDTH_CFG_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter be selected.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Downlink CCE allocation without ratio constraint	**DL_CCE_ALLO_NO_CONSTRAIN_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter	*VoNR*	Downlink CCE allocation without ratio constraint may not take effect when it is enabled together with Hybrid DSS Based on Asymmetric Bandwidth and the bandwidth of the NR-dedicated spectrum is less than 20 MHz. In this case, it is recommended that the **UE_PDCCH_FULL_BANDWIDTH_CFG_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter be selected.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Uplink RB reservation	**UL_RB_RSV_SW** option of the **NRCellAlgoSwitch.*VonrSwitch*** parameter	*VoNR*	It is not recommended that uplink RB reservation be enabled together with Hybrid DSS Based on Asymmetric Bandwidth. This is because if the reserved RBs are configured in the shared spectrum and some of the shared spectrum is occupied by LTE, interference will be generated on voice service UEs that occupy the reserved RBs in neighboring NR cells.

– Network sharing functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	RAN sharing with common carrier	**SHARED_FREQ** option of the **gNBSharingMode.*gNBMultiOpSharingMode*** parameter	*Multi-Operator Sharing*	When RAN sharing with common carrier is enabled together with Hybrid DSS Based on Asymmetric Bandwidth, the number of available RBs for an NR operator is calculated as follows: (Total number of RBs – Number of RBs shared with LTE) x RB allocation proportion configured for the operator.
NR FDD	Percentage-based dynamic RB sharing among operators	**RB_DYNAMIC_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*	When both percentage-based dynamic RB sharing among operators and PSD adaptation are enabled, the downlink cell throughput may decrease.

– Other functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Single DCI	**DL_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*Multi-Frequency Smart Aggregation*	The single downlink control information (DCI) function does not take effect for a UE if Hybrid DSS Based on Asymmetric Bandwidth is enabled in the PCell of the UE.
NR FDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag***	*High Speed Mobility*	When both High-speed Railway Superior Experience and Hybrid DSS Based on Asymmetric Bandwidth are enabled for a cell, the performance of Hybrid DSS Based on Asymmetric Bandwidth deteriorates as UEs in this high-speed cell move fast. In addition, UEs in this cell do not support two downlink additional DMRS positions as Hybrid DSS Based on Asymmetric Bandwidth is enabled, affecting the PDSCH demodulation performance of the UEs. Therefore, it is recommended that these two functions not be used together.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Compact bandwidth	**NRDUCell.*CustomizedBwConfigInd*** set to **CONFIG**	*Scalable Bandwidth*	After Hybrid DSS Based on Asymmetric Bandwidth is enabled, compact bandwidth supports only the following bandwidths: ● 14.8 MHz ● 19.6 MHz ● 19.8 MHz ● 24.8 MHz When the **NRDUCell.*CustomizedBwConfigInd*** parameter is set to **CONFIG** and the **NRDUCellTrp.*TxRxMode*** parameter is set to **16T16R** or **32T32R**, Hybrid DSS Based on Asymmetric Bandwidth cannot work with compact bandwidth.
Low-frequency NR TDD	Flexible spectrum access	**FLEX_SPCT_ACCESS_SW** option of the **NRDUCellCarrMgmt**.***MultiBandFlexSchSwitch*** parameter	*Uplink Flexible Spectrum Scheduling*	Flexible spectrum access (FSA) does not take effect in 16T16R or 32T32R NR FDD cells with Hybrid DSS Based on Asymmetric Bandwidth enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Cell load simulation	**NRDUCellSimulateLoad**.***TestMode***	None	Only FDD and subframe-level load simulation can be used in Hybrid DSS Based on Asymmetric Bandwidth scenarios. FDD load simulation is enabled when the **NRDUCellSimulateLoad**.***TestMode*** parameter is set to **FDD_SIMULATED_LOAD_MODE**. Subframe-level load simulation is enabled when the **NRDUCellSimulateLoad**.***TestMode*** parameter is set to **SUBFRM_SIMULATED_LOAD_MODE**.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	In Hybrid DSS Based on Asymmetric Bandwidth scenarios, if intra-band or inter-band CA is enabled and the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter is deselected, then a certain condition must be met. The condition varies as follows: If the number of RBs on the NR-dedicated spectrum is less than or equal to 25, the condition is as follows: **NRDUCellPucch**.***Format1RbNum*** + **NRDUCellPucch**.***Format3RbNum*** + **NRDUCellPucch**.***Format4RbNum*** + **NRDUCellPucch**.***CsiDedicatedRbNum*** + **NRDUCellPucch**.***Format4CsiDedicatedRbNum*** ≤ 14 If the number of RBs on the NR-dedicated spectrum is greater than 25 and less than or equal to 50, the
NR FDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
				condition is as follows: **NRDUCellPucch**.***Format1RbNum*** + **NRDUCellPucch**.***Format3RbNum*** + **NRDUCellPucch**.***Format4RbNum*** + **NRDUCellPucch**.***CsiDedicatedRbNum*** + **NRDUCellPucch**.***Format4CsiDedicatedRbNum*** < Number of RBs on the NR-dedicated spectrum – 15

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Huawei devices (including the eNodeB and gNodeB) must be used on the radio access network.

Base Station Models

- LTE
  3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, BBU5900A, BookBBU5901a, or BookBBU5901.
- NR
  3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, BBU5900A, BookBBU5901a, or BookBBU5901.

Boards

- LTE
  When the cell TX/RX mode is 2T2R, 2T4R, or 4T4R, only a UBBPd/UBBPe/UBBPg/UBBPh supports this function. **Table 4-24** lists the cell number

specifications of different baseband processing units after this function is enabled. Among main control boards, only a UMPTb/UMPTe/UMPTg/UMPTga/UMPTha/UMPThb supports this function.

When the cell TX/RX mode is 8T8R, only a UBBPe4/UBBPe6/UBBPe18/UBBPg/UBBPh supports this function. When Hybrid DSS Based on Asymmetric Bandwidth (single LTE cell) is enabled, the cell number specifications of the baseband processing units are the same as those before this function is enabled. When Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) is enabled, the cell number specifications of different baseband processing units are listed in **Table 4-25**. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTha/UMPThb supports this function.

When the cell TX/RX mode is 16T16R, FDD baseband processing units compatible with 16T16R massive MIMO support this function. The number of cells supported by each baseband processing unit is the same as that before this function is enabled. For details, see *LTE Technical Specifications of Baseband Processing Units* in *3900 & 5900 Series Base Station Product Documentation*. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTha/UMPThb supports this function.

When the cell TX/RX mode is 32T32R, FDD massive-MIMO-capable baseband processing units (except the UBBPex2) support this function. The number of cells supported by each baseband processing unit is the same as that before this function is enabled. For details, see *LTE Technical Specifications of Baseband Processing Units* in *3900 & 5900 Series Base Station Product Documentation*. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTha/UMPThb supports this function.

- NR

  When the cell TX/RX mode is 2T2R, 2T4R, or 4T4R, only a UBBPg2/UBBPg2a/UBBPg3/UBBPg3b/UBBPh supports this function. The cell number specifications of these baseband processing units are the same as those before this function is enabled. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTha/UMPThb supports this function.

  When the cell TX/RX mode is 8T8R, only a UBBPg/UBBPh supports this function. The cell number specifications of the baseband processing unit are the same as those before this function is enabled. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTha/UMPThb supports this function.

  When the cell TX/RX mode is 16T16R, FDD baseband processing units compatible with 16T16R massive MIMO support this function. The number of cells supported by each baseband processing unit is the same as that before this function is enabled. For details, see *NR Technical Specifications of Baseband Processing Units* in *3900 & 5900 Series Base Station Product Documentation*. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTha/UMPThb supports this function.

  When the cell TX/RX mode is 32T32R, FDD massive-MIMO-capable baseband processing units support this function. The number of cells supported by each baseband processing unit is the same as that before this function is enabled. For details, see *NR Technical Specifications of Baseband Processing Units* in *3900 & 5900 Series Base Station Product Documentation*. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTha/UMPThb supports this function.

Table 4-24 Cell number specifications of LTE baseband processing units (single LTE cell)

Working Mode	Baseband Processing Unit Model	Supported Cell Number Specifications (Compared with Those Before This Function Is Enabled)	Supported Spectrum Sharing Cell Number Specifications
LTE FDD or LM[a]	UBBPd5	Decrease or remain unchanged The cell number specifications decrease from six (2T2R) cells to three cells, or remain three (2T4R or 4T4R) cells.	A maximum of three cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth.
LTE FDD or LM	UBBPd6	Decrease The cell number specifications decrease from six (2T2R, 2T4R, or 4T4R) cells to three cells.	A maximum of three cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth.
LTE FDD or LM	UBBPe1	Remain unchanged The cell number specifications remain unchanged and are three (2T2R) cells.	A maximum of three cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth.
LTE FDD or LM	UBBPe2	Remain unchanged The cell number specifications remain unchanged and are three (2T2R, 2T4R, or 4T4R) cells.	A maximum of three cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth.
LTE FDD or LM	UBBPe3	Remain unchanged The cell number specifications remain unchanged and are six (2T2R) cells or three (2T4R or 4T4R) cells.	A maximum of three cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth, while the rest are common cells.
LTE FDD or LM	UBBPe4	Remain unchanged The cell number specifications remain unchanged and are six (2T2R, 2T4R, or 4T4R) cells or three (8T8R) cells.	A maximum of three cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth.

Working Mode	Baseband Processing Unit Model	Supported Cell Number Specifications (Compared with Those Before This Function Is Enabled)	Supported Spectrum Sharing Cell Number Specifications
LTE FDD or LM	UBBPe18	Remain unchanged The cell number specifications remain unchanged and are six (2T2R, 2T4R, or 4T4R) cells or three (8T8R) cells.	A maximum of three cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth.
LTE FDD, LM, or other working modes involving LTE FDD	UBBPe5	Remain unchanged • LTE FDD or LM The cell number specifications remain unchanged and are nine (2T2R, 2T4R, or 4T4R) cells. • Working mode combinations other than LM: The cell number specifications remain unchanged and are six (2T2R, 2T4R, or 4T4R) cells.	• LTE FDD or LM: A maximum of six cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth, while the rest are common cells. • Working mode combinations other than LM: A maximum of three cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth, while the rest are common cells.

Working Mode	Baseband Processing Unit Model	Supported Cell Number Specifications (Compared with Those Before This Function Is Enabled)	Supported Spectrum Sharing Cell Number Specifications
LTE FDD, LM, or other working modes involving LTE FDD	UBBPe6	Remain unchanged ● LTE FDD: The cell number specifications remain unchanged and are 12 (2T2R, 2T4R, or 4T4R) cells. ● LM (with **BBP.*SRT*** set to **DEFAULT**): The cell number specifications remain unchanged and are 12 (2T2R, 2T4R, or 4T4R) cells. ● LM (with **BBP.*SRT*** set to **NBIOT_ENHANCE**): The cell number specifications remain unchanged and are nine (2T2R, 2T4R, or 4T4R) cells. ● Working mode combinations other than LM: The cell number specifications remain unchanged and are nine (2T2R, 2T4R, or 4T4R) cells. ● LTE FDD: The cell number specifications remain unchanged and are six (8T8R) cells.	● LTE FDD: A maximum of six cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth, while the rest are common cells. ● LM (with **BBP.*SRT*** set to **DEFAULT**): A maximum of six cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth, while the rest are common cells. ● LM (with **BBP.*SRT*** set to **NBIOT_ENHANCE**): A maximum of six cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth, while the rest are common cells. ● Working mode combinations other than LM: A maximum of three cells can be enabled with Hybrid DSS Based on Asymmetric Bandwidth, while the rest are common cells.

Working Mode	Baseband Processing Unit Model	Supported Cell Number Specifications (Compared with Those Before This Function Is Enabled)	Supported Spectrum Sharing Cell Number Specifications
LTE FDD, LM, LN, or other working modes involving LTE FDD	UBBPg/ UBBPh	Remain unchanged	Same as the cell number specifications before this function is enabled
a: LM refers to LTE FDD and NB-IoT.			

Table 4-25 Cell number specifications of LTE baseband processing units (multiple LTE cells)

Working Mode	Baseband Processing Unit Model	Supported LTE Cell Number Specifications
LTE FDD	UBBPe4/UBBPe6/ UBBPe18	Three LTE sector split groups are supported, with two LTE cells in each group. A maximum of three LTE sector split groups support this function.
LTE FDD	UBBPg/UBBPh	The supported LTE sector split group specifications are the same as those before this function is enabled.

NOTICE

The number of cells configured on a baseband processing unit cannot exceed the specifications listed in **Table 4-24** or **Table 4-25**. If this condition is not met, NR indicators will be negatively affected, including the RRC connection setup success rate, QoS flow setup success rate, and user-perceived rate.

To prevent LTE cell activation failures caused by the cell activation sequence, cells with this function enabled must be bound to baseband processing units supporting this function. If cells are not bound to such baseband processing units, cell activation may fail when the baseband processing units supporting this function are fully occupied while other baseband processing units do not support this function.

When a UBBPh2/UBBPh3/UBBPh3a/UBBPh3b is used as the interface board to serve 32T32R cells using 2:1 or 3.2:1 CPRI compression and the co-carrier co-CPRI data function is enabled, cell number specifications are represented by an indicator calculated using the following formula: Number of LTE cells in a spectrum sharing cell group x Number of antennas. The sum of the values of this indicator for all spectrum sharing cell groups must be less than or equal to 192. If the number of cells has reached the corresponding specification, subsequent cell setups will fail. If the networking requirements are not met or the UBBPh capacity in terms of the number of cells in baseband combination mode is insufficient, cell setups will fail. For LTE cells, ALM-29240 Cell Unavailable and ALM-29248 RF Out of Service will be reported with "The BBP capability is insufficient" as the cause value. For NR cells, ALM-29870 NR DU Cell TRP Unavailable and ALM-29871 NR DU Cell TRP Capability Degraded will be reported with "The baseband capability is insufficient" as the cause value.

When a UBBPh3a is used as the interface board to serve 16T16R cells using 2:1 or 3.2:1 CPRI compression and the co-carrier co-CPRI data function is enabled, cell number specifications are represented by an indicator calculated using the following formula: Number of LTE cells in a spectrum sharing cell group x Number of antennas. The sum of the values of this indicator for all spectrum sharing cell groups must be less than or equal to 192. If the number of cells has reached the corresponding specification, subsequent cell setups will fail. If the networking requirements are not met or the UBBPh capacity in terms of the number of cells in baseband combination mode is insufficient, cell setups will fail. For LTE cells, ALM-29240 Cell Unavailable and ALM-29248 RF Out of Service will be reported with "The BBP capability is insufficient" as the cause value. For NR cells, ALM-29870 NR DU Cell TRP Unavailable and ALM-29871 NR DU Cell TRP Capability Degraded will be reported with "The baseband capability is insufficient" as the cause value.

Only a UBBPe or a later series baseband processing unit supports the "LTE and NR 3-ms frame offset difference" function.

RF Modules

All 3000 series and 5000 series RF modules that meet the following conditions support this function:

- The RF module is shared by LTE and NR.
- The following NR bandwidths are supported and can be configured: 15 MHz, 20 MHz, 25 MHz, 30 MHz, and 40 MHz.
- The following frequency bands are supported and can be configured: 700 MHz, 800 MHz, 850 MHz, 900 MHz, 1800 MHz, 2100 MHz, 2600 MHz, AWS, and PCS.
- The following TX/RX modes are supported and can be configured: 2T2R, 2T4R, 4T4R, 8T8R, 16T16R, and 32T32R.

  8T8R RF modules supporting this function include integrated 8T8R RRUs and 8T8R RRUs formed by combined 4T4R RRUs.

  Of all 16T16R RF modules, those supporting both LTE FDD and NR FDD support this function.

  Of all 32T32R RF modules, those supporting both LTE FDD and NR FDD support this function.

  32T32R RF modules support only one LTE and NR spectrum sharing cell group. **Table 4-26** lists the carrier number specifications before and after Hybrid DSS Based on Asymmetric Bandwidth is enabled in a frequency band.

Table 4-26 Carrier number specifications of 32T32R RF modules

RF Module Type	Original Carrier Number Specifications	Carrier Number Specifications After Hybrid DSS Based on Asymmetric Bandwidth Is Enabled
Single-band module	Two carriers in one band	Three carriers in one band
Dual-band module (AAU5768 and AAU5765 excluded)	Two carriers in one band, four carriers in two bands	Three carriers in one band, five carriers in two bands
Dual-band module (AAU5768 and AAU5765)	Three carriers in one band, four carriers in two bands	Four carriers in one band, five carriers in two bands

NOTICE

When the NR cell bandwidth is 15 MHz or 20 MHz, the RF module requirements for LTE FDD and NR Flash Dynamic Spectrum Sharing must also be met. For details, see *LTE FDD and NR Dynamic Spectrum Sharing*.

When NB-IoT needs to be deployed in a guard band of an LTE cell with Hybrid DSS Based on Asymmetric Bandwidth enabled, only the RRU3262, RRU3965, RRU5309, RRU5508 (700 MHz), RRU5509t (800 MHz), RRU5512 (700 MHz), RRU5512t (800 MHz), RRU5519et (800 MHz), and RRU5909 can be used.

The following RF modules support the "LTE and NR 3-ms frame offset difference" function in Hybrid DSS Based on Asymmetric Bandwidth scenarios:

- AAU578*xAB*, RRU592*xAB*, RRU552*xAB*, and RRU582*xAB*, where *x* is a digit and *A* and *B* stand for different letters. (*A* and *B* can each be omitted.)
- RRU5508 (02312XBW), RRU5901 (02312WPR), RRU5516 (02313XTQ), RRU5502 (02312BSJ), RRU5904 (02311UWT), RRU5502N (02313EYK), and RRU5509t (02312GEB)

Cells

NR SDL cells do not support Hybrid DSS Based on Asymmetric Bandwidth.

LTE and NR cells must meet all of the following requirements:

- Supported bandwidth

  3900 and 5900 series base stations:

  Supported bandwidths are classified into standard bandwidths and non-standard bandwidths.

  – Standard bandwidths: See **Table 4-27**.

Table 4-27 Standard cell bandwidths to which Hybrid DSS Based on Asymmetric Bandwidth can be applied

LTE Cell Bandwidth	NR Cell Bandwidth
10 MHz	15 MHz
10 MHz	20 MHz
15 MHz	20 MHz
20 MHz	25 MHz
20 MHz	30 MHz
20 MHz	40 MHz

– Non-standard bandwidths: The NR cell can also work on a non-standard bandwidth when the compact bandwidth function is enabled. **Table 4-28** lists the non-standard bandwidths supported by the NR cell and the corresponding LTE standard cell bandwidths.

Table 4-28 Non-standard bandwidths to which Hybrid DSS Based on Asymmetric Bandwidth can be applied

LTE Cell Bandwidth	NR Cell Bandwidth
10 MHz	14.8 MHz
15 MHz	19.6 MHz
15 MHz	19.8 MHz
20 MHz	24.8 MHz

NOTICE

The NR cell can work on a non-standard bandwidth only when the compact bandwidth function is enabled. When a non-standard bandwidth is used, in addition to the hardware requirements described in this section, the hardware requirements of compact bandwidth must also be met. For details about the hardware requirements of compact bandwidth, see related descriptions in *Scalable Bandwidth* in *5G RAN Feature Documentation*.

For details about the NR cell bandwidths supported in different frequency bands, see descriptions of cell bandwidth and compact bandwidth in *Scalable Bandwidth* in *5G RAN Feature Documentation*.

- The full spectrum of an LTE cell must be within that of the corresponding NR cell for Hybrid DSS Based on Asymmetric Bandwidth. Otherwise, certain issues may occur; for example, cells may span beyond the spectrum owned by an operator or the power may increase.

- TX/RX mode

  The TX/RX modes of LTE and NR cells must be the same and must be one of 2T2R, 2T4R, 4T4R, 8T8R, 16T16R, and 32T32R.
- LTE and NR cells must have the same operating frequency band, and the center frequency spacing between the LTE and NR cells must be an integer multiple of 300 kHz.
- The PSD of the LTE and NR cells must be the same. If the total power of the LTE and NR cells exceeds the maximum transmit power of the RF module, the power of the cells needs to be reduced or the RF module needs to be replaced.
- LTE and NR cells must be deployed in 1:1 co-coverage mode, and must be served by the same transmit channels of an RF module.
- When Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) is required, the two LTE cells must have different PCIs as well as different PCI mod 3 values to preferentially ensure LTE performance. If NR performance needs to be preferentially ensured, the two LTE cells must have different PCIs but the same PCI mod 3 value.
- When Hybrid DSS Based on Asymmetric Bandwidth needs to be used together with dynamic point selection (DPS) transmission mode, the following conditions must be met:
  - The TX/RX mode is not 8T8R, 16T16R, or 32T32R.
  - The transmit power configurations of activated cells are the same.

4.3.4 Networking

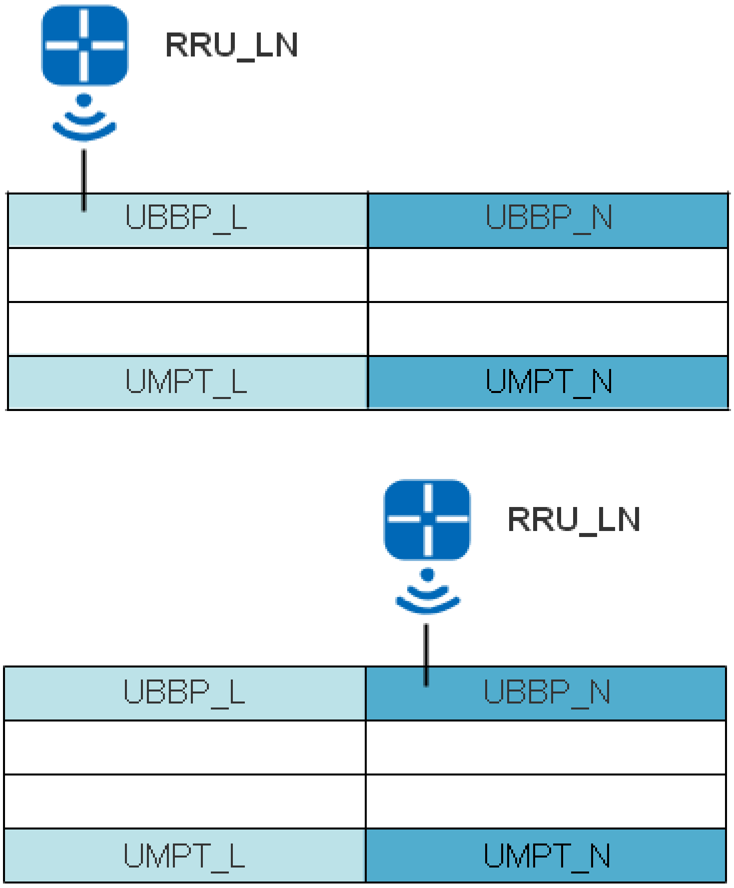
When the LTE and NR co-carrier co-CPRI data function is disabled (by setting the LTE parameter **LteNrSpctShrCellGrp.*LteNrCoCarrCoCpriDataSw*** to **OFF**):

- If the independent power configuration mode is used, LTE and NR must share a BBU.
- If the spectrum power sharing mode is used, LTE and NR must share a BBU. In addition, the **LNR_PWR_WITH_SPCT_OPT_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter must be selected to solve the power allocation conflicts caused by inconsistent LTE and NR latency when each uses their respective optical fibers over the CPRI port.

When the co-carrier co-CPRI data function is enabled (by setting the LTE parameter **LteNrSpctShrCellGrp.*LteNrCoCarrCoCpriDataSw*** to **ON**), only the following topologies are supported: intra-BBU single-fiber CPRI MUX topology in separate-MPT and co-MPT LTE/NR scenarios, and intra-board cold backup ring topology in co-MPT LTE/NR scenarios. These topologies do not pose special requirements on board installation positions. In an intra-BBU single-fiber CPRI MUX topology and an intra-board cold backup ring topology, LTE and NR cells can be established on the same or different baseband processing units. The following example illustrates the networking requirements using LTE and NR cells established on different baseband processing units.

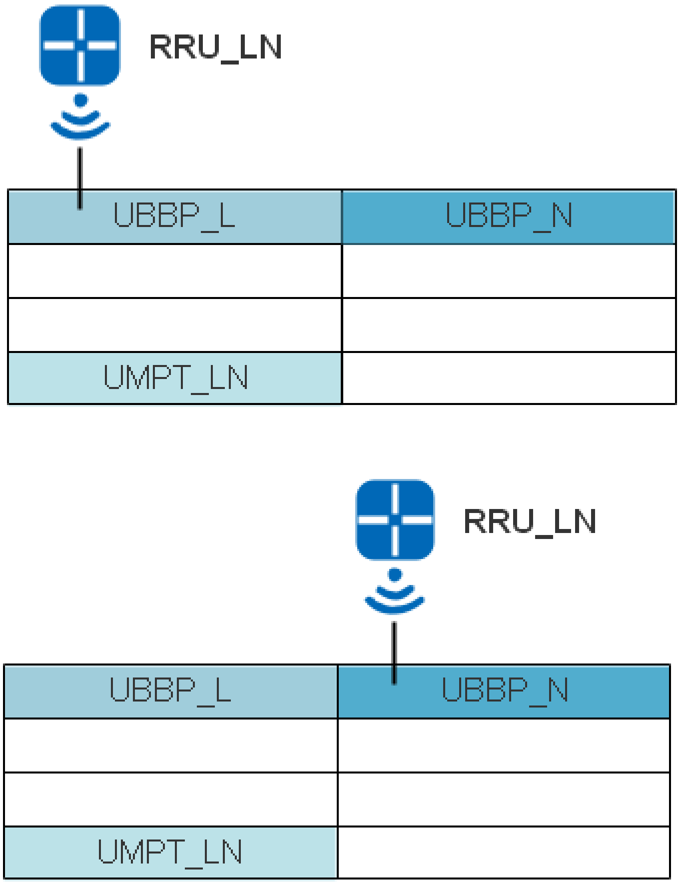
- Intra-BBU CPRI MUX topology in separate-MPT LTE/NR scenarios: LTE and NR share a BBU, and the LTE or NR baseband processing unit is connected directly to the RRU by optical fiber, as shown in **Figure 4-10**.

Figure 4-10 Intra-BBU CPRI MUX topology in separate-MPT LTE/NR scenarios



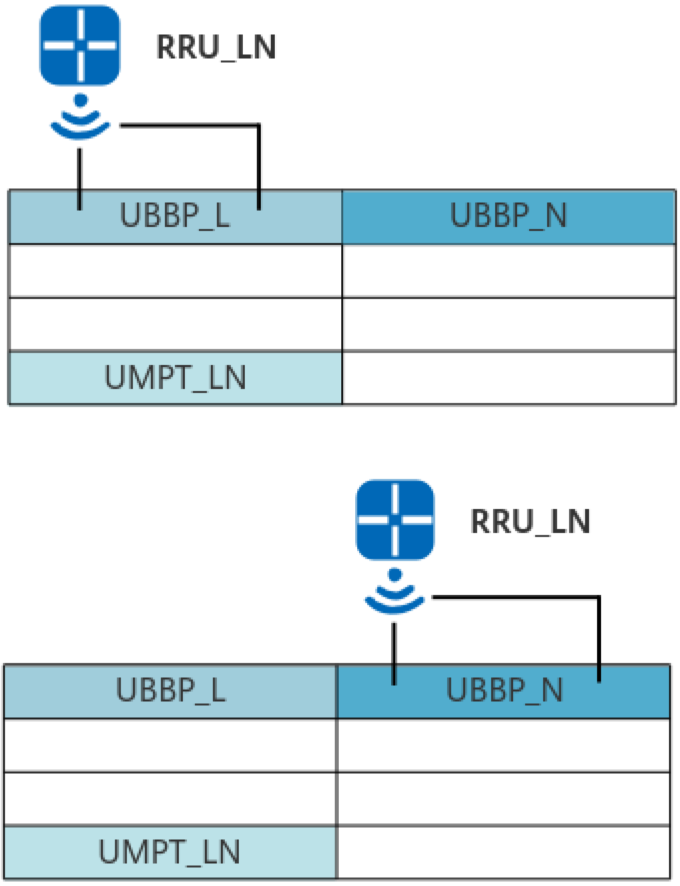
- Intra-BBU CPRI MUX topology in co-MPT LTE/NR scenarios: LTE and NR share a BBU, and the LTE or NR baseband processing unit is connected directly to the RRU by optical fiber, as shown in **Figure 4-11**.

Figure 4-11 Intra-BBU CPRI MUX topology in co-MPT LTE/NR scenarios



- Intra-board cold backup ring topology in co-MPT LTE/NR scenarios: LTE and NR share a BBU, and the LTE or NR baseband processing unit is connected directly to the RRU by optical fiber to form a ring topology, as shown in **Figure 4-12**.

Figure 4-12 Intra-board cold backup ring topology in co-MPT LTE/NR scenarios



NOTICE

In separate-MPT scenarios, the versions of NR and LTE must be V100R017C10 or later for Hybrid DSS Based on Asymmetric Bandwidth to take effect. In addition, if NR runs version *N*, LTE must run a version in the range of [*N* – 2, *N* + 2]. For example, if NR runs V100R019C10, the earliest version that LTE can run is V100R017C10.

4.3.5 Others

- This function takes effect only when MRFD-171221 Hybrid DSS Based on Asymmetric Bandwidth(LTE FDD) and MRFD-171261 Hybrid DSS Based on Asymmetric Bandwidth (NR) are both enabled.
- It is recommended that this function be enabled in all cells in the planned deployment area to reduce the interference caused by the near-far effect.
- This function requires LTE FDD and NR FDD networks to be time-synchronized. For details on how to configure time synchronization, see *Synchronization* in *eRAN Feature Documentation* and *Synchronization* in *5G RAN Feature Documentation*.
- NR UEs need to support the CRS rate matching function. This function allows the gNodeB to indicate the RE positions of the LTE CRSs, so that NR UEs can

avoid conflicts with the LTE CRSs. UEs support the CRS rate matching function when the *rateMatchingLTE-CRS* IE is included in the *BandNR* IE and the value of the *rateMatchingLTE-CRS* IE is "supported". For details, see section 5.1.4.2 "PDSCH resource mapping with RE level granularity" in 3GPP TS 38.214 V15.5.0. Only the NR UEs that support CRS rate matching on the live network support Hybrid DSS Based on Asymmetric Bandwidth. NR UEs that do not support CRS rate matching can access NR FDD cells enabled with Hybrid DSS Based on Asymmetric Bandwidth, but the uplink and downlink throughput of these NR UEs decrease.

- If certain parameters are modified for an NR cell engaged in Hybrid DSS Based on Asymmetric Bandwidth, the LTE parameters **CellRbReserve.*RbRsvStartIndex*** and **CellRbReserve.*RbRsvEndIndex*** must be modified accordingly for the LTE cell in the same spectrum sharing group as the NR cell. The new settings must ensure that the negotiation between the eNodeB and gNodeB will succeed. Otherwise, the NR cell will fail to be set up. These parameters include the following:
  – **COMM_PUCCH_RB_RES_OPT_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter, which controls common PUCCH RB resource optimization
  – **PUCCH_COMPRESSION_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter, which controls ultimate PUCCH compression
  – **NRDUCellPucch**.***CommonPucchResComprDegree*** parameter, which specifies the compression degree of common PUCCH RBs

4.4 Operation and Maintenance

4.4.1 Data Configuration

CAUTION

If the activation of this function is to be verified through signaling message tracing, you need to start signaling tracing on the MAE-Access before activating this function in LTE and NR cells. Otherwise, messages regarding the activation of this function cannot be observed after this function is activated.

4.4.1.1 Data Preparation

Table 4-29 and Table 4-31 describe the parameters used for function activation. Table 4-30 and Table 4-32 describe the parameters used for function optimization. This section does not describe parameters related to cell establishment.

The configuration of some parameters can result in automatic cell restarts during function activation and deactivation, and modifying the parameter settings of one RAT will cause both LTE and NR cells to automatically restart. As such, it is recommended that you deactivate cells prior to parameter configuration and reactivate them after parameter configuration is complete. This will prevent the cells from being repeatedly restarted during parameter configuration.

To prevent LTE cell activation failures due to insufficient baseband processing resources, cells enabled with this function must be bound to baseband processing units supporting this function. If cells are not bound to such baseband processing units, cell activation may fail when the baseband processing units supporting this function are fully occupied while other baseband processing units do not support this function.

Table 4-29 LTE parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloud Switch	**SpectrumCloud.*SpectrumCloudSwitch***	Set this parameter to **LTE_NR_SPECTRUM_SHR**.
Spectrum Cloud Enhancement Switch	**SpectrumCloud.*SpectrumCloudEnhSwitch***	Select the **LNR_SPECTRUM_SHR_ASYM_SW** option of this parameter.

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloud Enhancement Switch	**SpectrumCloud.*SpectrumCloudEnhSwitch***	Set this parameter based on the network plan. ● When SRS resource allocation to LTE is not required, select the **LTE_UE_SRS_NOT_CONFIG_SW** option of this parameter. Otherwise, deselect this option. The **LTE_UE_SRS_NOT_CONFIG_SW** option can be selected only when all of the following conditions are met: – The **SRSCfg.*SrsSubframeCfg*** parameter is set to **SC3**. – The **SrsSubframeRecfSwitch** option of the **CellAlgoSwitch.*SrsAlgoSwitch*** parameter is deselected. ● When the spectrum power sharing mode is used, and the LTE and NR baseband processing units need to be connected to the RRU through their respective optical fibers, select the **LNR_PWR_WITH_SPCT_OPT_SW** option. When the independent power configuration mode is used, LTE and NR can use their respective optical fiber over the CPRI port without the need of selecting this option.

Parameter Name	Parameter ID	Setting Notes
		● Select the **WBB_MBB_CONTROL_OPT_SW** option of this parameter when the simultaneous use with WBB is required.
LNR Spectrum Sharing Switch	**LteNrSpctShrCellGrp.*LteNrSpctShrSwitch***	Select the **LTE_CSI_RS_AVOID_POLICY_SW** option of this parameter. The **LTE_CSI_RS_AVOID_POLICY_SW** option cannot be selected when the LTE cell TX/RX mode is 8T8R.
LTE and NR Spectrum Sharing Cell Group ID[a]	**SpectrumCloud.*LteNrSpectrumShrCellGrpId***	Set this parameter based on the network plan.
Spectrum Sharing Mode	**SpectrumCloud.*SpctShrMode***	Set this parameter to **LTE_NR_PWR_DYN_SHR_WITH_SPCT** or **LTE_NR_PWR_INDEPENDENT** based on the network plan. When the LTE and NR co-carrier co-CPRI data function is enabled (the LTE parameter **LteNrSpctShrCellGrp.*LteNrCoCarrCoCpriDataSw*** is set to **ON**), this parameter can only be set to **LTE_NR_PWR_DYN_SHR_WITH_SPCT**.
LTE and NR Spectrum Sharing Cell Group ID[a]	**LteNrSpctShrCellGrp.*LteNrSpectrumShrCellGrpId***	Set this parameter based on the network plan. The value of the **LteNrSpctShrCellGrp.*LteNrSpectrumShrCellGrpId*** parameter must be the same as that of the **gNBDULteNrSpctShrCg.*LteSpctShrCellGrpId*** parameter.

Parameter Name	Parameter ID	Setting Notes
LTE Pri Res Ratio in LTE and NR Spct Shr	**LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio***	Set this parameter based on the network plan. The value of this parameter varies with the spectrum allocation policy, and the following lists the configuration suggestions for ensuring network performance when different spectrum allocation policies are adopted: ● If spectrum resources need to be preferentially allocated to LTE, it is recommended that this parameter be set to a value within the range of **51** to **80**. ● If spectrum resources need to be preferentially allocated to NR, it is recommended that this parameter be set to a value within the range of **20** to **49**. ● If spectrum resources need to be evenly allocated to LTE and NR, you are advised to set this parameter to **50**. If this parameter is set to a value less than 20 or greater than 80, the access success rate may decrease or the service drop rate may increase in high load scenarios. If this parameter is set to **0**, LTE UEs cannot access the network. Therefore, the value **0** is not recommended. If this parameter is set to **100**, NR UEs cannot access the network.

Parameter Name	Parameter ID	Setting Notes
		Therefore, the value **100** is not recommended.
DL SPS Restrict Ratio	**SpectrumCloud.*DlSpsRestrictRatio***	Set this parameter based on the network plan. The proportion of RBs allocated for LTE downlink semi-persistent scheduling equals (**SpectrumCloud.*DlSpsRestrictRatio*** x **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio***/100)%.
LTE and NR Co Carrier Co CPRI Data Switch	**LteNrSpctShrCellGrp.*LteNrCoCarrCoCpriDataSw***	Set this parameter based on the network plan. This parameter must be set to **ON** only when LTE and NR share the same CPRI optical fibers and the CPRI bandwidth is insufficient. If the cell TX/RX mode is 32T32R, this parameter can be set to **ON** only when the CPRI compression ratio is 2.2:1, 2:1, or 3.2:1. If the cell TX/RX mode is 16T16R, this parameter can be set to **ON** only when the CPRI compression ratio is 2:1 or 3.2:1.
Frame Offset[b]	**CellFrameOffset.*FrameOffset***	Set this parameter to a value confirmed by Huawei engineers.
FDD Frame Offset[b]	**ENodeBFrameOffset.*FddFrameOffset***	Set this parameter to a value confirmed by Huawei engineers.

Parameter Name	Parameter ID	Setting Notes
TA Offset	**CellFrameOffset.*TaOffset***	Ensure that the TA offset specified by this parameter on the LTE side is the same as the TA offset (specified by the **NRDUCell.*TaOffset*** parameter) on the NR side. Otherwise, the actual TA offset on the LTE or NR side is inconsistent with the configured value. That is, the following configuration requirements must be met: ● When the **CellFrameOffset.*TaOffset*** parameter is set to **0Ts**, the **NRDUCell.*TaOffset*** parameter must be set to **0Tc**. ● When the **CellFrameOffset.*TaOffset*** parameter is set to **400Ts**, the **NRDUCell.*TaOffset*** parameter must be set to **25600Tc**. ● When the **CellFrameOffset.*TaOffset*** parameter is set to **624Ts**, the **NRDUCell.*TaOffset*** parameter must be set to **39936Tc**. Before changing the value of this parameter, deactivate both the LTE and NR cells.

Parameter Name	Parameter ID	Setting Notes
a: In scenarios of a single LTE cell, an LTE spectrum sharing cell group can contain only one LTE cell. In scenarios of multiple LTE cells, an LTE spectrum sharing cell group can contain only the sector split cells of the same LTE FDD sector split group. b: If both the **CellFrameOffset.*FrameOffset*** and **ENodeBFrameOffset.*FddFrameOffset*** parameters are configured, the frame offset specified by the **CellFrameOffset.*FrameOffset*** parameter takes effect for the LTE cell.		

Table 4-30 LTE parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
LNR Spectrum Sharing Switch	**LteNrSpctShrCellGrp.*LteNrSpctShrSwitch***	The **LNR_RES_ALLOC_ADAPT_SW** option of this parameter together with the **LteNrSpctShrCellGrp.*LteNrSpctShrLtePriResRatio*** parameter determines the spectrum resource allocation mode for LTE and NR cells. For details, see **4.1.1.2.2 Flexible Spectrum Priority Mode**. When the function of inter-RAT preferential guarantee for GBR services is required, select the **GBR_PRI_ALLOC_SW** option of this parameter. When the SSB frequency-domain position is beyond the NR-dedicated spectrum, select the **LNR_RES_ALLOC_DIRECTION_OPT_SW** option (mandatory) and the **HDSS_DL_PRB_ALLOCATE_OPT_SW** option (recommended) of this parameter.

Parameter Name	Parameter ID	Setting Notes
LNR Spectrum Sharing Switch	**LteNrSpctShrCellGrp.*LteNrSpctShrSwitch***	Select the **LTE_NR_SRS_ALLOC_OPT_SW** option of this parameter when LTE and NR SRS resource allocation optimization is required based on the network plan. The preceding option can be selected only when the following conditions are all met: ● The cell TX/RX mode is not 32T32R. ● The **Cell.*HighSpeedFlag*** parameter must be set to **LOW_SPEED**. ● The **CellUlschAlgo.*UlSrsFreqSelSchSinrThld*** parameter must be set to **255**. ● The **NSA_DC_UL_PATH_SELECTION_SW** option of the **NsaDcMgmtConfig.*NsaDcAlgoSwitch*** parameter is deselected. ● The **NCellSrsMeasPara.*SrsAutoNCellMeasS-witch*** parameter is set to **OFF**. ● The **LTE_UE_SRS_NOT_CONFIG_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter is deselected.

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloud Extend Switch	**SpectrumCloud**.***SpectrumCloudExtSwitch***	Select the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of this parameter when Hybrid DSS Based on Asymmetric Bandwidth is enabled together with the LTE and NR flexible frame offset function based on the network plan.
Parameter Configuration Protection Switch	**eNodeBResModeAlgo**.***ParamCfgProtectSw***	Select the **DSS_CROSS_RAT_CFG_CHECK_SW** option of this parameter when "cross-RAT configuration check in DSS scenarios" is required based on the network plan.

Table 4-31 NR parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloudification Switch	**NRDUCellAlgoSwitch**.***SpectrumCloudSwitch***	Select the **LTE_NR_FDD_SPCT_SHR_SW** option of this parameter.
Spectrum Cloudification Enhance Switch	**NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch***	Select the **LTE_NR_FDD_SPCT_SHR_ASYM_SW** option of this parameter.

Parameter Name	Parameter ID	Setting Notes
NR Reserved RB Start Index	**gNBDULteNrSpctShrCg.** ***NrReservedRbStartIndex***	Set this parameter based on the network plan. This parameter must be set to a value corresponding to the frequency-domain position at either end of the NR spectrum. If the **gNBDULteNrSpctShrCg.** ***NrUlReservedRbStartIndex*** parameter is also configured, the uplink dedicated spectrum configuration takes the value of the **gNBDULteNrSpctShrCg.** ***NrUlReservedRbStartIndex*** parameter, and the downlink dedicated spectrum configuration still takes the value of the **gNBDULteNrSpctShrCg.** ***NrReservedRbStartIndex*** parameter.
NR Reserved RB End Index	**gNBDULteNrSpctShrCg.** ***NrReservedRbEndIndex***	Set this parameter based on the network plan. If the **gNBDULteNrSpctShrCg.** ***NrUlReservedRbEndIndex*** parameter is also configured, the uplink dedicated spectrum configuration takes the value of the **gNBDULteNrSpctShrCg.** ***NrUlReservedRbEndIndex*** parameter, and the downlink dedicated spectrum configuration still takes the value of the **gNBDULteNrSpctShrCg.** ***NrReservedRbEndIndex*** parameter.

Parameter Name	Parameter ID	Setting Notes
NR Uplink Reserved RB Start Index	**gNBDULteNrSpctShrCg.** ***NrUlReservedRbStartIndex***	Set this parameter based on the network plan. This parameter must be set to a value corresponding to the frequency-domain position at either end of the NR spectrum.
NR Uplink Reserved RB End Index	**gNBDULteNrSpctShrCg.** ***NrUlReservedRbEndIndex***	Set this parameter based on the network plan.
NR Spectrum Sharing Cell Group ID[a]	**NRDUCellSpctCloud.*NrSpctShrCellGrpId***	Set this parameter based on the network plan.
NR Spectrum Sharing Cell Group ID[a]	**gNBDULteNrSpctShrCg.** ***NrSpctShrCellGrpId***	Set this parameter based on the network plan.
LTE Spectrum Sharing Cell Group ID[a]	**gNBDULteNrSpctShrCg.** ***LteSpctShrCellGrpId***	Set this parameter based on the network plan. The value of the **gNBDULteNrSpctShrCg.** ***LteSpctShrCellGrpId*** parameter must be the same as that of the **LteNrSpctShrCellGrp.*LteNrSpectrumShrCellGrpId*** parameter.
FDD LNR Spectrum Sharing Switch	**gNBDULteNrSpctShrCg.** ***FddLteNrSpctShrSwitch***	Select the **LTE_CSI_RS_AVOID_SW** and **LTE_CRS_PORT_1_RM_SW** options of this parameter based on the network plan. These two options can be individually or both selected. The **LTE_CSI_RS_AVOID_SW** option cannot be selected when the NR cell TX/RX mode is 8T8R.

Parameter Name	Parameter ID	Setting Notes
SSB Frequency Position Describe Method	**NRDUCell.***SsbDescMethod*	It is recommended that this parameter be set to **SSB_DESC_TYPE_NARFCN** in NSA networking. In SA networking, set this parameter to **SSB_DESC_TYPE_GSCN**.
SSB Frequency Position	**NRDUCell.***SsbFreqPos*	Set this parameter to a value confirmed by Huawei engineers.
SSB Period	**NRDUCell.***SsbPeriod*	Set this parameter based on the network plan.
SIB1 Period	**NRDUCell.***Sib1Period*	Set this parameter to **MS40(40)**.
Format1 RB Number	**NRDUCellPucch.***Format1RbNum*	Set this parameter based on the network plan. For details about the configuration suggestions for different scenarios, see data preparation for PUCCH channel management in *Channel Management*.
Format3 RB Number	**NRDUCellPucch.***Format3RbNum*	Set this parameter based on the network plan. For details about the cells for which this parameter takes effect, see the parameter meaning. For details about the configuration suggestions for different scenarios, see data preparation for PUCCH channel management in *Channel Management*.

Parameter Name	Parameter ID	Setting Notes
Format3 CSI-dedicated RB Number	**NRDUCellPucch.*CsiDedicatedRbNum***	Set this parameter based on the network plan. The value **RB0** is not recommended. For details about the configuration suggestions for different scenarios, see data preparation for PUCCH channel management in *Channel Management*.
Format4 RB Number	**NRDUCellPucch.*Format4RbNum***	Set this parameter based on the network plan. For details about the cells for which this parameter takes effect, see the parameter meaning. For details about the configuration suggestions for different scenarios, see data preparation for PUCCH channel management in *Channel Management*.
Format4 CSI-dedicated RB Number	**NRDUCellPucch.*Format4CsiDedicatedRbNum***	Set this parameter based on the network plan. For details about the configuration suggestions for different scenarios, see data preparation for PUCCH channel management in *Channel Management*.

Parameter Name	Parameter ID	Setting Notes
PDCCH Algorithm Extension Switch	**NRDUCellPdcch.*PdcchAlgoExtSwitch***	Set this parameter based on the network plan. If the UE-specific PDCCH can span over the entire NR bandwidth, select the **UE_PDCCH_FULL_BANDWIDTH_CFG_SW** option. If the UE-specific PDCCH spans over the dedicated NR bandwidth, deselect the **UE_PDCCH_FULL_BANDWIDTH_CFG_SW** option. It is recommended that this option be deselected, which is the default value.

Parameter Name	Parameter ID	Setting Notes
Spectrum Sharing Start Symbol	**NRDUCellPdcch.*SpctShrStartSymbol***	When the **UE_PDCCH_FULL_BANDWIDTH_CFG_SW** option of the **NRDUCellPdcch.*PdcchAlgoExtSwitch*** parameter is selected, the configuration suggestions are as follows: ● Assume that one or two LTE CRS ports are configured. When NR traffic volume requirements are greater than LTE traffic volume requirements, set this parameter to **SYM1**; when NR traffic volume requirements are lower than LTE traffic volume requirements, set this parameter to **SYM2**. Otherwise, the proportion of CCE allocation failures increases in the cell with larger traffic volume requirements, affecting the average uplink and downlink UE throughputs. ● When four LTE CRS ports are configured, set this parameter to **SYM2**. When the **UE_PDCCH_FULL_BANDWIDTH_CFG_SW** option of the **NRDUCellPdcch.*PdcchAlgoExtSwitch*** parameter is deselected, set this parameter to **SYM0**.

Parameter Name	Parameter ID	Setting Notes
Occupied Symbol Number	**NRDUCellPdcch.*OccupiedSymbolNum***	Set this parameter to **2SYM** or **3SYM** when the **NRDUCellPdcch.*SpctShrStartSymbol*** parameter is set to **SYM0** or **SYM1**, and to **1SYM** when the **NRDUCellPdcch.*SpctShrStartSymbol*** parameter is set to **SYM2**. It is recommended that the **NRDUCellPdcch.*OccupiedSymbolNum*** parameter be set to **2SYM** to ensure sufficient resources for proper service operation when the following conditions are met: The NR cell bandwidth is 25 MHz or 30 MHz, the **UE_PDCCH_FULL_BANDWIDTH_CFG_SW** option of the NR parameter **NRDUCellPdcch.*PdcchAlgoExtSwitch*** is deselected, and the **NRDUCellCoreset.*CommonCtrlResStartSymbol*** parameter is set to an invalid value. There are no special requirements in other NR cell bandwidth scenarios.
Common Control Resource Start Symbol	**NRDUCellCoreset.*CommonCtrlResStartSymbol***	If the NR common PDCCH and UE-specific PDCCH are transmitted on the same symbols, set this parameter to an invalid value. If the NR common PDCCH and UE-specific PDCCH are transmitted on different symbols, it is recommended that this parameter be set to **SYM0**.

Parameter Name	Parameter ID	Setting Notes
Common Control Resource RB Number	**NRDUCellCoreset.*CommonCtrlResRbNum***	It is recommended that the **NRDUCellCoreset.*CommonCtrlResRbNum*** parameter be set to **RB24(24 RBs x 2 Symbols)** to ensure sufficient resources for services when the following conditions are met: The NR cell bandwidth is 25 MHz or 30 MHz, the **UE_PDCCH_FULL_BANDWIDTH_CFG_SW** option of the **NRDUCellPdcch.*PdcchAlgoExtSwitch*** parameter is deselected, and the **NRDUCellCoreset.*CommonCtrlResStartSymbol*** parameter is set to **SYM0**. In other scenarios, there is no special requirement.
Frame Offset[b]	**gNodeBParam.*FrameOffset***	Set this parameter to a value confirmed by Huawei engineers.
Frame Offset[b]	**gNBFreqBandConfig.*FrameOffset***	Set this parameter to a value confirmed by Huawei engineers.

Parameter Name	Parameter ID	Setting Notes
TA Offset	**NRDUCell.*TaOffset***	Ensure that the TA offset specified by this parameter on the NR side is the same as the TA offset (specified by the **CellFrameOffset.*TaOffset*** parameter) on the LTE side. Otherwise, the actual TA offset on the LTE or NR side is inconsistent with the configured value. That is, the following configuration requirements must be met: ● When the **CellFrameOffset.*TaOffset*** parameter is set to **0Ts**, the **NRDUCell.*TaOffset*** parameter must be set to **0Tc**. ● When the **CellFrameOffset.*TaOffset*** parameter is set to **400Ts**, the **NRDUCell.*TaOffset*** parameter must be set to **25600Tc**. ● When the **CellFrameOffset.*TaOffset*** parameter is set to **624Ts**, the **NRDUCell.*TaOffset*** parameter must be set to **39936Tc**. Before changing the value of this parameter, deactivate both the LTE and NR cells.
RB Reserve Mode[c]	**NRDUCellRbReserve.*RbRsvMode***	Set this parameter to **NB_GB_RB_RESERVED**.

Parameter Name	Parameter ID	Setting Notes
RB Reserve Type[c]	**NRDUCellRbReserve.*RbRsvType***	Set this parameter to **UPLINK_MODE** or **DOWNLINK_MODE** to reserve uplink and downlink RBs, respectively.
RB Reserve Start Index[c]	**NRDUCellRbReserve.*RbRsvStartIndex***	Set this parameter to a value confirmed by Huawei engineers. The recommended configuration is described in **4.1.1.3.2 NR Functions**.
RB Reserve End Index[c]	**NRDUCellRbReserve.*RbRsvEndIndex***	Set this parameter to a value confirmed by Huawei engineers. The recommended configuration is described in **4.1.1.3.2 NR Functions**.
Spectrum Cloudification Enhance Switch	**NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch***	Set this parameter based on the network plan. When Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) is enabled, select the **LTE_CRS_RATEMATCH_ALL_SYM_SW** option if LTE performance needs to be preferentially ensured, and deselect this option if NR performance needs to be preferentially ensured.
BWP Configuration Policy Switch	**NRDUCellAlgoSwitch.*BwpConfigPolicySwitch***	You are advised to select the **INIT_BWP_FULL_BW_SW** and **BWP0_DL_FULL_BW_SW** options of this parameter when the SSB frequency-domain position is beyond the NR-dedicated spectrum based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Downlink Schedule Algorithm Enhanced Switch	**NRDUCellDlSch**.***DlSchAlgoEnhSwitch***	Select the **HDSS_DL_SCH_BW_ADAPT_SW** option of this parameter based on the network plan.
HDSS Spectral Efficiency Coefficient	**NRDUCellDlSch**.***HdssSpctEffCoefficient***	It is recommended that this parameter be set to **100** when downlink scheduling bandwidth adaptation in HDSS scenarios is enabled based on the network plan.
HDSS Aperiodic CSI Interval Adapt Coeff	**NRDUCellDlSch**.***HdssAperCsiIntvlAdaptCoeff***	Set this parameter based on the network plan.
a: In the current version, an NR spectrum sharing cell group can contain only one NR cell. b: If both the **gNodeBParam.*FrameOffset*** and **gNBFreqBandConfig.*FrameOffset*** parameters are configured, the frame offset specified by the **gNBFreqBandConfig.*FrameOffset*** parameter takes effect for the NR cell. c: These four parameters need to be configured when Hybrid DSS Based on Asymmetric Bandwidth is used together with NB-IoT deployed in LTE guard band mode.		

Table 4-32 NR parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Downlink Additional DMRS Position	**NRDUCellPdsch.*DlAdditionalDmrsPos***	Set this parameter to **POS1**.
Spectrum Cloudification Enhance Switch	**NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch***	Select the **LTE_SS_PBCH_RM_OPT_SW** option of this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloudification Enhance Switch	**NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch***	According to the network plan, if Hybrid DSS Based on Asymmetric Bandwidth (single LTE cell) is enabled and the bandwidth of NR dedicated spectrum is greater than or equal to 10 MHz, select the **LTE_CRS_RATEMATCH_ALL_SYM_SW** (when the LTE spectrum sharing cell and neighboring LTE cells are configured with the same number of CRS ports) or **LTE_CRS_RM_ALL_SYM_4PORT_SW** (when the LTE spectrum sharing cell and neighboring LTE cells are configured with different numbers of CRS ports) option of this parameter. This is to reduce the interference of neighboring LTE cells on the NR spectrum sharing cell.
Spectrum Cloudification Enhance Switch	**NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch***	Select the **PWR_SPCT_DENSITY_ADAPT_SW** option of this parameter based on the network plan.
Spectrum Cloud Enhancement Switch	**NRDUCellAlgoSwitch.*SpectrumCloudEnhSwitch***	Select the **SSB_ADJ_SW** option of this parameter when the cell TX/RX mode is 8T8R, 16T16R, or 32T32R and four SSB beams are configured. In other scenarios, use the default setting of the **SSB_ADJ_SW** option.

Parameter Name	Parameter ID	Setting Notes
SSB Time Position	**NRDUCell**.*SsbTimePos*	Set this parameter as follows when time-domain SSB staggering is required based on the network plan: ● If the SSB is configured within the shared spectrum, set this parameter to **SSB23_PCI_MOD2_STAGGER**. ● If the SSB is configured within the NR-dedicated spectrum, set this parameter to **SSB_PCI_MOD3_STAGGER**.
Paging Scheduling Algorithm Switch	**NRDUCellPagingConfig**.*PagingSchAlgoSwitch*	Select the **PAGING_OSI_STAGGER_SW** option when paging-OSI staggering is required based on the network plan.
Downlink Schedule Algorithm Switch	**NRDUCellDlSch**.*DlSchAlgoSwitch*	Select the **TIME_DOMAIN_RMSI_STAGGER_SW** option when time-domain RMSI staggering is required based on the network plan.
LNR Spectrum Share Switch	**NRDUCellSpctShare**.*LnrSpectrumShareSwitch*	Select the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of this parameter when Hybrid DSS Based on Asymmetric Bandwidth is enabled together with the LTE and NR flexible frame offset function based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Config Interception Policy Switch	**gNBOamParam**.***CfgInterceptPolicySw***	Select the **DSS_CROSS_RAT_CFG_CHECK_SW** option of this parameter when "cross-RAT configuration check in DSS scenarios" is required based on the network plan.

4.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

This function must be activated or deactivated on both the LTE and NR sides. When Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) is required, parameters need to be configured for each LTE split cell on the LTE side.

Activation Command Examples (Single LTE Cell)

Activation command examples on the LTE side

```
//Deactivating the cell
DEA CELL: LocalCellId=0;
//Binding the cell that requires this function to a baseband processing unit supporting this function
MOD EUCELLSECTOREQM: LocalCellId=0, SectorEqmId=1, BaseBandEqmId=12;
//Setting the percentage of spectrum resources to be preferentially allocated to LTE MBB services, and turning on the LTE and NR co-carrier co-CPRI data switch
ADD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LteNrSpctShrLtePriResRatio=50, LteNrCoCarrCoCpriDataSw=ON;
//(Optional, configurable only when the cell TX/RX mode is not 8T8R) Configuring RE-level avoidance of the LTE CSI-RS by NR
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LteNrSpctShrSwitch=LTE_CSI_RS_AVOID_POLICY_SW-1;
//(Optional, required when the cell TX/RX mode is 8T8R) Configuring subframe-level avoidance of the LTE CSI-RS by NR
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LteNrSpctShrSwitch=LTE_CSI_RS_AVOID_POLICY_SW-0;
//Turning on the switch controlling Hybrid DSS Based on Asymmetric Bandwidth, adding the LTE cell to an LTE spectrum sharing cell group, setting the spectrum sharing mode (assuming that SRS resources need to be configured for LTE UEs), turning on the switch controlling power allocation optimization in spectrum power sharing mode, turning on the WBB/MBB control policy optimization switch, and setting the LTE downlink SPS restrict ratio
ADD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudSwitch=LTE_NR_SPECTRUM_SHR, SpectrumCloudEnhSwitch=LNR_SPECTRUM_SHR_ASYM_SW-1&LTE_UE_SRS_NOT_CONFIG_SW-0&LNR_PWR_WITH_SPCT_OPT_SW-1&WBB_MBB_CONTROL_OPT_SW-1, LteNrSpectrumShrCellGrpId=0, SpctShrMode=LTE_NR_PWR_DYN_SHR_WITH_SPCT, DlSpsRestrictRatio=60;
//Setting the frame offset and TA offset
ADD CELLFRAMEOFFSET: LocalCellId=0, FrameOffsetMode=CustomFrameOffset, FrameOffset=0,
```



```
TaOffset=0Ts;
//Activating the cell
ACT CELL: LocalCellId=0;
```

Activation command examples on the NR side

```
//Deactivating the cell
DEA NRCELL: NrCellId=0;
//Adding an LTE spectrum sharing cell group and an NR spectrum sharing cell group
ADD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0;
//Adding an NR cell to the NR spectrum sharing cell group
ADD NRDUCELLSPCTCLOUD: NrDuCellId=0, NrSpctShrCellGrpId=0;
//Setting the number of RBs for the NR PUCCH (using cells for which the NSA TDM pattern type strategy is
not set to LTE uplink preferred (NRDUCellAlgoSwitch.NsaTdmPatternTypeStrategy set to a value other than
LTE_UL_PREFERRED) as an example)
MOD NRDUCELLPUCCH: NrDuCellId=0, Format1RbNum=RB2, Format3RbNum=RB4,
CsiDedicatedRbNum=RB4, Format4RbNum=RB0, Format4CsiDedicatedRbNum=RB0;
//Configuring symbols occupied by the NR PDCCH
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=UE_PDCCH_FULL_BANDWIDTH_CFG_SW-0,
SpctShrStartSymbol=SYM0, OccupiedSymbolNum=2SYM;
//Setting the frame offset
MOD GNODEBPARAM: FrameOffset=0;
//Setting the TA offset
MOD NRDUCELL: NrDuCellId=0, FrequencyBand=N1, TaOffset=0TC;
//Turning on the switch controlling Hybrid DSS Based on Asymmetric Bandwidth
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudSwitch=LTE_NR_FDD_SPCT_SHR_SW-1,
SpectrumCloudEnhSwitch=LTE_NR_FDD_SPCT_SHR_ASYM_SW-1;
//Configuring the NR dedicated spectrum (using a 30 MHz NR cell as an example)
MOD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0, NrReservedRbStartIndex=0,
NrReservedRbEndIndex=51, NrUlReservedRbStartIndex=0, NrUlReservedRbEndIndex=52;
//Configuring the number of RBs for common control resources
MOD NRDUCELLCORESET: NrDuCellId=0, CommonCtrlResStartSymbol=SYM0, CommonCtrlResRbNum=RB24;
//Configuring the SSB frequency-domain position based on actual conditions (using SsbFreqPos set to
426860 as an example)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_FDD, SsbDescMethod=SSB_DESC_TYPE_NARFCN,
SsbFreqPos=426860;
//Configuring the SSB and SIB1 periods
MOD NRDUCELL: NrDuCellId=0, SsbPeriod=MS20, Sib1Period=MS40;
//(Optional, configurable only when the cell TX/RX mode is not 8T8R) Configuring RE-level avoidance of the
LTE CSI-RS by NR
MOD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0,
FddLteNrSpctShrSwitch=LTE_CSI_RS_AVOID_SW-1;
//(Optional, required when the cell TX/RX mode is 8T8R) Configuring subframe-level avoidance of the LTE
CSI-RS by NR
MOD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0,
FddLteNrSpctShrSwitch=LTE_CSI_RS_AVOID_SW-0;
//(Optional; required when Hybrid DSS Based on Asymmetric Bandwidth is used together with NB-IoT
deployed in LTE guard band mode) Configuring reserved RBs when NB-IoT is deployed
ADD NRDUCELLRBRESERVE: NrDuCellId=0, Index=0, RbRsvMode=NB_GB_RB_RESERVED,
RbRsvType=DOWNLINK_MODE, RbRsvStartIndex=0, RbRsvEndIndex=3;
ADD NRDUCELLRBRESERVE: NrDuCellId=0, Index=1, RbRsvMode=NB_GB_RB_RESERVED,
RbRsvType=UPLINK_MODE, RbRsvStartIndex=0, RbRsvEndIndex=3;
//(Optional; required when Hybrid DSS Based on Asymmetric Bandwidth is used together with downlink
scheduling bandwidth adaptation in HDSS scenarios) Configuring downlink scheduling bandwidth
adaptation in HDSS scenarios
MOD NRDUCELLDLSCH: NrDuCellId=0, HdssAperCsiIntvlAdaptCoeff=135,
DlSchAlgoEnhSwitch=HDSS_DL_SCH_BW_ADAPT_SW-1, HdssSpctEffCoefficient=100;
//(Optional; required when Hybrid DSS Based on Asymmetric Bandwidth is used together with downlink
scheduling bandwidth adaptation in HDSS scenarios) Configuring the range of RBs within the downlink
interference range, which must be the same as the shared spectrum
MOD NRDUCELLINTRFBAND: NrDuCellId=0, InterferenceBandIndex=0, StartRb=0, EndRb=106,
InterferenceDirection=DOWNLINK;
MOD NRDUCELLINTRFIDENT: NrDuCellId=0, IntrfIdentificationMethod=DL_MANUAL,
IntrfOptimizationMethod=DL_UE_LEVEL_INTRF_AVOID-1;
//Activating the cell
ACT NRCELL: NrCellId=0;
```

Activation Command Examples (Multiple LTE Cells)

Activation command examples on the LTE side

```
//Deactivating the cells
DEA CELL: LocalCellId=0;
DEA CELL: LocalCellId=3;
//Binding the cells that require this function to a baseband processing unit supporting this function
MOD EUCELLSECTOREQM: LocalCellId=0, SectorEqmId=1, BaseBandEqmId=12;
MOD EUCELLSECTOREQM: LocalCellId=3, SectorEqmId=1, BaseBandEqmId=12;
//Setting the percentage of spectrum resources to be preferentially allocated to LTE MBB services, and
turning off the LTE and NR co-carrier co-CPRI data switch
ADD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LteNrSpctShrLtePriResRatio=50,
LteNrCoCarrCoCpriDataSw=OFF;
//(Optional, configurable only when the cell TX/RX mode is not 8T8R) Configuring RE-level avoidance of the
LTE CSI-RS by NR
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSpctShrSwitch=LTE_CSI_RS_AVOID_POLICY_SW-1;
//(Optional, required when the cell TX/RX mode is 8T8R) Configuring subframe-level avoidance of the LTE
CSI-RS by NR
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSpctShrSwitch=LTE_CSI_RS_AVOID_POLICY_SW-0;
//Turning on the switch controlling Hybrid DSS Based on Asymmetric Bandwidth, adding the LTE cells to an
LTE spectrum sharing cell group, setting the spectrum sharing mode (assuming that SRS resources need to
be configured for LTE UEs), turning on the switch controlling power allocation optimization in spectrum
power sharing mode, turning on the WBB/MBB control policy optimization switch, and setting the LTE
downlink SPS restrict ratio
ADD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudSwitch=LTE_NR_SPECTRUM_SHR,
SpectrumCloudEnhSwitch=LNR_SPECTRUM_SHR_ASYM_SW-1&LTE_UE_SRS_NOT_CONFIG_SW-0&LNR_PWR_
WITH_SPCT_OPT_SW-1&WBB_MBB_CONTROL_OPT_SW-1, LteNrSpectrumShrCellGrpId=0,
SpctShrMode=LTE_NR_PWR_DYN_SHR_WITH_SPCT, DlSpsRestrictRatio=60;
ADD SPECTRUMCLOUD: LocalCellId=3, SpectrumCloudSwitch=LTE_NR_SPECTRUM_SHR,
SpectrumCloudEnhSwitch=LNR_SPECTRUM_SHR_ASYM_SW-1&LTE_UE_SRS_NOT_CONFIG_SW-0&LNR_PWR_
WITH_SPCT_OPT_SW-1&WBB_MBB_CONTROL_OPT_SW-1, LteNrSpectrumShrCellGrpId=0,
SpctShrMode=LTE_NR_PWR_DYN_SHR_WITH_SPCT, DlSpsRestrictRatio=60;
//Setting the frame offset and TA offset
ADD CELLFRAMEOFFSET: LocalCellId=0, FrameOffsetMode=CustomFrameOffset, FrameOffset=0,
TaOffset=0Ts;
ADD CELLFRAMEOFFSET: LocalCellId=3, FrameOffsetMode=CustomFrameOffset, FrameOffset=0,
TaOffset=0Ts;
//Activating the cells
ACT CELL: LocalCellId=0;
ACT CELL: LocalCellId=3;
```

Activation command examples on the NR side

```
//Deactivating the cell
DEA NRCELL: NrCellId=0;
//Adding an LTE spectrum sharing cell group and an NR spectrum sharing cell group
ADD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0;
//Adding an NR cell to the NR spectrum sharing cell group
ADD NRDUCELLSPCTCLOUD: NrDuCellId=0, NrSpctShrCellGrpId=0;
//Setting the number of RBs for the NR PUCCH (using cells for which the NSA TDM pattern type strategy is
not set to LTE uplink preferred (NRDUCellAlgoSwitch.NsaTdmPatternTypeStrategy set to a value other than
LTE_UL_PREFERRED) as an example)
MOD NRDUCELLPUCCH: NrDuCellId=0, Format1RbNum=RB2, Format3RbNum=RB4,
CsiDedicatedRbNum=RB4, Format4RbNum=RB0, Format4CsiDedicatedRbNum=RB0;
//Configuring symbols occupied by the NR PDCCH
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=UE_PDCCH_FULL_BANDWIDTH_CFG_SW-0,
SpctShrStartSymbol=SYM0, OccupiedSymbolNum=2SYM;
//Setting the frame offset
MOD GNODEBPARAM: FrameOffset=0;
//Setting the TA offset
MOD NRDUCELL: NrDuCellId=0, FrequencyBand=N1, TaOffset=0TC;
//Turning on the switch controlling Hybrid DSS Based on Asymmetric Bandwidth
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudSwitch=LTE_NR_FDD_SPCT_SHR_SW-1,
SpectrumCloudEnhSwitch=LTE_NR_FDD_SPCT_SHR_ASYM_SW-1;
//Configuring the NR dedicated spectrum (using a 30 MHz NR cell as an example)
MOD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0, NrReservedRbStartIndex=0,
```

```
NrReservedRbEndIndex=51, NrUlReservedRbStartIndex=0, NrUlReservedRbEndIndex=52;
//Configuring the number of RBs for common control resources
MOD NRDUCELLCORESET: NrDuCellId=0, CommonCtrlResStartSymbol=SYM0, CommonCtrlResRbNum=RB24;
//Configuring the SSB frequency-domain position based on actual conditions (using SsbFreqPos set to
426860 as an example)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_FDD, SsbDescMethod=SSB_DESC_TYPE_NARFCN,
SsbFreqPos=426860;
//Configuring the SSB and SIB1 periods
MOD NRDUCELL: NrDuCellId=0, SsbPeriod=MS20, Sib1Period=MS40;
//(Optional, configurable only when the cell TX/RX mode is not 8T8R) Configuring RE-level avoidance of the
LTE CSI-RS by NR
MOD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0,
FddLteNrSpctShrSwitch=LTE_CSI_RS_AVOID_SW-1;
//(Optional, required when the cell TX/RX mode is 8T8R) Configuring subframe-level avoidance of the LTE
CSI-RS by NR
MOD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0,
FddLteNrSpctShrSwitch=LTE_CSI_RS_AVOID_SW-0;
//Selecting the LTE_CRS_RATEMATCH_ALL_SYM_SW option (assuming that LTE performance needs to be
preferentially ensured)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SpectrumCloudEnhSwitch=LTE_CRS_RATEMATCH_ALL_SYM_SW-1;
//Activating the cell
ACT NRCELL: NrCellId=0;
```

Optimization Command Examples (Single LTE Cell)

Optimization command examples on the LTE side

```
//(Optional, required when NR performs CRS rate matching in the pattern of one LTE CRS port) Adjusting
the PA configuration of the cell
MOD CELLDLPCPDSCHPA: LocalCellId=0, PaPcOff=DB0_P_A;
//Turning on the LTE and NR shared spectrum allocation adaptation switch
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSpctShrSwitch=LNR_RES_ALLOC_ADAPT_SW-1;
//Turning on the switch controlling inter-RAT preferential guarantee for GBR services
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LteNrSpctShrSwitch=GBR_PRI_ALLOC_SW-1;
//Turning on the LTE and NR SRS resource allocation optimization switch
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSpctShrSwitch=LTE_NR_SRS_ALLOC_OPT_SW-1;
//(Optional, recommended only when the SSB frequency-domain position is beyond the NR-dedicated
spectrum) Selecting the HDSS_DL_PRB_ALLOCATE_OPT_SW and LNR_RES_ALLOC_DIRECTION_OPT_SW
options
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSpctShrSwitch=HDSS_DL_PRB_ALLOCATE_OPT_SW-1&LNR_RES_ALLOC_DIRECTION_OPT_SW-1;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Selecting the
LNR_3MS_FRAME_OFFSET_DIFF_SW option
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudExtSwitch=LNR_3MS_FRAME_OFFSET_DIFF_SW-1;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Configuring
the frame offset for the cell with the LNR_3MS_FRAME_OFFSET_DIFF_SW option selected
MOD CELLFRAMEOFFSET: LocalCellId=0, FrameOffsetMode=CustomFrameOffset, FrameOffset=0;
//Turning on the switch for "cross-RAT configuration check in DSS scenarios"
MOD ENODEBRESMODEALGO: ParamCfgProtectSw=DSS_CROSS_RAT_CFG_CHECK_SW-1;
```

Optimization command examples on the NR side

```
//Configuring the position of the downlink additional DMRS
MOD NRDUCELLPDSCH: NrDuCellId=0, DlAdditionalDmrsPos=POS1;
//Configuring CRS rate matching in the pattern of one LTE CRS port
MOD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0,
FddLteNrSpctShrSwitch=LTE_CRS_PORT_1_RM_SW-1;
//Selecting the LTE_CRS_RATEMATCH_ALL_SYM_SW option (recommended when the NR dedicated
spectrum is greater than or equal to 10 MHz)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SpectrumCloudEnhSwitch=LTE_CRS_RATEMATCH_ALL_SYM_SW-1;
//Turning on the LTE SS/PBCH rate matching optimization switch and PSD adaptation switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SpectrumCloudEnhSwitch=LTE_SS_PBCH_RM_OPT_SW-1&PWR_SPCT_DENSITY_ADAPT_SW-1;
//(Optional, required when the cell TX/RX mode is 8T8R, 16T16R, or 32T32R and four SSB beams are
```

```
configured) Turning on the SSB beam adjustment switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=SSB_ADJ_SW-1;
//(Optional, recommended only when the SSB frequency-domain position is beyond the NR-dedicated spectrum) Selecting the INIT_BWP_FULL_BW_SW and BWP0_DL_FULL_BW_SW options
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
BwpConfigPolicySwitch=INIT_BWP_FULL_BW_SW-1&BWP0_DL_FULL_BW_SW-1;
//(Optional, applicable only when the SSB is configured within the shared spectrum) Setting SsbTimePos to SSB23_PCI_MOD2_STAGGER
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=SSB_ADJ_SW-0;
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_FDD, SsbTimePos=SSB23_PCI_MOD2_STAGGER;
//(Optional, applicable only when the SSB is configured within the NR-dedicated spectrum) Setting SsbTimePos to SSB_PCI_MOD3_STAGGER
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=SSB_ADJ_SW-1;
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_FDD, SsbTimePos=SSB_PCI_MOD3_STAGGER;
//Enabling paging-OSI staggering
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, PagingSchAlgoSwitch=PAGING_OSI_STAGGER_SW-1;
//Enabling time-domain RMSI staggering
MOD NRDUCELLDLSCH: NrDuCellId=0, DlSchAlgoSwitch=TIME_DOMAIN_RMSI_STAGGER_SW-1;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Selecting the LNR_3MS_FRAME_OFFSET_DIFF_SW option
MOD NRDUCELLSPCTSHARE: NrDuCellId=0,
LnrSpectrumShareSwitch=LNR_3MS_FRAME_OFFSET_DIFF_SW-1;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Configuring the frame offset for the cell with the LNR_3MS_FRAME_OFFSET_DIFF_SW option selected
MOD GNBFREQBANDCONFIG: FrequencyBand=N3, FrameOffset=92160;
//Turning on the switch for "cross-RAT configuration check in DSS scenarios"
MOD GNBOAMPARAM: CfgInterceptPolicySw=DSS_CROSS_RAT_CFG_CHECK_SW-1;
```

Optimization Command Examples (Multiple LTE Cells)

Optimization command examples on the LTE side

```
//(Optional, required when NR performs CRS rate matching in the pattern of one LTE CRS port) Adjusting the PA configuration of the cell
MOD CELLDLPCPDSCHPA: LocalCellId=0, PaPcOff=DB0_P_A;
MOD CELLDLPCPDSCHPA: LocalCellId=3, PaPcOff=DB0_P_A;
//Turning on the LTE and NR shared spectrum allocation adaptation switch and the switch controlling inter-RAT preferential guarantee for GBR services
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSpctShrSwitch=LNR_RES_ALLOC_ADAPT_SW-1&GBR_PRI_ALLOC_SW-1&LTE_NR_SRS_ALLOC_OPT_SW-1
;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Selecting the LNR_3MS_FRAME_OFFSET_DIFF_SW option
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudExtSwitch=LNR_3MS_FRAME_OFFSET_DIFF_SW-1;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Configuring the frame offset for the cell with the LNR_3MS_FRAME_OFFSET_DIFF_SW option selected
MOD CELLFRAMEOFFSET: LocalCellId=0, FrameOffsetMode=CustomFrameOffset, FrameOffset=0;
//Turning on the switch for "cross-RAT configuration check in DSS scenarios"
MOD ENODEBRESMODEALGO: ParamCfgProtectSw=DSS_CROSS_RAT_CFG_CHECK_SW-1;
```

Optimization command examples on the NR side

```
//Configuring the position of the downlink additional DMRS
MOD NRDUCELLPDSCH: NrDuCellId=0, DlAdditionalDmrsPos=POS1;
//Configuring CRS rate matching in the pattern of one LTE CRS port
MOD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0,
FddLteNrSpctShrSwitch=LTE_CRS_PORT_1_RM_SW-1;
//Turning on the LTE SS/PBCH rate matching optimization switch and PSD adaptation switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SpectrumCloudEnhSwitch=LTE_SS_PBCH_RM_OPT_SW-1&PWR_SPCT_DENSITY_ADAPT_SW-1;
//Enabling 3D coverage pattern
//(Optional, required when the cell TX/RX mode is 8T8R, 16T16R, or 32T32R and four SSB beams are configured) Turning on the SSB beam adjustment switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=SSB_ADJ_SW-1;
//Setting the coverage scenario of broadcast beams (using SCENARIO_202 as an example)
MOD NRDUFDDCELLTRPBEAM: NrDuCellTrpId=0, CoverageScenario=SCENARIO_202;
//(Optional, applicable only when the SSB is configured within the shared spectrum) Setting SsbTimePos to SSB23_PCI_MOD2_STAGGER
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=SSB_ADJ_SW-0;
```



```
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_FDD, SsbTimePos=SSB23_PCI_MOD2_STAGGER;
//(Optional, applicable only when the SSB is configured within the NR-dedicated spectrum) Setting
SsbTimePos to SSB_PCI_MOD3_STAGGER
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=SSB_ADJ_SW-1;
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_FDD, SsbTimePos=SSB_PCI_MOD3_STAGGER;
//Enabling paging-OSI staggering
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, PagingSchAlgoSwitch=PAGING_OSI_STAGGER_SW-1;
//Enabling time-domain RMSI staggering
MOD NRDUCELLDLSCH: NrDuCellId=0, DlSchAlgoSwitch=TIME_DOMAIN_RMSI_STAGGER_SW-1;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Selecting the
LNR_3MS_FRAME_OFFSET_DIFF_SW option
MOD NRDUCELLSPCTSHARE: NrDuCellId=0,
LnrSpectrumShareSwitch=LNR_3MS_FRAME_OFFSET_DIFF_SW-1;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Configuring
the frame offset for the cell with the LNR_3MS_FRAME_OFFSET_DIFF_SW option selected
MOD GNBFREQBANDCONFIG: FrequencyBand=N3, FrameOffset=92160;
//Turning on the switch for "cross-RAT configuration check in DSS scenarios"
MOD GNBOAMPARAM: CfgInterceptPolicySw=DSS_CROSS_RAT_CFG_CHECK_SW-1;
```

Deactivation Command Examples (Single LTE Cell)

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Deactivation command examples on the LTE side

```
//Deactivating the cell
DEA CELL: LocalCellId=0;
//Turning off the switch controlling Hybrid DSS Based on Asymmetric Bandwidth
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudSwitch=OFF,
SpectrumCloudEnhSwitch=LNR_SPECTRUM_SHR_ASYM_SW-0;
//Removing the LTE cell from the LTE spectrum sharing cell group
RMV SPECTRUMCLOUD: LocalCellId=0;
RMV LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0;
//Activating the cell
ACT CELL: LocalCellId=0;
```

Deactivation command examples on the NR side

```
//Deactivating the cell
DEA NRCELL: NrCellId=0;
//Turning off the switch controlling Hybrid DSS Based on Asymmetric Bandwidth
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudSwitch=LTE_NR_FDD_SPCT_SHR_SW-0,
SpectrumCloudEnhSwitch=LTE_NR_FDD_SPCT_SHR_ASYM_SW-0;
//Removing the NR cell from the NR spectrum sharing cell group
RMV NRDUCELLSPCTCLOUD: NrDuCellId=0;
RMV GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0;
//Activating the cell
ACT NRCELL: NrCellId=0;
```

Deactivation Command Examples (Multiple LTE Cells)

Deactivation command examples on the LTE side

```
//Deactivating the cells
DEA CELL: LocalCellId=0;
DEA CELL: LocalCellId=3;
//Turning off the switch controlling Hybrid DSS Based on Asymmetric Bandwidth
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudSwitch=OFF,
SpectrumCloudEnhSwitch=LNR_SPECTRUM_SHR_ASYM_SW-0;
MOD SPECTRUMCLOUD: LocalCellId=3, SpectrumCloudSwitch=OFF,
SpectrumCloudEnhSwitch=LNR_SPECTRUM_SHR_ASYM_SW-0;
//Removing the LTE cells from the LTE spectrum sharing cell group
RMV SPECTRUMCLOUD: LocalCellId=0;
RMV SPECTRUMCLOUD: LocalCellId=3;
RMV LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0;
```

```
//Activating the cells
ACT CELL: LocalCellId=0;
ACT CELL: LocalCellId=3;
```

Deactivation command examples on the NR side

```
//Deactivating the cell
DEA NRCELL: NrCellId=0;
//Turning off the switch controlling Hybrid DSS Based on Asymmetric Bandwidth
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudSwitch=LTE_NR_FDD_SPCT_SHR_SW-0,
SpectrumCloudEnhSwitch=LTE_NR_FDD_SPCT_SHR_ASYM_SW-0;
//Removing the NR cell from the NR spectrum sharing cell group
RMV NRDUCELLSPCTCLOUD: NrDuCellId=0;
RMV GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0;
//Activating the cell
ACT NRCELL: NrCellId=0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Whether Hybrid DSS Based on Asymmetric Bandwidth has taken effect can be queried using MML commands or by tracing signaling messages. The TA offsets that take effect on the LTE and NR sides can be queried using MML commands.

When Hybrid DSS Based on Asymmetric Bandwidth (multiple LTE cells) is enabled, activation verification needs to be performed in both LTE cells.

Using MML Commands

Hybrid DSS Based on Asymmetric Bandwidth takes effect when the following conditions on both the eNodeB and gNodeB sides are met:

- The value of the **Spectrum Sharing Status** parameter in the output of the eNodeB MML command **DSP LTENRSPCTSHRCELLGRP** is **DYNAMIC_SHARING**.
- The value of the **Spectrum Sharing Status** parameter in the output of the gNodeB MML command **DSP GNBDULTENRSPCTSHRCG** is **Dynamic Sharing**.

Run the following commands to observe the TA offsets that take effect on the LTE and NR sides:

- On the eNodeB side, run the **DSP LTENRSPCTSHRCELLGRP** command. **Actual TA Offset** in the command output indicates the actually effective TA offset in the current LTE spectrum sharing cell group.
- On the gNodeB side, run the **DSP GNBDULTENRSPCTSHRCG** command. **Actual TA Offset** in the command output indicates the actually effective TA offset in the current NR spectrum sharing cell group.

Tracing Signaling Messages

NOTICE

Before activating this function in LTE and NR cells, start signaling tracing on the MAE-Access. Otherwise, messages regarding the activation of this function cannot be observed after this function is activated.

Hybrid DSS Based on Asymmetric Bandwidth takes effect when the following conditions on both the eNodeB and gNodeB sides are met:

Perform the following steps to start eNodeB message tracing on the MAE-Access:

Step 1 Log in to the MAE-Access. Choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. In the navigation tree of the displayed window, expand **LTE** > **Application Layer** > **Inter-RAT Huawei-Proprietary Interface Trace**.

Step 2 In the displayed dialog box, select an NE and set related parameters. Then, click **Finish** to start an inter-RAT Huawei-proprietary interface tracing task.

Step 3 After this function is activated in LTE and NR cells, messages are traced over the inter-RAT Huawei-proprietary interface. View the value of the **lnss-group-status-change-info** > **nr-ul-cell-information/nr-dl-cell-information** > **spectrum-sharing-status** IE in the **LNR_INTER_RAT_SPECTRUM_CHANGE_INDICATION** message. The value is **ss-dynamic-sharing**.

----End

Perform the following steps to start gNodeB message tracing on the MAE-Access:

Step 1 Log in to the MAE-Access. Choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. In the navigation tree of the displayed window, expand **NR** > **Application Layer** > **Inter-RAT Huawei-Proprietary Interface Trace**.

Step 2 In the displayed dialog box, select an NE and set related parameters. Then, click **Finish** to start an inter-RAT Huawei-proprietary interface tracing task.

Step 3 After this function is activated in LTE and NR cells, messages are traced over the inter-RAT Huawei-proprietary interface. View the value of the **lnss-group-status-change-info** > **nr-ul-cell-information/nr-dl-cell-information** > **spectrum-sharing-status** IE in the **LNR_INTER_RAT_SPECTRUM_CHANGE_INDICATION** message. The value is **ss-dynamic-sharing**.

----End

4.4.3 Network Monitoring

This function increases cell downlink throughput as follows:

- Average downlink throughput of an NR cell = **N.ThpVol.DL.Cell**/ **N.ThpTime.DL.Cell**
- Average downlink throughput of an LTE cell = **L.Thrp.bits.DL**/ **L.Thrp.Time.Cell.DL.HighPrecision**

5 DSS/Hybrid DSS Performance Boost

To further improve spectrum utilization when LTE FDD and NR share spectrum, DSS/Hybrid DSS Performance Boost is introduced. This feature includes the following functions: LTE PDCCH overhead optimization, LTE CRS overhead optimization, LTE polarization-based power assistance, optimized scheduling of LTE and NR UEs, DSS preferential-resource percentage adaptation, DSS inter-RAT PDCCH spatial multiplexing, DSS PDCCH resource sharing optimization, and LNR independent configuration of uplink and downlink resource percentages. For details about these functions (excluding LNR independent configuration of uplink and downlink resource percentages), see *LTE FDD and NR Dynamic Spectrum Sharing*. **Table 5-1** describes whether the individual functions are compatible with LTE FDD and NR Flash Dynamic Spectrum Sharing and Hybrid DSS Based on Asymmetric Bandwidth.

Table 5-1 Functions in DSS/Hybrid DSS Performance Boost

Function	LTE FDD and NR Flash Dynamic Spectrum Sharing	Hybrid DSS Based on Asymmetric Bandwidth
LTE PDCCH overhead optimization	Compatible	Incompatible
LTE CRS overhead optimization	Compatible	Compatible
LTE polarization-based power assistance	Compatible	Compatible
Optimized scheduling of LTE and NR UEs	Compatible	Compatible
DSS preferential-resource percentage adaptation	Compatible	Compatible
DSS inter-RAT PDCCH spatial multiplexing	Compatible	Compatible
DSS PDCCH resource sharing optimization	Compatible	Compatible

Function	LTE FDD and NR Flash Dynamic Spectrum Sharing	Hybrid DSS Based on Asymmetric Bandwidth
LNR independent configuration of uplink and downlink resource percentages	Incompatible	Compatible

5.1 Principles

The principles of the functions (excluding LNR independent configuration of uplink and downlink resource percentages) in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Principles" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

LNR independent configuration of uplink and downlink resource percentages

When the **LNR_INDEP_UL_DL_RES_RATIO_SW** option of the LTE parameter **SpectrumCloud**.***SpectrumCloudExtSwitch*** is selected, **the uplink and downlink LTE-preferred resource percentages can be configured independently**. The uplink percentage is specified by the **SpectrumCloud**.***UlLtePreferredResRatio*** parameter and the downlink percentage is specified by the **SpectrumCloud**.***DlLtePreferredResRatio*** parameter.

This function requires that the NR SSB be configured in the NR-dedicated spectrum.

5.2 Network Analysis

5.2.1 Benefits

The benefits of the functions (excluding LNR independent configuration of uplink and downlink resource percentages) in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Benefits" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

LNR independent configuration of uplink and downlink resource percentages

This function enhances the flexibility in configuring uplink and downlink resources for LTE and NR.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.2.2 Impacts**.

The network impacts of the functions (excluding LNR independent configuration of uplink and downlink resource percentages) in DSS/Hybrid DSS Performance

Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Impacts" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

LNR independent configuration of uplink and downlink resource percentages

When this function is enabled, the impacts are as follows:

- A higher uplink LTE-preferred resource percentage indicates that more resources are allocated to LTE and fewer resources to NR in the uplink.
- A higher downlink LTE-preferred resource percentage indicates that more resources are allocated to LTE and fewer resources to NR in the downlink.

5.3 Requirements

5.3.1 Licenses

The license requirements of the functions in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Licenses" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

The prerequisite functions of the functions (excluding LNR independent configuration of uplink and downlink resource percentages) in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Prerequisite Functions" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

LNR independent configuration of uplink and downlink resource percentages requires the following functions.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Hybrid DSS Based on Asymmetric Bandwidth	**SpectrumCloud.*SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR** and the **LNR_SPECTRUM_SHR_ASYM_SW** option of the **SpectrumCloud.*SpectrumCloudEnhSwitch*** parameter selected	*LTE FDD and NR Hybrid Dynamic Spectrum Sharing*	LNR independent configuration of uplink and downlink resource percentages can be enabled only when Hybrid DSS Based on Asymmetric Bandwidth is enabled.

5.3.2.2 Mutually Exclusive Functions

The mutually exclusive functions of the functions (excluding LNR independent configuration of uplink and downlink resource percentages) in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Mutually Exclusive Functions" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

LNR independent configuration of uplink and downlink resource percentages has no mutually exclusive functions.

5.3.2.3 Function Impacts

The function impacts of the functions (excluding LNR independent configuration of uplink and downlink resource percentages) in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Function Impacts" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

LNR independent configuration of uplink and downlink resource percentages has no function impacts.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- LTE CRS overhead optimization

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the base station model requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.

- LTE polarization-based power assistance

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the base station model requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.

- Optimized scheduling of LTE and NR UEs

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the base station model requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.

- DSS preferential-resource percentage adaptation

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the base station model requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.

- DSS inter-RAT PDCCH spatial multiplexing

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the base station model requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. In addition, LampSite base stations cannot be used.

- DSS PDCCH resource sharing optimization

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the base station model requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. In addition, LampSite base stations cannot be used.

- LNR independent configuration of uplink and downlink resource percentages

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the base station model requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.

For the base station model requirements of Hybrid DSS Based on Asymmetric Bandwidth, see **4.3.3 Hardware**.

Boards

- LTE CRS overhead optimization

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the board requirements of both Hybrid DSS Based on Asymmetric Bandwidth and LTE CRS dynamic muting must be met. For the board requirements of CRS dynamic muting, see the related descriptions in *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*.

- LTE polarization-based power assistance

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, only UBBPe and later series boards can be used among the boards supporting Hybrid DSS Based on Asymmetric Bandwidth.

- Optimized scheduling of LTE and NR UEs

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the board requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.
- DSS preferential-resource percentage adaptation

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the board requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.
- DSS inter-RAT PDCCH spatial multiplexing

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the board requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.
- DSS PDCCH resource sharing optimization

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the board requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.
- LNR independent configuration of uplink and downlink resource percentages

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, only UBBPe and later series boards can be used among the boards supporting Hybrid DSS Based on Asymmetric Bandwidth.

For the board requirements of Hybrid DSS Based on Asymmetric Bandwidth, see **4.3.3 Hardware**.

RF Modules

- LTE CRS overhead optimization

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the RF module requirements of both Hybrid DSS Based on Asymmetric Bandwidth and LTE CRS dynamic muting must be met. For the RF module requirements of CRS dynamic muting, see the related descriptions in *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*.
- LTE polarization-based power assistance

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the RF module requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.
- Optimized scheduling of LTE and NR UEs

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the RF module requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.
- DSS preferential-resource percentage adaptation

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the RF module requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.
- DSS inter-RAT PDCCH spatial multiplexing

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the RF module requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.

- DSS PDCCH resource sharing optimization

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the RF module requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.
- LNR independent configuration of uplink and downlink resource percentages

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the RF module requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.

For the RF module requirements of Hybrid DSS Based on Asymmetric Bandwidth, see **4.3.3 Hardware**.

Cells

- LTE CRS overhead optimization

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. In addition, the TX/RX mode of each LTE cell must be 4T4R or 8T8R, the TX/RX mode of the NR cell must be 4T4R, 8T8R, or 4T, and the **Cell.*CrsPortNum*** parameter must be set to **CRS_PORT_4** for each LTE cell.
- LTE polarization-based power assistance

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. In addition, the cell TX/RX mode must be 4T4R and the **Cell.*CrsPortNum*** parameter must be set to **CRS_PORT_4** for each LTE cell.
- Optimized scheduling of LTE and NR UEs

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.
- DSS preferential-resource percentage adaptation

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.
- DSS inter-RAT PDCCH spatial multiplexing

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. In addition, the TX/RX mode of each LTE cell must be 4T4R, 8T8R, 16T16R, or 32T32R, the TX/RX mode of the NR cell must be 4T, 4T4R, 8T8R, 16T16R, or 32T32R, there cannot be more than two LTE split cells, and the **Cell.*CrsPortNum*** parameter must be set to **CRS_PORT_4** for each LTE cell.
- DSS PDCCH resource sharing optimization

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. In addition, the following requirements must be met:
  - On the NR side:

    - The NR cell bandwidth does not exceed 30 MHz.
    - **NRDUCell**.***NrDuCellNetworkingMode*** is set to **NORMAL_CELL**.
  - On the LTE side:
    - **Cell**.***MultiRruCellFlag*** is set to **BOOLEAN_FALSE** or **Cell**.***MultiRruCellMode*** is set to a value other than **SFN**.
    - **CellDlpcPdschPa.*****PaPcOff*** is set to a value other than **DB_6_P_A** and **DB_4DOT77_P_A**.
    - **Cell.*****CrsPortNum*** is set to **CRS_PORT_4**. Alternatively, **Cell.*****CrsPortNum*** is set to **CRS_PORT_2**, **NRDUCELLPDCCH**.***SpctShrStartSymbol*** is set to **SYM2**, and **NRDUCELLPDCCH**.***OccupiedSymbolNum*** is set to **1SYM**.
- LNR independent configuration of uplink and downlink resource percentages

  To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met.

  This function requires that the NR SSB be configured in the NR-dedicated spectrum.

For the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth, see **4.3.3 Hardware**.

5.3.4 Others

The other requirements of the functions in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Others" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Data preparation for the functions (excluding LNR independent configuration of uplink and downlink resource percentages) in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Data Preparation" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

LNR independent configuration of uplink and downlink resource percentages

Table 5-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-2 Parameter used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Spectrum Cloud Extend Switch	**SpectrumCloud**.***SpectrumCloudExtSwitch***	LNR_INDEP_UL_DL_RES_RATIO_SW	Select this option when LNR independent configuration of uplink and downlink resource percentages is required based on the network plan.
UL LTE-Preferred Res Ratio	**SpectrumCloud**.***UlLtePreferredResRatio***	None	Set this parameter based on the network plan.
DL LTE-Preferred Res Ratio	**SpectrumCloud**.***DlLtePreferredResRatio***	None	Set this parameter based on the network plan.

5.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Configuration using MML commands for the functions (excluding LNR independent configuration of uplink and downlink resource percentages) in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Using MML Commands" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

LNR independent configuration of uplink and downlink resource percentages

Activation Command Examples

```
//Enabling LNR independent configuration of uplink and downlink resource percentages and setting the uplink LTE-preferred resource percentage
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudExtSwitch=LNR_INDEP_UL_DL_RES_RATIO_SW-1, UlLtePreferredResRatio=50, DlLtePreferredResRatio=50;
```

Deactivation Command Examples

```
//Disabling LNR independent configuration of uplink and downlink resource percentages
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudExtSwitch=LNR_INDEP_UL_DL_RES_RATIO_SW-0;
```


5.4.1.3 Using the MAE-Deployment

Configuration using the MAE-Deployment for the functions in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Using the MAE-Deployment" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

5.4.2 Activation Verification

Activation verification of the functions (excluding LNR independent configuration of uplink and downlink resource percentages) in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Activation Verification" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

LNR independent configuration of uplink and downlink resource percentages

After enabling LNR independent configuration of uplink and downlink resource percentages, observe the average number of available LTE downlink RBs and the average number of available LTE uplink RBs in the spectrum sharing cell. This function has taken effect if the average number of available LTE downlink RBs (indicated by **L.ChMeas.PRB.DL.Actual.Avail**) increases with the value of **SpectrumCloud**.***DlLtePreferredResRatio*** and the average number of available LTE uplink RBs (indicated by **L.ChMeas.PRB.UL.Actual.Avail**) increases with the value of **SpectrumCloud**.***UlLtePreferredResRatio***.

5.4.3 Network Monitoring

Network monitoring of the functions in DSS/Hybrid DSS Performance Boost used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Network Monitoring" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6 DSS and NR Flexible Refarming

To further improve spectrum utilization when LTE FDD and NR share spectrum, DSS and NR Flexible Refarming is introduced. When this function is enabled in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the shared spectrum but not the NR-dedicated spectrum will be refarmed. For details about refarming of the shared spectrum to and from NR, see *LTE FDD and NR Dynamic Spectrum Sharing*.

6.1 Principles

The principles of DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Principles" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.2 Network Analysis

6.2.1 Benefits

The benefits of DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Benefits" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

The network impacts of DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Impacts" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.3 Requirements

6.3.1 Licenses

The license requirements of DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Licenses" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

The prerequisite functions of DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Prerequisite Functions" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.3.2.2 Mutually Exclusive Functions

The mutually exclusive functions of DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Mutually Exclusive Functions" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.3.2.3 Function Impacts

The function impacts of DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Function Impacts" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the base station model requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. For details, see **4.3.3 Hardware**.

Boards

To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the board requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. For details, see **4.3.3 Hardware**.

RF Modules

To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the RF module requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. For details, see **4.3.3 Hardware**.

Cells

To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. In addition, the cell TX/RX mode cannot be 16T16R or 32T32R. For the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth, see **4.3.3 Hardware**.

The shared spectrum of an LTE spectrum sharing cell will not be refarmed to NR if each of its co-coverage inter-frequency neighboring cells meets any of the following conditions:

- The cell is a downlink-only cell (with **Cell.*WorkMode*** set to **DL_ONLY**). For details, see *Cell Management*.
- The cell is a licensed assisted access (LAA) cell (with **Cell.*WorkMode*** set to **LAA**). For details, see *Licensed Assisted Access (TDD)*.
- The cell is an eMTC-only cell (with the **EMTC_ONLY_CELL_SWITCH** option of the **CellEmtcAlgo.*EmtcAlgoSwitch*** parameter selected). For details, see *eMTC*.
- Access to the cell is barred.

6.3.4 Others

The other requirements of DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Others" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Data preparation for DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Data Preparation" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.4.1.2 Using MML Commands

Configuration using MML commands for DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Using MML Commands" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.4.1.3 Using the MAE-Deployment

Configuration using the MAE-Deployment for DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Using the MAE-Deployment" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.4.2 Activation Verification

Activation verification of DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Activation Verification" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

6.4.3 Network Monitoring

Network monitoring of DSS and NR Flexible Refarming used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Network Monitoring" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

7 LTE and NR Uplink Spectrum Pooling

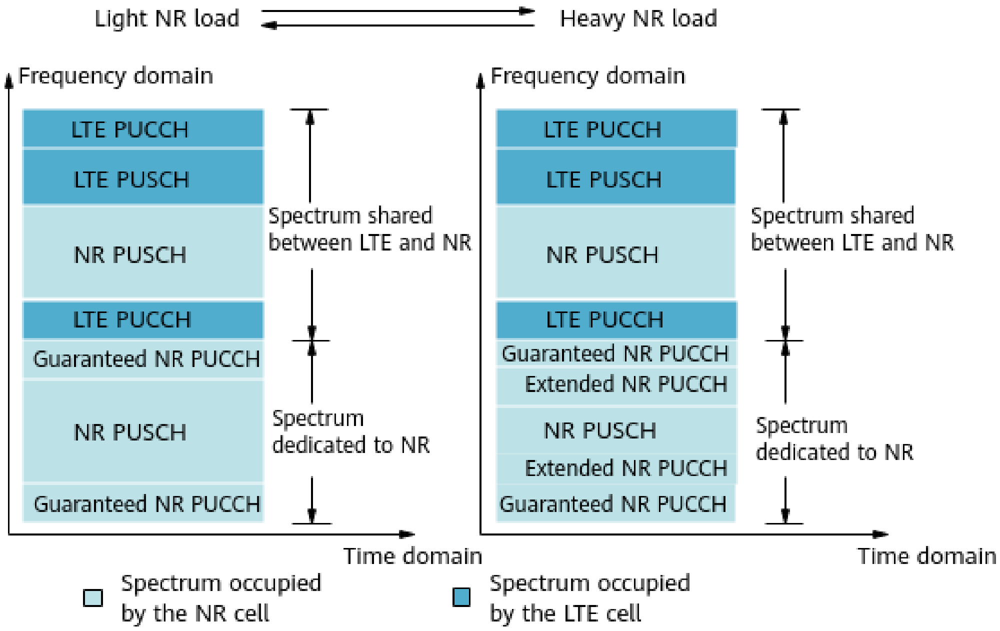
LTE and NR Uplink Spectrum Pooling aims to enhance uplink spectrum utilization when LTE FDD and NR share spectrum. In Hybrid DSS Based on Asymmetric Bandwidth scenarios, the functions (excluding LNR PUCCH RB resource adaptation) in LTE and NR Uplink Spectrum Pooling involve the shared spectrum but not the NR-dedicated spectrum. For details about uplink spectrum pooling in the shared spectrum, see *LTE FDD and NR Dynamic Spectrum Sharing*.

7.1 Principles

The principles of the functions (excluding LNR PUCCH RB resource adaptation) in LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details about LTE and NR Uplink Spectrum Pooling, see the corresponding "Principles" section in *LTE FDD and NR Dynamic Spectrum Sharing*. This section describes only LNR PUCCH RB resource adaptation.

Figure 7-1 shows how **LNR PUCCH RB resource adaptation** works in Hybrid DSS Based on Asymmetric Bandwidth scenarios. LNR PUCCH RB resource adaptation enables the NR PUCCH to utilize both guaranteed RBs and extended RBs. Both guaranteed RBs and extended RBs are deployed in the NR-dedicated spectrum. Extended RBs are adjusted adaptively based on the number of NR UEs and the load. **Under low-load conditions, guaranteed RBs alone are sufficient for the PUCCH to meet service demands. This leaves more available RBs for the NR PUSCH, and thereby increases the uplink UE throughput. As the load increases, extended RBs are also needed for the PUCCH to accommodate additional service demands.**

Figure 7-1 LNR PUCCH RB resource adaptation



Guaranteed and extended NR PUCCH RBs are configured as follows:

- Guaranteed NR PUCCH RBs

  The number of guaranteed NR PUCCH RBs varies depending on the networking as follows:

  – NSA networking: Number of guaranteed NR PUCCH RBs = **NRDUCellPucch**.***Format1RbNum*** + **NRDUCellPucch**.***Format3RbNum*** + **NRDUCellPucch**.***Format4RbNum*** + **NRDUCellPucch**.***CsiDedicatedRbNum*** + **NRDUCellPucch**.***Format4CsiDedicatedRbNum***

  – NSA and SA hybrid networking and SA networking: Number of guaranteed NR PUCCH RBs = **NRDUCellPucch**.***Format1RbNum*** + **NRDUCellPucch**.***Format3RbNum*** + **NRDUCellPucch**.***Format4RbNum*** + **NRDUCellPucch**.***CsiDedicatedRbNum*** + **NRDUCellPucch**.***Format4CsiDedicatedRbNum*** + Number of NR common PUCCH RBs

  The **NRDUCellPucch**.***Format3RbNum*** and **NRDUCellPucch**.***Format4RbNum*** parameters need to be set only when downlink intra-band CA (controlled by the **INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter) or downlink intra-FR inter-band CA (controlled by the **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter) is enabled. The number of NR common PUCCH RBs depends on the setting of the **COMM_PUCCH_RB_RES_OPT_SW** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter. For details about the parameters involved in the preceding formulas and the number of NR common PUCCH RBs, see descriptions of the PUCCH in *Channel Management* in *5G RAN Feature Documentation*.

- Extended NR PUCCH RBs

The number of extended NR PUCCH RBs is dynamically adjusted based on the load.

LNR PUCCH RB resource adaptation is controlled by the **LNR_PUCCH_RB_RES_ADAPT_SW** option of the **NRDUCellGigaBand**.***GigaBandCellAlgoSw*** parameter.

To improve uplink resource utilization, you are advised to turn on the switch for format-3 and format-4 ACK RB resource adaptation (controlled by the **F3_AND_F4_ACK_RB_RES_ADAPT_SW** option of the **NRDUCellPucch**.***PucchAlgoExtSwitch*** parameter).

7.2 Network Analysis

7.2.1 Benefits

The benefits of LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Benefits" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.2.3 Function Impacts**.

The network impacts of LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Impacts" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

7.3 Requirements

7.3.1 Licenses

The license requirements of LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Licenses" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

In Hybrid DSS Based on Asymmetric Bandwidth scenarios, LTE and NR Uplink Spectrum Pooling requires the following functions, in addition to the functions required when it is used in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Prerequisite Functions" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	UL BWP0 Configuration Policy	**NRDUCellBwp**.***UlBwp0ConfigPolicy*** set to **DEFAULT_DEDICATED_BANDWIDTH**	*LTE FDD and NR Hybrid Dynamic Spectrum Sharing*	LTE and NR Uplink Spectrum Pooling can be enabled in Hybrid DSS Based on Asymmetric Bandwidth scenarios only when **UL BWP0 Configuration Policy** is set to **DEFAULT_DEDICATED_BANDWIDTH**.

7.3.2.2 Mutually Exclusive Functions

The mutually exclusive functions of LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Mutually Exclusive Functions" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

7.3.2.3 Function Impacts

The function impacts of LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Function Impacts" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the base station model requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. For details, see **4.3.3 Hardware**.

Boards

To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, only UBBPe and later series boards can be used among the boards supporting Hybrid DSS Based on Asymmetric Bandwidth.

RF Modules

To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the RF module requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. For details, see **4.3.3 Hardware**.

Cells

To enable this function in Hybrid DSS Based on Asymmetric Bandwidth scenarios, the cell requirements of Hybrid DSS Based on Asymmetric Bandwidth must be met. For details, see **4.3.3 Hardware**.

If **NRDUCellSulRelation**.***SulType*** is set to **DYN_SUL**, **NRDUCellGigaBand**.***LnrUlRbEstCorrectCoeff*** must be set to **0**.

7.3.4 Others

The other requirements of LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios are the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Others" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

Data preparation for LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Data Preparation" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

7.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Configuration using MML commands for the functions (excluding LNR PUCCH RB resource adaptation) in LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Using MML Commands" section in *LTE FDD and NR Dynamic Spectrum Sharing*. This section describes only LNR PUCCH RB resource adaptation.

Activation Command Examples

Activation command examples on the NR side

```
//Enabling LNR PUCCH RB resource adaptation
DEA NRCELL: NrCellId=1;
MOD NRDUCELLPUCCH: NrDuCellId=1, Format1RbNum=RB2, Format3RbNum=RB2,
CsiDedicatedRbNum=RB2, Format4RbNum=RB0, Format4CsiDedicatedRbNum=RB0,
PucchAlgoSwitch=COMM_PUCCH_RB_RES_OPT_SW-1;
MOD NRDUCELLGIGABAND: NrDuCellId=1, GigaBandCellAlgoSw=LNR_PUCCH_RB_RES_ADAPT_SW-1,
LnrPucchRbAdjProtDuration=30;
ACT NRCELL: NrCellId=1;
//Turning on the switch for format-3 and format-4 ACK RB resource adaptation
MOD NRDUCELLPUCCH: NrDuCellId=1, PucchAlgoExtSwitch=F3_AND_F4_ACK_RB_RES_ADAPT_SW-1;
```

Deactivation Command Examples

Deactivation command examples on the NR side

```
//Disabling LNR PUCCH RB resource adaptation
DEA NRCELL: NrCellId=1;
MOD NRDUCELLPUCCH: NrDuCellId=1, Format1RbNum=RB2, Format3RbNum=RB4,
CsiDedicatedRbNum=RB4, Format4RbNum=RB0, Format4CsiDedicatedRbNum=RB0,
PucchAlgoSwitch=COMM_PUCCH_RB_RES_OPT_SW-0;
MOD NRDUCELLGIGABAND: NrDuCellId=1, GigaBandCellAlgoSw=LNR_PUCCH_RB_RES_ADAPT_SW-0;
ACT NRCELL: NrCellId=1;
//Turning off the switch for format-3 and format-4 ACK RB resource adaptation
MOD NRDUCELLPUCCH: NrDuCellId=1, PucchAlgoExtSwitch=F3_AND_F4_ACK_RB_RES_ADAPT_SW-0;
```

7.4.1.3 Using the MAE-Deployment

Configuration using the MAE-Deployment for LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Using the MAE-Deployment" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

7.4.2 Activation Verification

Activation verification of LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Activation Verification" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

7.4.3 Network Monitoring

Network monitoring of LTE and NR Uplink Spectrum Pooling used in Hybrid DSS Based on Asymmetric Bandwidth scenarios is the same as in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see the corresponding "Network Monitoring" section in *LTE FDD and NR Dynamic Spectrum Sharing*.

8 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

11 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- Feature parameter description documents in *eRAN Feature Documentation*
  - *Carrier Aggregation*
  - *Uplink Scheduling*
  - *Downlink Scheduling*
  - *Physical Channel Resource Management*
  - *VoLTE*
  - *MIMO*
  - *Massive MIMO Enhancements (FDD)*
  - *eMIMO (FDD)*
  - *UL CoMP*
  - *DRX and Signaling Control*
  - *Intra-RAT Mobility Load Balancing*
  - *LCS*
  - *Extended CP*
  - *Random Access Control*
  - *SFN*
  - *Cell Management*
  - *Flexible Bandwidth based on Overlap Carriers (FDD)*
  - *GSM and LTE Buffer Zone Optimization*
  - *UL Refarming Zero Bufferzone*
  - *UMTS and LTE Zero Bufferzone*
  - *CDMA and LTE Zero Bufferzone*
  - *Uplink Coordinated Scheduling*
  - *CSPC*
  - *Relay*
  - *eMTC*
  - *NB-IoT Basics (FDD)*

- *Extended Cell Range*
- *Superior Uplink Coverage (FDD)*
- *eMBMS*
- *Virtual 4T4R (FDD)*
- *DL CoMP (FDD)*
- *GSM and LTE Dynamic Power Sharing*
- *Super Combined Cell (FDD)*
- *Energy Conservation and Emission Reduction*
- *TDM eICIC (FDD)*
- *ICIC*
- *Short TTI (FDD)*
- *Smart 8T8R (FDD)*
- *PTT*
- *On-Demand TX Power Allocation Under EME*
- *Dedicated Carrier for TM9*
- *Seamless Intra-Band Carrier Joining (FDD)*
- *Compact Bandwidth (FDD)*
- *High Speed Mobility*
- *CPRI Compression*
- *pRRU Uplink Interference Suppression*
- *RAN Sharing*
- *Synchronization*
- *Low-Band Booster*
- *Physical Resource Slicing (FDD)*
- *LumiLink Solution*
- *LTE Flexible Spectrum (FDD)*

- Feature parameter description documents in *5G RAN Feature Documentation*
  - *RuralLink Solution*
  - *Flexible Bandwidth (FDD)*
  - *Carrier Aggregation*
  - *CA Experience Improvement*
  - *Downlink Scheduling*
  - *Channel Management*
  - *Energy Conservation and Emission Reduction*
  - *Power Control*
  - *Multi-Operator Sharing*
  - *Beam Management*
  - *VoNR*
  - *Network Slicing*
  - *Resource Control in NSA Networking*

  - *CoMP*
  - *Indoor Coverage*
  - *Cell Combination*
  - *Hyper Cell*
  - *Scalable Bandwidth*
  - *Super Uplink*
  - *High Speed Mobility*
  - *CPRI Compression*
  - *Cell Management*
  - *Synchronization*
  - *Smart Coverage (FDD)*
  - *Smart 8T8R (FDD)*
  - *Turbo Coverage (FDD)*
  - *Distributed Antenna Solutions (FDD)*
  - *URLLC*
  - *RedCap*
  - *Interference Avoidance*
  - *LumiLink Solution*
  - *5G Networking and Signaling*
  - *Multi-Frequency Smart Aggregation*
  - *Uplink Flexible Spectrum Scheduling*
  - *QCI-based Resource Control*
  - *Green Site*
- *Dynamic Power Sharing Between LTE Carriers*
- *LTE and NR Power Allocation*
- *LTE Spectrum Coordination*
- *GSM and LTE Spectrum Concurrency*
- *UMTS and LTE Dynamic Power Sharing*
- *UMTS and LTE Spectrum Sharing*
- *UMTS and LTE Spectrum Sharing Based on DC-HSDPA*
- *EPC-based NSA Performance Enhancement*
- *LTE and NR Zero Bufferzone*
- *UL and DL Decoupling*